S G A 7

Exposé VI

# FORMAL DEFORMATION THEORY

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## 1. Formal existence theorem

We recall the notion of cofibered category (S.G.A.1.IV).
Let $\underline{C}$ be a fixed category. A cofibered category $\underline{A}$ over $\underline{C}$ is, by definition, a category $\underline{A}$ together with a functor $P : \underline{A} \to \underline{C}$ such that

Ax.1. For any map $f$ in $\underline{C}$ and any object $a$ in $\underline{A}$ with $P(a)$ = the source of $f$, there exists an $f$-morphism $\alpha : a \to b$ which is cocartesian (i.e. for any $f$-morphism $\alpha' : a \to b'$ there exists a unique $T$-morphism $\tau : b \to b'$ in $\underline{A}$ such that $\alpha' = \tau \cdot \alpha$ where $T$ = the target of $f$).

Ax.2. A composit of cocartesian morphism in $\underline{A}$ is again cocartesian.

The following properties of a cofibered category $P : \underline{A} \to \underline{C}$ follow immediately from the definitions:

(cancellation) Let $a \overset{\alpha}{\underset{\beta}{\rightrightarrows}} a'$ be any two cocartesian maps such that $P(\alpha) = P(\beta)$. If there exists a cocartesian map $b \xrightarrow{\gamma} a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

(factorization) Given cocartesian maps $a \xrightarrow{\alpha} a'$, $a \xrightarrow{\beta} a''$, there exists a cocartesian map $a' \xrightarrow{\gamma} a''$ such that $\gamma\alpha = \beta$ if and only if there exists $f : P(a') \to P(a'')$ in $\underline{C}$ such that $f \cdot P(\alpha) = P(\beta)$. Furthermore $\gamma$ is unique.

Let $\underline{A}$ be a cofibered category over $\underline{C}$. For each object $S$ in $\underline{C}$ we denote by $\underline{A}(S)$ the subcategory of $\underline{A}$ whose objects are the objects $a$ in $\underline{A}$ such that $P(a) = S$, and $\mathrm{Hom}_{\underline{A}(S)}(a,a') = \{\alpha \in \mathrm{Hom}_{\underline{A}}(a,a') \mid P(\alpha) = \mathrm{id}_S\}$.
A cofibered groupoid over $\underline{C}$ is a cofibered category $\underline{A}$ over $\underline{C}$ such that $\underline{A}(S)$

is a groupoid for every object $S$ in $\underline{C}$. We recall that a groupoid is a category in which every morphism is an isomorphism. Thus, a cofibered groupoid over $\underline{C}$ is a cofibered category $P : \underline{A} \to \underline{C}$ such that every map in $\underline{A}$ is cocartesian. Furthermore, for every map $\alpha \in F\ell(\underline{A})$, $\alpha$ is an isomorphism in $\underline{A}$ if and only if $P(\alpha)$ is an isomorphism.

Throughout the rest of this section, we fix a complete noetherian local ring $\Lambda$ with the residue field $k$, and we set $\underline{C}_\Lambda$ to be the category of artinian local $\Lambda$-algebras with the same residue field $k$, where the maps are local homomorphisms over $\Lambda$. From 1.2 on, a cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is always assumed to have the property that $\underline{A}(k) = \{e\}$, where $\{e\}$ denotes the category in which $\mathrm{Hom}_{\{e\}}(a, b)$ consists of one and only one element for any two objects $a, b$ in $\{e\}$. A map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called simply a $\underline{C}_\Lambda$-map if $p(a') = p(a)$ and $p(\alpha) = 1_{p(a)}$. A map $a' \xrightarrow{\alpha} a$ on $\underline{A}$ will be called surjective, by abuse of language, if $p(\alpha) : p(a') \longrightarrow p(a)$ is surjective in $\underline{C}_\Lambda$.

Ex.1.1.(a) Let $F : \underline{C} \longrightarrow$ (categories) be a functor. Define the category $\underline{F}$ as follows: an object in $\underline{F}$ is a pair $(S,a)$ where $S \in \mathrm{ob}(\underline{C})$ and $a \in \mathrm{ob}F(S)$, and a map $(S,a) \to (S',a')$ is a pair $(f,u')$ where $f : S \to S'$ is a map in $\underline{C}$ and $u' : F(f)a \to a'$ is a map in $F(S')$. Composition of two maps $(S,a) \xrightarrow{(f,u')} (S',a') \xrightarrow{(g,u'')} (S'',a'')$ is given by $(gf,\ u''\cdot(F(g)u')) : (S,a) \to (S'',a'')$. With respect to the functor $P : \underline{F} \to \underline{C}$ given by $(S,a) \mapsto S$, $\underline{F}$ becomes a cofibered category over $\underline{C}$. If $F(S)$ is a groupoid for every $S \in \mathrm{ob}(\underline{C})$, then $\underline{F}$ is a cofibered groupoid over $\underline{C}$. In particular, a functor $F : \underline{C} \to$ (groups) yields a cofibered group, and a functor $F : \underline{C} \to$ (Ens) yields a cofibered discrete groupoid. We also note that a natural transformation $\varphi : F \to G$, where $F,G : \underline{C} \to$ (categories) are functors, induces a cocartesian functor $\varphi : \underline{F} \to \underline{G}$ over $\underline{C}$.

Ex.1.1.(b) Let $X$, $Y$ be schemes over $\Lambda$ together with a fixed

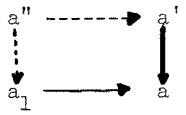
isomorphism $X \underset{\Lambda}{\otimes} k \xleftrightarrow{\tau} Y \underset{\Lambda}{\otimes} k$ . Then the functor $\mathrm{Isom}_{X,Y} : \underline{C}_\Lambda \to$ (groupoids) given by $\mathrm{Isom}_{X,Y}(S)$ = the set of isomorphisms $X \underset{\Lambda}{\otimes} S \xleftrightarrow{\sigma} Y \underset{\Lambda}{\otimes} S$ such that $\sigma \underset{\Lambda}{\otimes} k = \tau$ **yields a cofibered groupoid** $P : \underline{\mathrm{Isom}}_{X,Y} \to \underline{C}_\Lambda$ .

Ex.1.1.(c) Let $X_0$ be a fixed scheme over $k$ , and let $\underline{M}_{X_0}$ be the category consisting of deformations of $X_0$ over $\underline{C}_\Lambda$ (see §4). Thus an object in $\underline{M}_{X_0}$ is a pair $(R,X)$ where $R$ is in $\underline{C}_\Lambda$ and $X$ is a deformation of $X_0$ to $R$ . Then the functor $P : \underline{M}_{X_0} \to \underline{C}_\Lambda$ given by $(R,X) \rightsquigarrow R$ defines a cofibered groupoid over $\underline{C}_\Lambda$ .

Ex.1.1.(d) For any groupoid $X$ we denote by $[X]$ the discrete groupoid consisting of the isomorphism classes of objects in $X$ . If $P : \underline{F} \to \underline{C}_\Lambda$ is a cofibered groupoid, we obtain a functor $[\underline{F}] : \underline{C}_\Lambda \to$ (Ens) given by $[\underline{F}](S) = [\underline{F}(S)]$ , and in turn a cofibered discrete groupoid, which is, by abuse of notation, denoted by $[\underline{F}]$ .

Definition 1.2. *A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called semi-homogeneous if the following properties hold:*

(1) *every diagram*




*of solid arrows in $\underline{A}$ where $a' \to a$ is surjective can be completed into a commutative diagram including dotted arrows.*

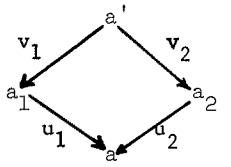
(2) *Let $R \underset{k}{\times} k[\varepsilon] \underset{\pi_2}{\overset{\pi_1}{\rightrightarrows}} R$ be the projection maps, where $k[\varepsilon]$ is the ring of dual numbers over $k$ , and let $a' \underset{b'}{\overset{\alpha}{\rightrightarrows}} a$ be $\pi_1$-maps in $\underline{A}$ . Then there exists a $\underline{C}_\Lambda$-map $\gamma : a' \to b'$ with $\alpha = \beta \cdot \gamma$ if and only if $(\pi_2)_*(a') \approx (\pi_2)_*(b')$ .*

Remark 1.3.(a) A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). We note that if $\underline{A}$ is semi-homogeneous. Then so is $[\underline{A}]$ , and in particular 1.2(2) entails the canonical map

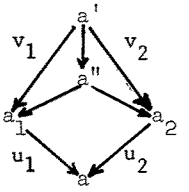
$$[\underline{A}(R \underset{k}{\times} k[\varepsilon])] \longrightarrow [\underline{A}(R)] \times [\underline{A}(k[\varepsilon])]$$

to be bijective.

Remark 1.3.(b) Consider a commutative diagram




in a cofibered category $\underline{A}$ over $\underline{C}_{\wedge}$ . Set $R = P(a)$ , $R' = P(a')$ and $R_i = P(a_i)$ for $i = 1, 2$ . If we set $a''$ to be the "direct image" under the map $(P(v_1),P(v_2)) : R' \to R_1 \underset{R}{\times} R_2$ , we obtain a commutative diagram

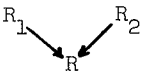



Thus 1.2(1) can be replaced by

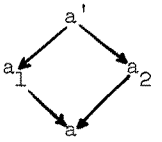
(1)': every diagram



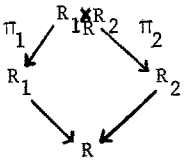

in $\underline{A}$ above




in $\underline{C}_{\wedge}$ , where $R_2 \to R$ is surjective can be completed to a commutative diagram




above




where $\pi_i$ $(i = 1,2)$ are projections.

Remark 1.3.(c) If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_{\wedge}$ , then $\underline{A}(k[\varepsilon])]$ is canonically provided with a vector space structure over $k$ , where $k[\varepsilon]$ is the ring of dual numbers over $k$ . Indeed, let $\underline{V}$ be the category of

finite-dimensional vector spaces over $k$ . Then any functor $H : \underline{\underline{V}} \to (Ens)$ preserving the product is canonically factored through

$$\begin{array}{ccc} \underline{\underline{V}} & \xrightarrow{H} & (Ens) \\ & \searrow^{\tilde{H}} \quad \nearrow & \\ & \underline{\underline{V}} & \end{array}$$

where $\underline{\underline{V}} \to (Ens)$ is the forgetful functor. On the other hand, the functor $G : \underline{\underline{V}} \longrightarrow \underline{C}_\wedge$ given by $W \longmapsto k[W] = k \oplus W$ , with $W^2 = 0$ , transforms products in $\underline{\underline{V}}$ into fibre products over $k$ in $\underline{C}_\wedge$ . If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\wedge$ , then $[\underline{A}(k[V] \underset{k}{\times} k[W])] \longrightarrow [\underline{A}(k[V])] \times [\underline{A}(k[W])]$ is bijective by 1.3(a), and the functor $V \longmapsto [\underline{A}(k[V])]$ commutes with products, hence the result.

Definition 1.4. (1) Let $R \in ob(\underline{C}_\wedge)$ . A small extension of $R$ in $\underline{C}_\wedge$ is a surjective map $R' \xrightarrow{f} R$ in $\underline{C}_\wedge$ such that $(\operatorname{Ker} f)^2 = 0$ and $\operatorname{Ker} f \simeq k$ . In other words, $R' \xrightarrow{f} R$ in $\underline{C}_\wedge$ is a small extension of $R$ if

$$0 \longrightarrow k \xrightarrow{t} R' \xrightarrow{f} R \longrightarrow 0$$

is exact for some $t \in R'$ with $t^2 = 0$ .

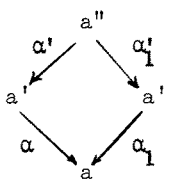
(2) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$ , and let $a$ be an object in $\underline{A}$ . A small prolongation of $a$ is an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ such that $P(\alpha)$ is a small extension in $\underline{C}_\wedge$ . An arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called "versal for small prolongations of $a$ " if, for every small prolongation $a_1' \longrightarrow a$ in $\underline{A}$ , there exists a commutative diagram

$$\begin{array}{ccc} a' & \longrightarrow & a_1' \\ & {\scriptstyle\alpha}\searrow \quad \swarrow & \\ & a & \end{array}$$

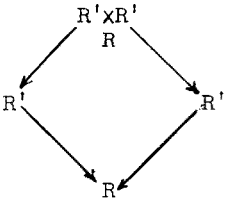
Proposition 1.5. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\wedge$ .
(1) Let $R \underset{k}{\times} k[\varepsilon] \underset{\pi_2}{\overset{\pi_1}{\rightrightarrows}} R$ be projection maps in $\underline{C}_\wedge$ , and let $a' \xrightarrow{\alpha} a$

be a $\pi_1$-map. Then there exists a map $a \xrightarrow{\beta} a'$ such that $\alpha\beta = 1_a$ if and only if there exists a map $a \longrightarrow (\pi_2)_* a'$ .

(2) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$ , and consider a commutative diagram




above

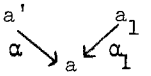



Define the canonical map $\tilde{f} : R' \underset{R}{\times} R' \longrightarrow k[\mathrm{Ker}\, f]$ by $\tilde{f}(r_1', r_2') = r_1'(0)+(r_1'-r_2')$ . If there exists a map $a' \longrightarrow (\tilde{f})_*(a")$ , then there exists a map $\gamma: a' \longrightarrow a$ such that $\alpha = \alpha_1 \cdot \gamma$ .

Proof (1) The part "only if" is clear. Now assume that there exists a map $\varphi : a \longrightarrow (\pi_2)_* a'$ . If we set $h : R \longrightarrow R \underset{k}{\times} k[\varepsilon]$ by $h = 1 \times P(\varphi)$ , then the composite map $R \xrightarrow{h} R \underset{h}{\times} k[\varepsilon] \xrightarrow{\pi_1} R$ is the identity, and therefore we get maps $a \underset{\alpha_1}{\overset{\gamma}{\rightleftarrows}} f_* a$ such that $\alpha_1 \cdot \gamma = 1_a$ where $P(\gamma) = h$ and $P(\alpha_1) = \pi_1$ . On the other hand, $\pi_2 \cdot h = P(\varphi)$ and hence there exists a $\pi_2$ -map $h_* a \longrightarrow (\pi_2)_* a'$ . This means that $(\pi_2)_*(h_* a) = (\pi_2)_* a'$ , and therefore by 1.2(2) we have a $\underline{C}_\Lambda$-map $u : h_* a \longrightarrow a'$ such that $\alpha \cdot u = \alpha_1$ . Define $\beta : a \longrightarrow a'$ by $\beta = u \cdot \gamma$ . Then $\alpha \cdot \beta = \alpha \cdot u \cdot \gamma = \alpha_1 \cdot \gamma = 1_a$ .

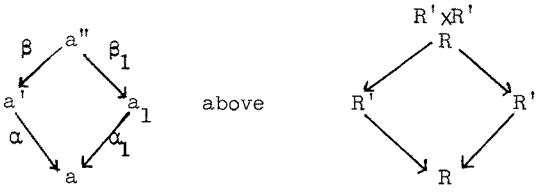
(2) The map $1 \times \tilde{f} : R' \underset{R}{\times} h' \longrightarrow R' \underset{k}{\times} k[\mathrm{Ker}\, f]$ is an isomorphism compatible with the projections on the first factor, and $\pi_2' \cdot (1 \times \tilde{f}) = \tilde{f}$ where $\pi_2' : R' \underset{k}{\times} k[\mathrm{Ker}\, f] \longrightarrow k[\mathrm{Ker}\, f]$ is the projection map. We note that $k[\mathrm{Ker}\, f] \overset{k}{\simeq} k[\varepsilon]$ since $R' \xrightarrow{f} R$ is a small extension. Therefore, if there exists a map $a' \longrightarrow (\tilde{f})_* a" = (\pi_2)_* [(1 \tilde{f})_* a"]$ , then by (1) above there exists a map $\gamma' : a' \longrightarrow a"$ such that $\alpha' \cdot \gamma' = 1_{a'}$ . Define $\gamma : a' \longrightarrow a$ by $\gamma = \alpha_1' \cdot \gamma'$ . We then have $\alpha_1 \cdot \gamma = \alpha$ .

Corollary 1.6. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$ and consider a diagram



in $\underline{A}$ where $\alpha_1$ is a small prolongation. Assume that $a$ is versal for small prolongations of $e$. Then there exists a map $a' \xrightarrow{\rho} a_1$ such that $\alpha = \alpha_1\rho$ in $\underline{A}$ if and only if there exists a map $f : P(a') \to P(a_1)$ such that $P(\alpha) = P(\alpha_1)\cdot f$ in $\underline{C}_\Lambda$.

Proof: Assume that $P(\alpha) = P(\alpha_1)\cdot f$ where $f : P(a') \to P(a_1)$. Replacing $a'$ by its direct image under $f$, we may and shall assume that $P(\alpha) = P(\alpha_1)$. Set $R' = P(a_1)$, $R = P(a)$. Since $\underline{A}$ is semi-homogeneous, we get a commutative diagram




Since $a$ is versal for small prolongations of $e$, so is $a'$ and therefore our statement follows from 1.5(2).

Definition 1.7. For each object $R$ in $\underline{C}_\Lambda$ we put $\bar{\Omega}_R = M/(m_\Lambda R + M^2)$ where $m_\Lambda$ $M$ are the maximal ideals of $\Lambda$, $R$ respectively. It is a finite-dimensional vector space over $k$, and $R \longmapsto \bar{\Omega}_R$ defines a functor $\bar{\Omega} : \underline{C}_\Lambda \to \underline{V}$ where $\underline{V}$ is the category of finite-dimensional vector spaces over $k$. If $\underline{A}$ is a cofibered category over $\underline{C}_\Lambda$, then the composite functor $\underline{A} \xrightarrow{P} \underline{C}_\Lambda \to \underline{V}$ is, by abuse of notation denoted by $\bar{\Omega}$ again.

Proposition 1.8. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$. Then, given an object $a$ in $\underline{A}$ versal for small prolongations of $e$, there exists $a' \to a$ such that

(i) $a'$ is versal for small prolongations of $a$

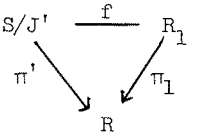
(ii) $p(a') \longrightarrow p(a)$ is surjective; and the kernel is annihilated by the maximal ideal of $p(a)$ .

(iii) $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

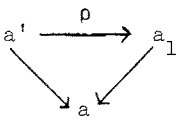
Proof: Set $R = P(a)$ and consider an exact sequence $0 \to J \to S \to R \to 0$ where $S$ is a formal power-series ring over $\Lambda$ in a finite number of variables, and $J \subset M^2$ where $M$ is the maximal ideal of $S$ . Let $\Sigma$ be the set of ideals $J_1$ in $S$ such that (i) $J \supset J_1 \supset MJ$ and (ii) there exists $\pi_1$-morphism $a_1 \to a$ in $\underline{A}$ where $\pi_1 : S/J_1 \to R$ is the canonical map. The property 1.2(1) of $\underline{A}$ together with the fact that $J/MJ$ is a finite dimensional vector space over $k$ entails that there exists a smallest ideal in the family $\Sigma$ . Let it be $J'$ and choose a $\pi'$-morphism $a' \overset{\mathfrak{a}}{\to} a$ where $\pi' : S/J' \to R$ is the canonical map. Note that $\overline{\Omega}_{a'} \to \overline{\Omega}_a$ is an isomorphism since $J' \subset J \subset M^2$ . We claim that $a' \overset{\mathfrak{a}}{\to} a$ is versal for small prolongations of $a$ . Indeed, let $a_1 \to a$ be a small prolongation. If we put $R_1 = P(a_1)$ , we obtain an exact commutative diagram

$$\begin{array}{ccccccc} 0 & \to & J/MJ & \to & S/MJ & \to & R \to 0 \\ & & \downarrow h & & \downarrow h_1 & & \| \\ 0 & \to & k & \to & R_1 & \to & R \to 0 \end{array}$$

If $h = 0$ , then $\pi_1 : R_1 \to R$ admits a cross-section so that $h_1$ induces $f : S/J' \to R_1$ . If $h \neq 0$ , then $h_1 : S/MJ \to R_1$ is surjective and hence $h_1$ induces $f : S/J' \to R_1$ by the definiton of the ideal $J'$ . Therefore in either cases we obtain a commutative diagram




Since $a$ is versal for small prolongations of $e$ , it follows from 1.6 that we obtain a commutative diagram




Let $P : \underline{A} \to \underline{C}_\Lambda$ be a cofibered category (in groupoids). It is then clear that we obtain a cofibered category $pro(P) : pro(\underline{A}) \to pro(\underline{C}_\Lambda)$ (in groupoids) **(pour la définition de la catégorie de pro-objets $pro(\underline{A})$ , voir A. Grothendieck, Sém. Bourbaki 195).**

$$\begin{array}{ccc} \underline{A} & \longrightarrow & pro(\underline{A}) \\ {\scriptstyle P}\downarrow & & \downarrow{\scriptstyle pro(P)} \\ \underline{C}_\Lambda & \longrightarrow & pro(\underline{C}_\Lambda) \end{array}$$

Now let $\hat{\underline{C}}_\Lambda$ be the full subcategory of pro-$\underline{C}_\Lambda$ in which $ob(\hat{\underline{C}}_\Lambda) = \{(R_i)_{i\in\underline{N}}$ , $R_i \in ob(\underline{C}_\Lambda) \mid \varphi_i : R_i \to R_{i-1}$ is surjective for all $i$ and induces isomorphisms $\underline{m}_i/\underline{m}_\Lambda+\underline{m}_i^2 \to \underline{m}_{i-1}/\underline{m}_\Lambda+\underline{m}_{i-1}^2$ for all sufficiently large $i\}$ where $\underline{N}$ stands for the ordered category of natural numbers, and $\underline{m}_i$ is the maximal ideal of $R_i$ . We set $\hat{\underline{A}}$ to be the full subcategory of pro-$\underline{A}$ in which

$$ob(\hat{\underline{A}}) = \{a \in ob(pro(\underline{A})) \mid pro(P)\ (a) \in ob(\hat{\underline{C}}_\Lambda)\}$$

We then obtain a cofibered category $\hat{P} : \hat{\underline{A}} \to \hat{\underline{C}}_\Lambda$ (in groupoids) with a commutative diagram

$$\begin{array}{ccc} \underline{A} & \hookrightarrow & \hat{\underline{A}} \\ {\scriptstyle P}\downarrow & & \downarrow{\scriptstyle \hat{P}} \\ \underline{C}_\Lambda & \hookrightarrow & \hat{\underline{C}}_\Lambda \end{array}$$

We note that the category $\hat{\underline{C}}_\Lambda$ is canonically equivalent to the category of complete local $\Lambda$-algebras of formally finite type over $\Lambda$ with the fixed residue field $k$ . Henceforth we shall make such identification.

Definition 1.9. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ . An object $\xi$ in $\hat{\underline{A}}$ is called a quasi-initial object of $\hat{\underline{A}}$ if for every object $a$ in $\hat{\underline{A}}$ there exists a map $\xi \to a$ in $\hat{\underline{A}}$ . A quasi-initial object $\xi$ of $\hat{\underline{A}}$ is called minimal

if the morphism $\xi : h_R \to [\hat{A}]$ induces a bijection $\xi(k[\varepsilon]): \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon]) \to [A(k[\varepsilon])]$, where $R = P(\xi)$. An object $\xi$ in $\hat{A}$ is called a versal formal object of $A$ if every surjective map $\alpha : a' \to a$ in $A$ induces a surjective map $\mathrm{Hom}_{\hat{A}}(\xi,a') \to \mathrm{Hom}_{\hat{A}}(\xi,a)$.

A versal formal object $\xi$ of $A$ is called minimal if the morphism $\xi : h_R \to [\hat{A}]$ induces a bijection $\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon]) \to [A(k[\varepsilon])]$ where $R = P(\xi)$.

Proposition 1.10. Let $A$ be a cofibered groupoid over $C_\Lambda$.

(i) every (minimally) versal formal object of $A$ is a quasi-initial (minimal) object of $\hat{A}$.

(ii) If $\xi$ is an object in $\hat{A}$ such that $\tilde{\xi}(k[\varepsilon]): \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon]) \to [A(k[\varepsilon])]$ is injective, where $P(\xi) = R$, then every endomorphism of $\xi$ in $\hat{A}$ is an automorphism. In particular a quasi-initial minimal object of $\hat{A}$, if it exists, is unique up to (non-canonical) isomorphism.

Proof: (i) let $\xi$ be a versal formal object of $A$. We must show that, for any object $a$ in $\hat{A}$, we have a map $\xi \to a$. By the definition of $\hat{A}$, there exists a sequence of surjective maps in $A$ : $\cdots \to a_n \xrightarrow{\alpha_n} a_{n-1} \to \cdots \to a_1 \xrightarrow{\alpha_1} a_0$ such that $a = \varprojlim_n a_n$. Assume that there exists maps $\varphi_i : \xi \to a_i$ such that $\alpha_i\varphi_i = \varphi_{i-1}$ for all $i < n$. Since $\xi$ is a versal formal object of $A$ and $\alpha_n : a_n \to a_{n-1}$ is surjective in $A$, we obtain a map $\xi \xrightarrow{\varphi_n} a_n$ such that $\alpha_n\varphi_n = \varphi_{n-1}$. If we set $\varphi = \varprojlim_n \varphi_n$, we have $\varphi : \xi \to a$.

(ii) Let $\xi \xrightarrow{\alpha} \xi$ be a map in $\hat{A}$. Then for every $\lambda \in \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon])$ we have $\lambda_*\xi = \lambda_*\alpha_*\xi \simeq (\lambda\cdot P(\alpha))_*\xi$ and hence by the hypothesis on $\xi$ we have that $\lambda = \lambda\cdot P(\alpha)$ for all $\lambda \in \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon])$, and therefore $P(\alpha) : R \to R$ is surjective. However it is trivial to see that every surjective endomorphism of a noetherian ring is always an automorphism and hence $P(\alpha)$ is an automorphism of $R$ and therefore $\alpha$ is an automorphism of $\xi$.

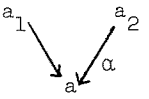
Theorem 1.11. (Schlessinger) Let $A$ be a cofibered groupoid over $C_\Lambda$. Then $A$

admits a minimally-versal formal object if and only if

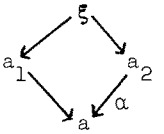
(i) $\underline{A}$ is semi-homogeneous

(ii) $[\underline{A}(k[\epsilon])]$ is a finite-dimensional vector space over $k$ (cf. 1.3(c)).

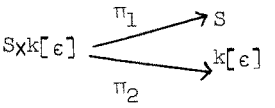
Proof; (Necessity) Let $\xi$ in $\hat{\underline{A}}$ be a minimally-versal formal object of $\underline{A}$, and consider a diagram




in $\underline{A}$ where $\alpha$ is surjective. Since $\xi$ is a quasi-initial object of $\hat{\underline{A}}$, there exists a map $\xi \to a_1$. Since $\alpha$ is surjective, the composit map $\xi \to a_1 \to a$ can be lifted to $a_2$ i.e., we obtain a commutative diagram

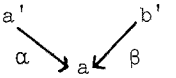



which proves the condition 1.2(1). We now prove the condition 1.2(2). Let

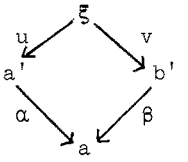



be the projections,

and let




be $\pi_1$-maps in $\underline{A}$. By the definition of $\xi$, we obtain a commutative diagram




and in particular we have $\pi_1 \cdot P(u) = \pi_1 \cdot P(v)$. Assume now that $(\pi_2)_* a' \approx (\pi_2)_* b'$. We then have $\pi_2 \cdot P(u) = \pi_2 \cdot P(v)$ because of the minimality of

$\xi$ , and hence $P(u) = \pi_1 P(u) \times \pi_2 P(u) = \pi_1 P(v) \times \pi_2 P(v) = P(v)$ .
Since $\underline{A}$ is a cofibered groupoid, we obtain a unique isomorphism $a' \xrightarrow{\sigma} b'$ in $\underline{A}(S \times k[\varepsilon])$ such that $\sigma u = v$ . Then $\beta\sigma u = \alpha u$ entails that $\beta\sigma = \alpha$ since $\underline{A}$ is a cofibered category.

(Sufficiency) Let $c_1, c_2, \ldots, c_n$ be objects in $\underline{A}(k[\varepsilon])$ which form a k-basis of the vector space $[\underline{A}(k[\varepsilon])]$ , and choose $a_1$ in $\underline{A}(k[\varepsilon] \underset{k}{\times} \ldots \underset{k}{\times} k[\varepsilon])$ such that $(\pi_j)_* a_1 \approx c_j$ . Then $\tilde{a}(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \to [\underline{A}(k[\varepsilon])]$ is bijective, where $R_1 = P(a_1)$ . In particular $a_1 \to e$ is versal for small prolongations of $e$ . Choose $a_2 \to a_1$ which is versal for small prolongations of $a_1$ such that $\overline{\Omega}_{a_2} \to \overline{\Omega}_{a_1}$ is bijective. We then choose $a_3 \to a_2$ , versal for small prolongations of $a_2$ such that $\overline{\Omega}_{a_3} \to \overline{\Omega}_{a_2}$ , is bijective etc., and set $\xi = \varprojlim_n a_n$ . We claim that $\xi$ is a minimally-versal formal object of $\underline{A}$ . Indeed, since $\overline{\Omega}_{a_n} \to \overline{\Omega}_{a_{n-1}}$ is bijective for all $n$ , $\overline{\Omega}_\xi \to \overline{\Omega}_{a_1}$ is bijective so that $\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon] \to [\underline{A}(k[\varepsilon])]$ is bijective. Now consider a diagram

$$\begin{array}{ccc} & & a' \\ & & \downarrow \alpha \\ \xi & \xrightarrow{h} & a \end{array}$$

where $\alpha$ is surjective in $\underline{A}$ . We must show that $h$ can be lifted to $a'$ . For this, one may assume, by induction, that $a'$ is a small prolongation of $a$ . Since $P(a)$ is an artinian ring, $h : \xi \to a$ is factored as a composit map $\xi \to a_q \longrightarrow a$ for some $q$ . Since $\underline{A}$ is semi-homogeneous, we get a diagram

$$\begin{array}{ccccc} & & a'' & \longrightarrow & a' \\ & & \downarrow \alpha' & & \downarrow \alpha \\ \xi & \longrightarrow & a_q & \longrightarrow & a \end{array}$$

where $\alpha'$ is also a small prolongation. We then obtain a commutative diagram

$$\begin{array}{ccccc} & & & & a'' \\ & & \nearrow & & \downarrow \alpha' \\ \xi & \longrightarrow & a_{q+1} & \longrightarrow & a_q \end{array}$$

and therefore we are done.

Remark 1.12. Let

$$\cdots\cdots \longrightarrow a_n \longrightarrow a_{n-1} \longrightarrow \cdots \longrightarrow a_2 \longrightarrow a_1 \longrightarrow a_0 = e$$

be an infinite sequences of arrows in $\underline{A}$ such that

$$\begin{cases} a_i \text{ is versal for small prolongations of } a_{i-1} \\ P(a_i) \longrightarrow P(a_{i-1}) \text{ is surjective} \\ \overline{\Omega}_{a_i} \longrightarrow \overline{\Omega}_{a_{i-1}} \text{ is bijective for } i \gg 0 \end{cases}$$

Then the above proof shows that $\varprojlim a_i$ is a versal formal object of $\underline{A}$ .

Remark 1.13. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ . If $\underline{A}$ has the property 1.11(i) and (ii), then so does $[\underline{A}]$ , and $\xi \in \mathrm{ob}(\hat{\underline{A}})$ is a minimally-versal formal object of $\underline{A}$ if and only if it is so for $[\underline{A}]$ by 1.10(ii).

Remark 1.14. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ , admitting a minimally-versal formal object $\xi$ . Then an invariant of $R = P(\xi)$ , which is a complete local $\Lambda$-algebra of formally finite type over $\Lambda$ , is automatically an invariant of $\underline{A}$ over $\underline{C}_\Lambda$ . Therefore, for example, we shall say that $\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$ if $R$ is (formally) smooth over $\Lambda$ i.e., a formal power-series algebra over $\Lambda$ , or we say that $\underline{A}$ is irreducible over $\underline{C}_\Lambda$ if $\mathrm{Spec}(R)$ is irreducible etc. In the next section we introduce the notion of "smoothness" of a functor $\underline{A} \to \underline{B}$ over $\underline{C}_\Lambda$ , where $\underline{A}$, $\underline{B}$ are cofibered categories in groupoids over $\underline{C}_\Lambda$ . It turns out (cf. 2.4) that "$\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$" is equivalent to " $P : \underline{A} \to \underline{C}_\Lambda$ is a smooth functor" , and therefore no confusion arises. We also define the dimension of $\underline{A}$ over $\underline{C}_\Lambda$ by $\dim_{\underline{C}_\Lambda} \underline{A} = \dim k \underset{\Lambda}{\otimes} R$ . For this we note the following fact: Let $\underline{C}_k$ be the category of artinian local k-algebras with the same residue field $k$ . Then $\underline{C}_k \to \underline{C}_\Lambda$ is a fully-faithful imbedding, whose (left) adjoint functor $\underline{C}_\Lambda \to \underline{C}_k$ is given by $R \to k \underset{\Lambda}{\otimes} R$ . Let $\underline{A}|\underline{C}_k$ be the pull-back of $P : \underline{A} \to \underline{C}_\Lambda$ under $\underline{C}_k \to \underline{C}_\Lambda$ . Then it is trivial to see that if $\xi$ is a minimally-versal formal object for $\underline{A}$ over $\underline{C}_\Lambda$ , then $k \underset{\Lambda}{\otimes} \xi$ is a minimally-versal formal object for $\underline{A}|\underline{C}_k$ over $\underline{C}_k$ , where $\xi \to k \underset{\Lambda}{\otimes} \xi$ is a cocartesian arrow above $R \to k \underset{\Lambda}{\otimes} R$ . It follows that $\dim_{\underline{C}_\Lambda} \underline{A} = \dim_{\underline{C}_k} (\underline{A}|\underline{C}_k)$ .

1.15. In the remaining of this paragraph, we biefly explain how to modify the preceeding theory in order to allow for residue field extensions.

Let $\Lambda$ and $K$ be complete noetherian local rings, with $K$ a $\Lambda$-algebra. We assume that

(a) the natural map from $\Lambda$ to $K$ is a local homomorphism;

(b) the residue field $K'$ of $K$ is an extension of finite type of the residue field $k$ of $\Lambda$ .

The most interesting case is when $K = K'$ and $K$ is a finite inseparable extension of $k$ .

We denote by $C^{\wedge}_{\Lambda,K}$ the category whose objects are the complete noetherian local rings $R$ , provided with a structure of $\Lambda$-algebra and with a $\Lambda$-epimorphism $s : R \longrightarrow K$ .
We denote by $C_{\Lambda,K}$ the subcategory of $C^{\wedge}_{\Lambda,K}$ consisting of the $(R, s)$ such that $\mathrm{Ker}(s)$ is a $R$-module of finite length. The category $C^{\wedge}_{\Lambda,K}$ can be identified with a full subcategory of $\mathbf{Pro}(C_{\Lambda,K})$ .

If $K$ is artinian, so is any object in $C_{\Lambda,K}$ .

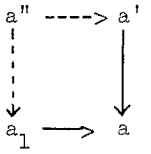
For any finite dimensional vector space $V$ over $K'$ we denote by $D_V(K)$ the algebra of dual numbers over $K$ defined by $V$ ; its elements are of the form $k + v\,\epsilon$ (with $k \in K$ , $v \in V$ and $\epsilon^2 = 0$ : $(k + v\,\epsilon)(k' + v'\,\epsilon) = (k\,k') + (k\,v' + k'\,v)\epsilon$ ). The map $s : D_V(K) \to K$ : $s(k + v\,\epsilon) = k$ , turns $D_V(K)$ into an object of $C_{\Lambda,K}$ . The role played before by $k[\epsilon]$ will be played here by $D_{K'}(K)$ .

As explained in 1.8, any cofibered groupoid $\underline{A}$ over $C_{\Lambda,K}$ extends in a natural way to a cofibered category $\underline{A}^{\wedge}$ over $C^{\wedge}_{\Lambda,K}$ . A map in $\underline{A}$ or $\underline{A}^{\wedge}$ will be called surjective if the corresponding homomorphism of algebras in $C_{\Lambda,k}$ or in $C^{\wedge}_{\Lambda,K}$ is.

We assume that $\underline{A}(K) = \{e\}$ .

Definition 1.16. A cofibered groupoïd $\underline{A}$ over $C_{\wedge,K}$ is called semi-homogeneous if the following properties hold:

(I) every diagram




of solid arrows in $\underline{A}$ , where $a' \longrightarrow a$ is surjective, can be completed into a commutative diagram including dotted arrows;

(II) For any $R$ , the natural map

$$[\underline{A}(R \times_K D_{K'}(K))] \to [\underline{A}(R) \times \underline{A}(D_{K'}(K))] \simeq [\underline{A}(R)] \times [\underline{A}(D_{K'}(K))]$$

is bijective.

If $\underline{A}$ is semi-homogeneous, so is $[\underline{A}]$ .

1.17. Let us first consider the restriction of $\underline{A}$ to the full subcategory of $C_{\wedge,K}$ whose objects are the $D_V(K)$ . Let $\Omega$ be the vector space $\Omega^1_{K/\wedge} \otimes_K K'$ , and $d : K \to \Omega$ the derivative. A map $f : D_V(K) \to D_W(K)$ can be written:

$$f(k + v\,\varepsilon) = k + (u(v) + D\,d\,k)$$

with $u \in \mathrm{Hom}_{K'}(V, W)$ and $D \in \mathrm{Hom}_{K'}(\Omega, W)$ . If we identify $f$ with $(u, D)$ , the composition law is

$$(u, D) \circ (u', D') = (uu', D + u\,D') \quad .$$

The condition (II) entails

(II') the functor from $K'$-vector spaces to sets

$$V \longmapsto [\underline{A}(D_V(K))]$$

commutes with products.

Lemma 1.18. Assume condition (II') is satisfied by $\underline{A}$ . Then

(1) $[\underline{A}(D_{K'}(K))]$ carries a unique vector space structure, such that $[\underline{A}(D_V(K))] \simeq V \otimes_{K'} [\underline{A}(D_{K'}(K))]$ as a functor in $V$ .

(2) There is a linear map $\gamma : \mathrm{Hom}(\Omega, K') \to [\underline{A}(D_{K'}(K))]$ such that, for any $D \in \mathrm{Hom}(\Omega, K')$ ,

$$[\underline{A}((k, D))] : [\underline{A}(D_{K'}(K))] \to [\underline{A}(D_{K'}(K))] \text{ is } x \longmapsto x + \gamma(D) .$$

(3) For any $(u, D) : D_V(K) \longrightarrow D_W(K)$ , the map

$(u, D) : [\underline{A}(D_V(K))] \to [\underline{A}(D_W(K))]$ i.e.

$(u, D) : V \otimes [\underline{A}(D_{K'}(K))] \to W \otimes [\underline{A}(D_{K'}(K))]$ is

$$x \longmapsto u(x) + \gamma(D) .$$

Let us introduce the notations

$F(V) = [\underline{A}(D_V(K))]$

$A = F(K')$

$F(u,D) : F(V) \to F(W)$ = the map induced by $(u,D) : D_V(K) \to D_W(K)$

$F(D) : F(V) \to F(V) : F(D) = F(1, D)$ .

By (II') and the fact that $K'$ is a $K'$-vector space object in the category of $K'$-vector spaces, $F(K')$ is a $K'$-vector space in (Ens), hence a $K'$-vector space, and (1) is clear. Let us now identify $F(V)$ and $V \otimes A$ .

We deduce from the identity

$$(u, D) = (1, D) \circ (u, 0) \text{ that}$$

$$F(u, D) = F(D) \circ (u \otimes 1_A) .$$

The functoriality of $F$ amounts to

$$F(0) = \mathrm{Id}$$

$$F(D) \circ F(D') = F(D + D')$$

$$(u \otimes 1) \circ F(D) = F(u\, D) \circ (u \otimes 1_A)$$

For any $D : \Omega \to V$ and any $n \in N$ , if $i$ and $p$ are the maps

$$i ; V \hookrightarrow V \oplus K'^n \qquad p : V \oplus K'^n \longrightarrow K^n ,$$

then $p\ i\ D = 0$ and the following diagramm is commutative

$$\begin{array}{ccccc} V \otimes A & \longrightarrow & (V \oplus K'^{n}) \otimes A & \longrightarrow & K'^{n} \otimes A \\ \downarrow F(D) & & \downarrow F(i,D) & & \| \\ V \otimes A & \longrightarrow & (V \oplus K'^{n}) \otimes A & \longrightarrow & K'^{n} \otimes A \end{array}$$

Let $x : K^{n} \to V$ be a map, $x_i = x(e_i)$, and let $a_i$ be a family of elements of $A$. We have $(1_V + x) \circ i \circ D = D$. Hence, if

$$F(i\,D)\ (\Sigma\ e_i\ a_i) = \Sigma\ e_i\ a_i + a \qquad (a \in V \otimes H)\ , \text{ then}$$

$$F(D)\ (\Sigma\ x_i\ a_i) = \Sigma\ x_i\ a_i + a$$

and $a$ does not depend on the $x_i$. Hence,

$$F(D)\ (y) = y + G(D) \qquad , \text{ with}$$

$$G(D + D') = G(D) + G(D')$$

$$u\ G(D) = F(u\ D) \qquad (\text{for any } u : V \to W)\ .$$

One easily checks that any such $G$ is defined, for some $\gamma \in \Omega \otimes A$, by the formula

$$G(D) = D \otimes 1_A\ (\gamma)\ ,$$

as promised in (2) (3).

Definition 1.19. (1) An object $\xi$ in $\hat{A}$ is called versal formal object if every surjective map $\alpha : a' \longrightarrow a$ in $A$ induces a surjective map $\mathrm{Hom}_{\hat{A}}(\xi, a') \longrightarrow \mathrm{Hom}_{\hat{A}}(\xi, a)$.

(2) A formal versal object $\xi$ in $\hat{A}$, with image $(R, s)$ in $C^{\hat{}}_{\Lambda,K}$, is called minimal if for any two maps $\varphi_1, \varphi_2 : R \rightrightarrows D_{K'}(K)$, if $\varphi_1(\xi)$ is isomorphic to $\varphi_2(\xi)$, then there exists a derivation $D$ of $K$ into $K'$ such that $(1, D)\ (\varphi_1) = \varphi_2$, i.e. $\varphi_2 - \varphi_2 = D\ s$.

Theorem 1.20. If $A$ is semi-homogeneous and if

$$\dim_{K'} ([A(D_{K'}(K))]) < \infty\ ,$$

then $A$ admits a minimally versal formal object, and any two of them are

isomorphic.

With the notations of 1.18, let $H^*$ be a subspace of $[\underline{A}(D_{K'}(K))]$, supplementary to the image of $\gamma$, and let $H$ be the dual of $H^*$. We define $R_o = D_H(K)$. Let $\xi_o$ be an object in $\underline{A}(R_o)$, whose image in $[\underline{A}(R_o)] \simeq H \otimes [\underline{A}(D_{K'}(K))] \simeq \mathrm{Hom}(H^*, [\underline{A}(D_{K'}(K))])$ is the inclusion of $H^*$ in $[\underline{A}(D_{K'}(K))]$. Then

(1) The maps $\mathrm{Hom}_{C_{\Lambda,K}}(D_H(K), D_V(K)) \longrightarrow [\underline{A}(D_V(K))]$ are surjective.

(2) The minimality condition 1.19(2) is valid for $(R_o, \xi_o)$.

Choose now some $S \in \hat{C}_{\Lambda,K}$, formally smooth over $\Lambda$, and an epimorphism $t : S \longrightarrow R_o$ such that

$$\Omega_{R_o/\Lambda} \otimes K' \xrightarrow{\sim} \Omega_{S/\Lambda} \otimes K' .$$

If $M$ is the maximal ideal of $S$, we inductively define, for $n \geq 1$, $R_n = S/I_n$ and $\xi_n \in \mathrm{Ob}(\underline{A}(R_n))$ by the properties that

(a) $I_{n+1}$ is the intersection of the ideals $I : I_n M \subset I \subset I_n$, such that $\xi_n$ extends to an object $\xi$ in $\underline{A}(S/I)$.

(b) $\xi_{n+1}$ is an extension of $\xi_n$.

If $R = \varprojlim R_n$ and $\xi = \varprojlim \xi_n$, then one checks that $(R, \xi)$ is a minimally-versal formal object, and the only one up to isomorphism.

## 2. Prorepresentable cofibered groupoids.

Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, admitting a minimally-versal formal object $\xi$. In general we can not recover $\underline{A}$ (up to $\underline{C}_\Lambda$-equivalence) out of $\xi$ alone. One has to know when two maps $R \underset{g}{\overset{f}{\rightrightarrows}} S$ yield a $\underline{C}_\Lambda$-isomorphism $\xi(f) \longrightarrow \xi(g)$, where $R = P(\xi)$. As an application of 1.11 and 1.12 we now consider the question of "recovering $\underline{A}$" up to $\underline{C}_\Lambda$-equivalence. Indeed, we

shall have a quite satisfactory answer applicable to a wide class of cofibered groupoids.

A number of cofibered groupoids which usually occur in practice enjoy a slightly stronger property than those of semi-homogeneous ones, as it will be defined explicitly in 2.5. To investigate them it is convenient to rephrase what we have done so far in terms of "smooth functors". All the notations and terminologies in the previous section remain in force throughout in this section.

Definition 2.1. Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We say that $\varphi$ is smooth if every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram

$$(*) \qquad \begin{array}{ccc} \varphi(a') & \longrightarrow & b' \\ & \searrow^{\varphi(\alpha)} & \downarrow \beta \\ & & \varphi(a) \end{array}$$

for some $\alpha : a' \longrightarrow a$ in $\underline{A}$. We say that $\varphi$ is minimally smooth if it is smooth and $\varphi(k[\varepsilon]) : [\underline{A}(k[\varepsilon])] \longrightarrow [\underline{B}(k[\varepsilon])]$ is bijective.

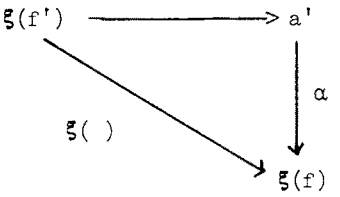
Remark 2.2(a) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We note that " $\varphi$ is smooth" $\Leftrightarrow$ "every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed into a commutative diagram $(*)$ such that $\varphi(a') \longrightarrow b'$ is a $\underline{C}_\Lambda$-isomorphism" $\Leftrightarrow$ "every small prolongation $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed into a commutative diagram $(*)$ ".

Remark 2.2(b) For any cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$, the natural functor $\underline{A} \longrightarrow [\underline{A}]$ is minimally smooth.

Remark 2.2(c) For each object $R$ in $\hat{\underline{C}}_\Lambda$, we set $h_R : \underline{C}_\Lambda \longrightarrow (\mathrm{Ens})$ to be the functor defined by $h_R(S) = \mathrm{Hom}_{\Lambda\text{-alg}}(R, S)$ for all $S$ in $\underline{C}_\Lambda$. It associates a cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$, and a map $R \longrightarrow S$ in $\underline{C}_\Lambda$

induces a functor $\underline{h}_S \longrightarrow \underline{h}_R$ of cofibered groupoids. The functor $\underline{h}_S \longrightarrow \underline{h}_R$ induced by a map $R \longrightarrow S$ in $\underline{C}_\wedge$ is smooth if and only if $S$ is a formal power-series ring over $R$ .

Remark 2.2(d) Given an object $R$ in $\hat{\underline{C}}_\wedge$ , let us denote by $(R, \underline{C}_\wedge)$ the category of arrows in $\hat{\underline{C}}_\wedge$ with the source $R$ and a target in $\underline{C}_\wedge$ . Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$ . Given $\xi \in \mathrm{ob}\,\hat{\underline{A}}$ with $P(\xi) = R$ , let $\underline{A}' \longrightarrow (R, \underline{C}_\wedge)$ be the pull-back of $\underline{A} \longrightarrow \underline{C}_\wedge$ under the canonical functor $(R, \underline{C}_\wedge) \longrightarrow \underline{C}_\wedge$ and choose a section $\xi^\#$ such that $\xi^\#(1_R) = \xi$ . (This simply means that we choose, for each map $R \xrightarrow{f} S$ , a f-map $\xi \longrightarrow \xi(f)$ ). We then obtain a functor $\xi^\# : \underline{h}_R \longrightarrow \underline{A}$ over $\underline{C}_\wedge$ by sitting $(R \xrightarrow{f} S) \longmapsto \xi(f)$ . Of course this functor does not depend on the choice of $\xi^\#$ up to natural equivalence of functors. Now " $\xi^\# : \underline{h}_R \longrightarrow \underline{A}$ is (minimally) smooth" means that every surjective map $a' \xrightarrow{\alpha} \xi(f)$ in $\underline{A}$ can be completed into a commutative diagram




(and $\underline{h}_R(k[\varepsilon]) \longrightarrow \underline{A}(k[\varepsilon])$ is bijective), which means precisely that $\xi$ is a (minimally) versal formal object of $\underline{A}$ over $\underline{C}_\wedge$ . Thus we conclude that the existence of a (minimally) versal formal object of $\underline{A}$ is equivalent to the existence of a (minimally) smooth functor $\underline{h}_R \longrightarrow \underline{A}$ over $\underline{C}_\wedge$ for some $R \in \mathrm{ob}(\hat{\underline{C}}_\wedge)$ .

The following general properties on smoothness are trivial to verify.

Proposition 2.3. Let $\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\varphi} \underline{C}$ be functors of cofibered groupoids over $\underline{C}_\wedge$ .

(a) If $\varphi$, $\psi$ are both smooth, then so is $\psi.\varphi$ .

(b) If $\psi.\varphi$ is smooth and $\varphi$ is surjective on isomorphism classes of objects,

then $\psi$ is smooth.

Remark 2.4. Let $\underline{A}$ be a cofibered groupoids over $\underline{C}_\Lambda$, admitting a versal formal object $\xi$. This means, by 2.2(d), that $\xi : \underline{h}_R \to \underline{A}$ is smooth, where $R = P(\xi)$. Therefore we have, " $\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$ (c.f. 1.13) i.e., R is a formal power-series algebra over $\Lambda$ " $\Leftrightarrow$ "the composite functor $\underline{h}_R \xrightarrow{\xi} \underline{A} \xrightarrow{P} \underline{C}_\Lambda$ $(= \underline{h}_\Lambda)$ is smooth" $\Leftrightarrow$ " $\underline{A} \xrightarrow{P} \underline{C}_\Lambda$ is a smooth functor" by 2.3(b).

We now consider a stronger property than being semi-homogeneous.

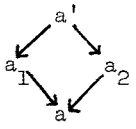
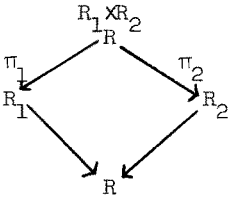
Definition 2.5. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. It is called homogeneous if every diagram

$$\begin{array}{ccc} & & a' \\ & & \downarrow \\ b' & \longrightarrow & a \end{array}$$

where $a' \to a$ is surjective admits a fibre-product $a' \underset{a}{\times} b'$ in $\underline{A}$ such that

$$P(a' \underset{a}{\times} b') = P(a') \underset{P(a)}{\times} P(b') .$$

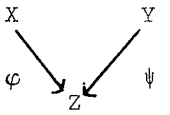
Remark 2.6(a) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

(*) $\quad$  in $\underline{A}$ above the diagram 

in $\underline{C}_\Lambda$ where $\pi_i$ are the projections. If a fibre-product $a_1 \underset{a}{\times} a_2$ exists in $\underline{A}$, we obtain a unique map $\varphi : a' \to a_1 \underset{a}{\times} a_2$ compatible with the projections. If $P(a_1 \underset{a}{\times} a_2) = R_1 \underset{R}{\times} R_2$, then $P(\varphi)$ is the identity map on $R_1 \underset{R}{\times} R_2$ and therefore $\varphi$ is a $\underline{C}_\Lambda$-isomorphism i.e., $a' = a_1 \underset{a}{\times} a_2$. In other words, if $\underline{A}$ is a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, every diagram (*) defines a fibre-product $a_1 \underset{a}{\times} a_2$ provided $R_1 \to R$ is surjective in $\underline{C}_\Lambda$. This fact entails immediately

that a homogeneous cofibered groupoid is, a fortiori, semi-homogeneous.

Remark 2.6(b) Consider a diagram



of categories and functors. The ordinary fibre-product $X \underset{Z}{\times} Y$ is a category in which an object is a pair $(x,y)$ such that $\varphi(x) = \psi(y)$, and a map $(x,y) \to (x', y')$ consists of a pair $\alpha : x \to x'$, $\beta : y \to y'$ such that $\varphi(\alpha) = \psi(\beta)$. The fibre-product $X \overset{2}{\underset{Z}{\times}} Y$ in the sense of 2-categories is a category in which an object is a triple $(x,y;\gamma)$, where $\varphi(x) \overset{\gamma}{\to} \psi(y)$ is an isomorphism, and a map $(x,y;\gamma) \to (x',y';\gamma')$ consists of a pair $x \overset{\alpha}{\to} x'$ $y \overset{\beta}{\to} y'$ with a commutative diagram

$$\begin{array}{ccc} \varphi(x) & \xrightarrow{\varphi(\alpha)} & \varphi(x') \\ {\scriptstyle\gamma}\downarrow & & \downarrow{\scriptstyle\gamma'} \\ \psi(y) & \xrightarrow[\psi(\beta)]{} & \psi(y') \end{array}$$

One can rephrase the notions of (semi)homogeneous cofibered groupoid or smooth functor in terms of fibre-product of categories (in the sense of 2-category). Now let $P : \underline{A} \to \underline{C}_{/\Lambda}$ be a cofibered category, and consider an object $R_1 \underset{R}{\times} R_2$ in $\underline{C}_{/\Lambda}$. If we choose a direct image functor $(\pi_i)_* : \underline{A}(R_1 \underset{R}{\times} R_2) \to \underline{A}(R_i)$, where $\pi_i : R_1 \underset{R}{\times} R_2 \to R_i$ is the projection for $i = 1,2$, we obtain a functor $\underline{A}(R_1 \underset{R}{\times} R_2) \to \underline{A}(R_1) \overset{2}{\underset{\underline{A}(R)}{\times}} \underline{A}(R_2)$. It is clear from the definition of cofibered category that this functor, up to canonical natural equivalence, does not depend on the choice of direct image functors. It is now clear from the definition that a cofibered groupoid $\underline{A}$ over $\underline{C}_{/\Lambda}$ is homogeneous over $\underline{C}_{/\Lambda}$ if and only if $\underline{A}(R_1 \underset{R}{\times} R_2) \to \underline{A}(R_1) \overset{2}{\underset{\underline{A}(R)}{\times}} \underline{A}(R_2)$ is an equivalence of categories whenever $R_1 \to R$ is surjective in $\underline{C}_{/\Lambda}$, and $\underline{A}$ is semi-homogeneous if and only if

$$[\underline{A}(R \underset{R}{\times} R_2)] \longrightarrow [\underline{A}(R_1) \overset{2}{\underset{\underline{A}(R)}{\times}} \underline{A}(R_2)]$$

is surjective whenever $R_1 \longrightarrow R$ is surjective in $\underline{C}_\Lambda$, and is bijective whenever $R = k$ and $R_1 = k[\varepsilon]$. Now let $\varphi : \underline{A} \to \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. For each map $R' \to R$ in $\underline{C}$, $\varphi$ induces a functor $\underline{A}(R') \to \underline{A}(R) \underset{\underline{B}(R)}{\overset{2}{\times}} \underline{B}(R')$ uniquely determined up to a (canonical) natural equivalence. It is again trivial to see that $\varphi$ is smooth if and only if the functor $A(R') \to \underline{A}(R) \underset{\underline{B}(R)}{\overset{2}{\times}} \underline{B}(R')$ is surjective on isomorphism classes of objects, whenever $R' \to R$ is surjective.

Remark 2.6(c). If $\underline{A}$ is a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, the functor $A(k[V \times W]) \to \underline{A}(k[V]) \times \underline{A}(k[W])$ is an equivalence of categories for any finite-dimensional vector spaces $V$, $W$ over $k$ (We recall our convention $\underline{A}(k) = \{e\}$). In particular, we have

$$\mathrm{Aut}_{\underline{A}(k[V\times W])}(1_{V\times W}) \xrightarrow{\sim} \mathrm{Aut}_{\underline{A}(k[V])}(1_V) \times \mathrm{Aut}_{\underline{A}(k[W])}(1_W)$$

where $1_V$ is the direct image of $e$ under the unique map $k \to k[V]$ in $\underline{C}_\Lambda$. It follows that if $\underline{A}$ is homogeneous over $\underline{C}_\Lambda$, $\mathrm{Aut}(1_\varepsilon)$ as well as $[\underline{A}(k[\varepsilon])]$ are canonically provided with vector space structures over $k$.

Remark 2.6(d). For any object $R$ in $\hat{\underline{C}}_\Lambda$, the cofibered (discrete) groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$ is homogeneous.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. However, the homogeneity of $\underline{A}$ does not imply in general the homogeneity of $[\underline{A}]$. Indeed we have

Proposition 2.7. *Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$. The following statements are equivalent.*

(i) *$[\underline{A}]$ is homogeneous i.e. $[\underline{A}(R' \underset{R}{\times} S)] \longrightarrow [\underline{A}(R')] \underset{[\underline{A}(R)]}{\times} [\underline{A}(S)]$ is bijective for every small extension $R' \to R$.*

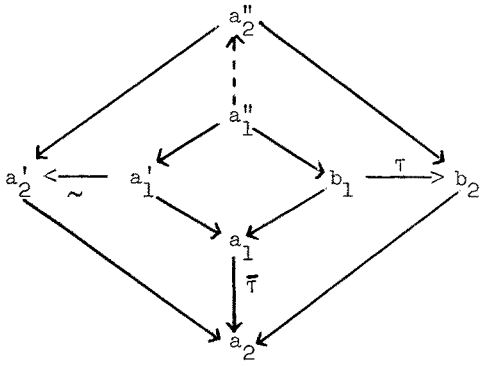
(ii) *For every small prolongation $a' \to a$ in $\underline{A}$, the map*

$$\mathrm{Aut}_{\underline{A}(R')}(a') \longrightarrow \mathrm{Aut}_{\underline{A}(R)}(a)$$

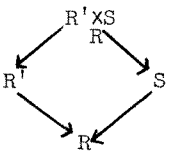
is surjective, where $R' = P(a')$ , $R = P(a)$ .

Proof. (i) $\Rightarrow$ (ii) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$ and let $a' \to a$ be a f-map in $\underline{A}$ . Given $\tau \in \mathrm{Aut}_{\underline{A}(R)}(a)$ , we consider $a' \underset{a}{\times} a'$ and $a' \underset{a}{\times} a'_\tau$ where $a'_\tau \longrightarrow a$ denotes the composite map $a' \longrightarrow a \xrightarrow{\tau} a$ . Since $[\underline{A}(R' \underset{R}{\times} R')] \longrightarrow [\underline{A}(R')] \underset{[\underline{A}(R)]}{\times} [\underline{A}(R')]$ is bijective, we conclude the existence of an isomorphism $a' \underset{a}{\times} a' \xrightarrow{u} a' \underset{a}{\times} a'_\tau$ , and $u$ induces, via the projection on the second factor, an isomorphism $a' \xrightarrow{\tau'} a'$ which is a lifting of $\tau$ .

(ii) $\Rightarrow$ (i) Let $a''_i$ $(i=1,2)$ be objects in $\underline{A}(R' \underset{R}{\times} S)$ which have the same image under $[\underline{A}(R' \underset{R}{\times} S)] \to [\underline{A}(R')] \underset{[\underline{A}(R)]}{\times} [\underline{A}(S)]$ . We then obtain a diagram of solid arrows




above




where $\bar{\tau} ; a_1 \to a_2$ is the induced isomorphism from $\tau$ so that the square box on the lower right corner is commutative. If the condition (ii) is verified, one can always replace the given isomorphism $a'_1 \xrightarrow{\sim} a'_2$ so that the square box on the left corner is also commutative. Since $a''_i = a'_i \underset{a_i}{\times} b_i$ for $i = 1, 2$ by 2.6(a), we obtain the isomorphism $a''_1 \xrightarrow{\sim} a''_2$ . This proves that $[\underline{A}(R' \underset{R}{\times} S)] \longrightarrow [\underline{A}(R')] \underset{[\underline{A}(R)]}{\times} [\underline{A}(S)]$ is injective, and hence is bijective since $\underline{A}$ is homogeneous by hypothesis.

Now let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$ , and let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$ . We have a canonical map

$\tilde{f} : R' \underset{R}{\times} R' \longrightarrow k[\mathrm{Ker}\ f]$ given by $(r_1', r_2') \longrightarrow r_1'(0) + (r_1' - r_2')$ , and $1 \times \tilde{f} : R' \underset{R}{\times} R' \longrightarrow R' \underset{k}{\times} k[\mathrm{Ker}\ f]$ is an isomorphism. Given any two f-maps $a_i \to a$ $(i=1,2)$ in $\underline{A}$ , $a_1 \underset{a}{\times} a_2$ is an object over $R' \underset{R}{\times} R'$ and in turn we may consider $\tau(a_1, a_2) = (\tilde{f})_*(a_1 \underset{a}{\times} a_2)$ which is an object in $\underline{A}(k[\mathrm{Ker}\ f])$ . Since $1 \times \tilde{f} : R' \underset{R}{\times} R' \longrightarrow R' \underset{k}{\times} k[\mathrm{Ker}\ f]$ is an isomorphism, we have the isomorphism $a_1 \underset{a}{\times} a_2 \simeq a_1 \times \tau(a_1, a_2)$ compatible with the projections on the first factor. The statement 1.5(2) takes a more definite form for a homogeneous groupoid.

Lemma 2.8. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\wedge$ , and $a_i \xrightarrow{\alpha_i} a$ $(i = 1, 2)$ f-maps in $\underline{A}$ . Then

(1) There exists $\rho : a_1 \to a_2$ in $\underline{A}$ with $\alpha_2 \cdot \rho = \alpha_1$ if and only if there exists an arrow $a_1 \to \tau(a_1, a_2)$ in $\underline{A}$ .

(2) There exists $\rho : a_1 \to a_2$ with $\alpha_2 \cdot \rho = \alpha_1$ such that $P(\rho) = \mathrm{id}_{R'}$ if and only if $\tau(a_1, a_2) = 0$ in $[\underline{A}(k[\mathrm{Ker}\ f])]$ .

Proof: We prove both at the same time. There exists $\rho : a_1 \to a_2$ in $\underline{A}$ with $\alpha_2 \cdot \rho = \alpha_1$ (such that $P(\rho) = \mathrm{id}_{R'}$) $\Leftrightarrow$ there exists $h : a_1 \to a_1 \underset{a}{\times} a_2$ in $\underline{A}$ with $\mathrm{pr}_1 \circ h = \mathrm{id}_{a_1}$ (such that $P(h) =$ diagonal map) $\Leftrightarrow$ there exists $g : a_1 \to a_1 \times \tau(a_1, a_2)$ in $\underline{A}$ with $\mathrm{pr}_1 \circ g = \mathrm{id}_{a_1}$ (such that $P(g) : R' \to R' \underset{k}{\times} k[\mathrm{Ker}\ f]$ is given by $r' \to (r', r'(0))$ $\Leftrightarrow$ there exists an arrow $\gamma : a_1 \to \tau(a_1, a_2)$ in $\underline{A}$ (such that $P(\gamma) : R' \to k[\mathrm{Ker}\ f]$ is given by $r' \to r'(0)$) . However, there exists an arrow $\gamma : a_1 \to \tau(a_1, a_2)$ in $\underline{A}$ such that $P(\gamma) : R' \to k[\mathrm{Ker}\ f]$ is factored through $R' \to k \to k[\mathrm{Ker}\ f]$ if and only if $\tau(a_1, a_2) = 0$ .

Proposition 2.9. (a) Let $\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$ be functors of cofibered groupoids over $\underline{C}_\wedge$ . Assume that

(i) $\underline{B}$ is semi-homogeneous and $\underline{C}$ is homogeneous

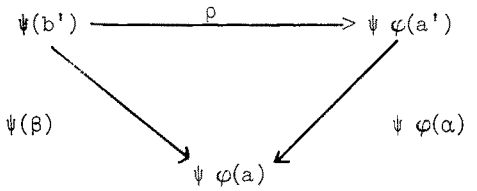
(ii) $\psi(k[\epsilon]) : [\underline{B}(k[\epsilon])] \longrightarrow [\underline{C}(k[\epsilon])]$ is injective.

Then the composite functor $\psi \circ \varphi$ is smooth if and only if $\varphi$ and $\psi$ are both

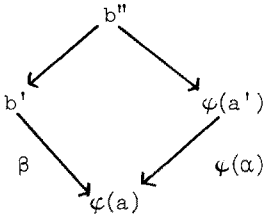
smooth.

(b) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\wedge$. If $\underline{A}$ is semi-homogeneous and $\underline{B}$ is homogeneous, then $\hat{\varphi} : \hat{\underline{A}} \to \hat{\underline{B}}$ is smooth (i.e., for any surjective map $b' \xrightarrow{\beta} \hat{\varphi}(a)$ in $\hat{\underline{B}}$, there exists $\alpha : a' \to a$ in $\hat{\underline{A}}$ and a $\hat{\underline{C}}_\wedge$-isomorphism $\rho : \hat{\varphi}(a') \xrightarrow{\sim} b'$ such that $\beta \cdot \rho = \varphi(\alpha)$).

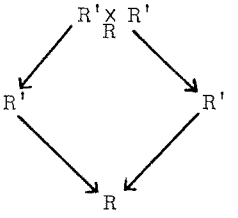
Proof: (a) Assume that the composite functor $\psi \circ \varphi$ is smooth. If $\varphi$ is smooth, then so is $\psi$ by 2.3(b), and therefore it suffices to show that $\varphi$ is smooth. Let a small prolongation $b' \xrightarrow{\beta} \varphi(a)$ in $\underline{B}$ be given. Since the composite functor $\varphi \circ \psi$ is smooth, there exists an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ and a $\underline{C}_\wedge$-isomorphism $\rho : \psi(b') \longrightarrow \psi\ \varphi(a')$ with a commutative diagram




Since $\underline{B}$ is semi-homogeneous, we can choose a commutative diagram

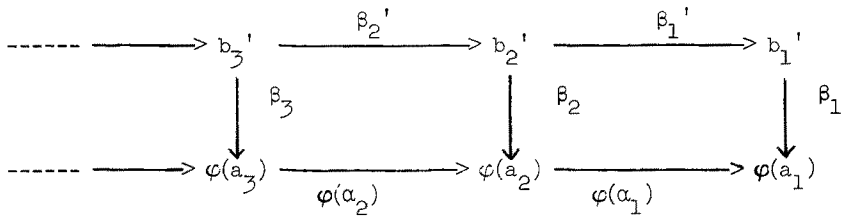



above



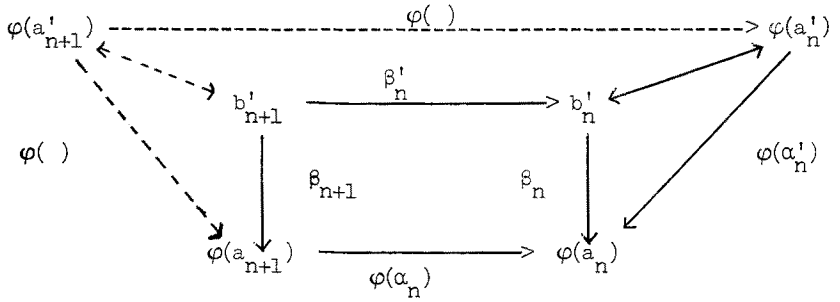

where $R = P(a)$, $R' = P(b') = P(a')$. Since $\underline{C}$ is homogeneous, $\psi(b'') = \psi(b') \underset{\psi\,\varphi(a)}{\times} \psi\varphi(a')$ by 2.6(a), and it follows from 2.8(2) that $0 = \tau(\psi\ \varphi(a'), \psi(b')) = (\tilde{f})_* \psi(b'')$, and hence $(\tilde{f})_*(b'') = 0$ be the hypothesis (ii). It follows from 1.5(2) that there exists a map $\gamma : b' \longrightarrow \varphi(a')$ such that $\beta = \varphi(\alpha) \cdot \gamma$.

(b) Let a surjective map $\beta : b' \to \varphi(a)$ in $\hat{\underline{B}}$ be given. This means that we are given a sequence of commutative diagrams

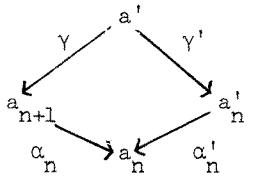


in $\underline{B}$ , where $\alpha_v$ , $\beta_v$ , $\beta_v'$ are surjective for all $v$ , such that $\beta = \varprojlim_v \beta_v$ .

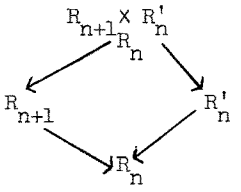
To show that $\hat{\varphi}$ is smooth, it suffices to prove that a commutative diagram below of solid arrows in which $\alpha_n$ , $\beta_n$ , $\beta_{n+1}$ , $\beta_n'$ are surjective can be completed into a commutative diagram including dotted arrows.



Now $\underline{A}$ is, by hypotheses, semi-homogeneous and $\alpha_n : a_n' \to a_n$ is surjective, and therefore there exists a commutative diagram

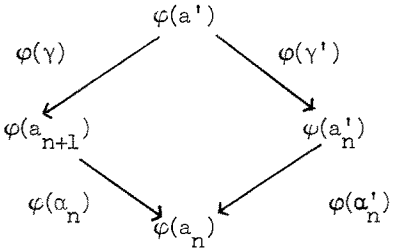


in $\underline{A}$ over

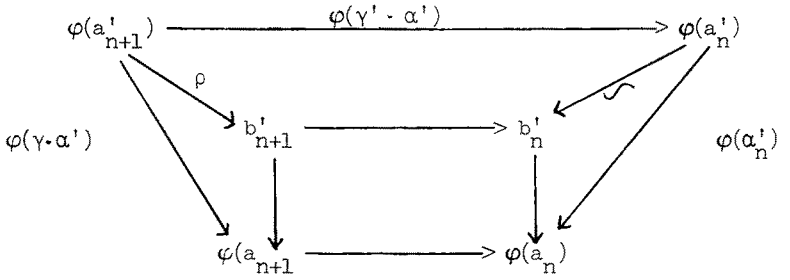


in $\underline{C}_\wedge$ .

The hypothesis that $\underline{B}$ is homogeneous entails that



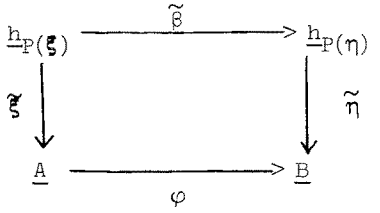
is a diagram of a fibre-product in $\underline{B}$ , and therefore we obtain a unique map $\delta : b'_{n+1} \to \varphi(a')$ in $\underline{B}$ with an appropriate commutativity. Since $\beta_{n+1}$ , $\beta'_n$ are both surjective, so is $\delta$ and therefore, since $\varphi$ is smooth, there exists $a'_{n+1} \xrightarrow{\alpha'} a'$ in $\underline{A}$ and a $\underline{C}_\Lambda$-isomorphism $\varphi(a'_{n+1}) \overset{\rho}{\underset{\sim}{\to}} b'_{n+1}$ such that $\delta \cdot \rho = \varphi(\alpha')$ . We then obtain the commutative diagram




Corollary 2.10. (a) Let $\varphi : \underline{A} \to \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$ . Let $\xi$ be a versal formal object of $\underline{A}$ , and $\eta$ a minimally-versal formal object of $\underline{B}$ . If $\underline{B}$ is homogeneous, then for any map $\beta : \eta \to \varphi(\xi)$ in $\hat{\underline{B}}$ , $P(\xi)$ is a formal power-series ring over $P(\eta)$ through $P(\beta) : P(\eta) \to P(\varphi(\xi)) = P(\xi)$ .

(b) Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$ , and let $\xi$ , $\eta$ be versal formal objects of $\underline{A}$ . If $\eta$ is minimal, then for any map $\alpha : \eta \to \xi$ in $\hat{\underline{A}}$ , $P(\xi)$ is a formal power-series ring over $P(\eta)$ through $P(\alpha) : P(\eta) \to P(\xi)$ .

Proof: (a) A map $\beta : \eta \to \varphi(\xi)$ induces a commutative diagram




Since $\tilde{\xi}$ , $\varphi$ are smooth, $\tilde{\eta} \cdot \tilde{\beta} = \varphi \cdot \tilde{\xi}$ is smooth. On the other hand,

$\mathfrak{N}(k[\epsilon])$ : $\underline{h}_{P(\eta)}(k[\epsilon]) \to [\underline{B}(k[\epsilon])]$ is an isomorphism and therefore $\mathfrak{F}$ is smooth by 2.9(a), which means that $P(\xi)$ is a formal power-series ring over $P(\eta)$ .

(b) follows from (a) by setting $\underline{A} = \underline{B}$ .

Finally we consider the question of "representability" of a cofibered groupoid. Let $\underline{C}$ be a fixed category. Each object $R$ in $\underline{C}$ defines a functor $h_R : \underline{C} \to (\mathrm{Ens})$ given by $h_R(S) = \mathrm{Hom}_{\underline{C}}(R,S)$ , and in turn we obtain a functor $h : \underline{C}^0 \to \underline{\mathrm{Hom}}(\underline{C},(\mathrm{Ens}))$ which is fully-faithful. A "cocartegory in $\underline{C}$ " is a diagram

$$h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} \longrightarrow h_{R_1} \underset{s_2}{\overset{s_1}{\underset{\longleftarrow}{\overset{\longrightarrow}{\epsilon}}}} h_{R_0} \qquad \text{with } s_i \cdot \epsilon = \text{identity for } i = 1,2$$

(where $h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} = (h_{R_1}) \underset{s_1}{\times} \underset{h_{R_0}}{\times} \underset{s_2}{\times} (h_{R_1})$) such that

(i) $c$ is associative and is compatible with $s_i$ for $i = 1,2$

(ii) $$h_{R_1} \xrightarrow{\sim} h_{R_0} \underset{h_{R_0}}{\times} h_{R_1} \xrightarrow{\epsilon \times 1} h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} \xrightarrow{c} h_{R_1}$$

$$h_{R_1} \xrightarrow{\sim} h_{R_1} \underset{h_{R_0}}{\times} h_{R_0} \xrightarrow{1 \times \epsilon} h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} \xrightarrow{c} h_{R_1}$$

are both identities.

If $(R_0, R_1, s_1, s_2, \epsilon, c)$ is a cocategory in $\underline{C}$ , then for an object $S$ in $\underline{C}$ , we can define a category $h_{(R_0, R_1, \ldots)}(S)$ by setting ob $h_{(R_0, R_1, \ldots)}(S) = \mathrm{Hom}_{\underline{C}}(R_0, S)$ , and Fl $h_{(R_0, R_1, \ldots)}(S) = \mathrm{Hom}_{\underline{C}}(R_1, S)$ in which the composition rule of arrows is given by $c : h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} \to h_{R_1}$ . Then $h_{(R_0, R_1, \ldots)} : \underline{C} \to (\text{categories})$ is a functor and in turn defines a cofibered category $\underline{h}_{(R_0, R_1, \ldots)}$ over $\underline{C}$ . For instance, any object $R$ in $\underline{C}$ defines a discrete cocategory $\underline{h}_R = \underline{h}_{(R, R, s_1, s_2, \epsilon, c)}$ by setting $s_1, s_2, \epsilon, c$ to be all equal to the identity map. A homomorphism $(R_0, R_1, \ldots) \to (R_0', R_1', \ldots)$

of cocategories in $\underline{C}$ is a pair of maps $h_{R_0} \to h_{R_0'}$ , $h_{R_1} \to h_{R'_1}$ in $\underline{C}$ such that the induced map

$$h_{(R_0,\ R_1,\ldots)}(S) \to h_{(R_0',\ R_1',\ldots)}(S)$$

is a functor for every object $S$ in $\underline{C}$ . For instance, each cocategory $(R_0, R_1,\ldots)$ is provided with a homomorphism $\epsilon : \underline{h}_{R_0} \to \underline{h}_{(R_0,\ R_1,\ldots)}$ of cocategories. A cogroupoid in $\underline{C}$ is by definition a cocategory $(R_0, R_1,\ldots)$ in $\underline{C}$ such that $h_{(R_0,\ R_1,\ldots)}(S)$ is a groupoid for every object $S$ in $\underline{C}$ . In other words, a cogroupoid in $\underline{C}$ is a cocategory $(R_0, R_1,\ldots)$ in $\underline{C}$ such that $\underline{h}_{(R_0,\ R_1,\ldots)}$ is a cofibered category in groupoids over $\underline{C}$ .

In practice, a cocategory or cogroupoid occur in a following fashion: Let $P : \underline{F} \to \underline{C}$ be a cofibered category, and let $\xi$ be a fixed object in $\underline{F}$ . Set $R_0 = P(\xi)$ and let $(R_0, \underline{C})$ be the category of arrows in $\underline{C}$ with the source $R_0$ . Let $P : \underline{F}' \to (R_0, \underline{C})$ be the pull-back of $\underline{F}$ under the canonical functor $(R_0, \underline{C}) \to \underline{C}$ , and choose a cocartesian section $\xi^{\#}$ of $\underline{F}'$ over $(R_0, \underline{C})$ such that $\xi^{\#}(id_{R_0}) = \xi$ . (This simply means that we choose, for each arrow $R_0 \xrightarrow{f} S$ in $\underline{C}$ , a cocartesian f-map $\xi \to \xi^{\#}(f)$ ). Define the functor $\mathrm{Hom}_{\xi^{\#}} : (R_0 \amalg R_0, \underline{C}) \to (\mathrm{Ens})$ given by

$$X = (f_1, f_2) \longmapsto \mathrm{Hom}_{\underline{F}(S)}(\xi^{\#}(f_1), \xi^{\#}(f_2)) \quad \text{where}$$

$S$ = the target of $f_1$ ( = the target of $f_2$ ). We note that this functor does not depend on the choice of $\xi^{\#}$ up to (canonical) natural equivalence. Henceforth, by abuse of notation, we shall simply write $\xi$ for $\xi^{\#}$ . If this functor is representable by an object $X_0 = (R_0 \underset{s_2}{\overset{s_1}{\rightrightarrows}} R_1)$ in $(R_0 \quad R_0, \underline{C})$ i.e. if there exists $R_0$-map $\varphi : \xi(s_1) \to \xi(s_2)$ in $\underline{F}$ such that the induced map $\tilde{\varphi} : \mathrm{Hom}_{(R_0 \amalg R_0,\ \underline{C})}(X_0, X) \to \mathrm{Hom}_{\xi}(X)$ is bijective for all $X$ in $(R_0 \amalg R_0, \underline{C})$ , then the object $X_0$ defines a cocategory $(R_0, R_1,\ldots)$ in $\underline{C}$ in a natural way. Indeed, a triple of arrows

$$R_0 \overset{u}{\underset{w}{\overset{v}{\rightrightarrows}}} S$$

in $\underline{C}$ defines a composition

$$\mathrm{Hom}_{\underline{F}(S)}(\xi(v), \xi(w)) \times \mathrm{Hom}_{\underline{F}(S)}(\xi(u), \xi(v)) \to \mathrm{Hom}_{\underline{F}(S)}(\xi(u), \xi(w))$$

and in turn yields the composition rule

$$h_{R_1}(S) \underset{h_{R_0}(S)}{\times} h_{R_1}(S) \to h_{R_1}(S)$$

depending functorially on $S$ . Consequently the functor $\mathrm{Hom}_\xi$ defines the cocategory $(R_0, R_1, \ldots)$ in $\underline{C}$ and in turn defines a cofibered category $\underline{h}_{(R_0, R_1, \ldots)}$ in $\underline{C}$ . By definition, $\mathrm{ob}\, \underline{h}_{(R_0, R_1, \ldots)} = \coprod_S \mathrm{Hom}_{\underline{C}}(R_0, S)$ (disjoint union for $S \in \mathrm{ob}(\underline{C})$ ), and an arrow $(R_0 \overset{f}{\to} S) \to (R_0 \overset{f'}{\to} S')$ is a pair $S \overset{h}{\to} S'$ and $R_1 \overset{u}{\to} S'$ in $\underline{C}$ such that $u \cdot s_1 = h \cdot f$ and $u \cdot s_2 = f'$ , and the structural functor $P : \underline{h}_{(R_0, R_1, \ldots)} \to \underline{C}$ is given by $(R_0 \overset{f}{\to} S) \longmapsto S$ . Now the pair $(\xi, \varphi)$ defines the functor

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \ldots)} \to \underline{F} \text{ over } \underline{C} \text{ by}$$

$$(R_0 \overset{f}{\to} S) \longmapsto \xi(f)$$

$$[(R_0 \overset{f}{\to} S) \xrightarrow{(h,u)} (R_0 \to S')] \longmapsto \text{the composit map } \xi(f) \xrightarrow{h_*} \xi(hf) \xrightarrow{u(\varphi)} \xi(f')$$

We note that, for each object $S$ in $\underline{C}$ , the functor $(\xi,\varphi)(S)\ \underline{h}_{(R_0,R_1,\ldots)}(S) \to \underline{F}(S)$ is fully-faithful by virtue of the definition of $(R_0 \rightrightarrows R_1)$ coming from the representability of the functor $\mathrm{Hom}_\xi$ . Consequently the functor $(\xi,\varphi) : \underline{h}_{(R_0,R_1,\ldots)} \to \underline{F}$ is an $\underline{C}$-equivalence if and only if $\xi$ is a quasi-initial object of $\underline{F}$ over $\underline{C}$ i.e., for every object $a$ in $\underline{F}$ there exists a cocartesian arrow $\xi \to a$ . Instead of this functor $\mathrm{Hom}_\xi$ one may also consider the functor

$$\mathrm{Isom}_\xi : (R_0 \amalg R_0, \underline{C}) \to (\mathrm{Ens})$$

given by $X = (f_1,f_2) \longmapsto \mathrm{Isom}_{\underline{F}(S)}(\xi(f_1),\xi(f_2))$ where $S$ is the target of $f_1$ ( = the target of $f_2$ ). If this functor is representable by an object $(R_0 \rightrightarrows R_1)$ in $(R_0 \amalg R_0, \underline{C})$ , then it defines a cogroupoid $(R_0,R_1,\ldots)$ in $\underline{C}$ together with the functor $(\xi,\varphi) : \underline{h}_{(R_0,R_1,\ldots)} \to \underline{F}$ , which is, when $\underline{F}$ is a cofibered groupoid, a $\underline{C}$-equivalence if and only if $\xi$ is a quasi-initial object of $\underline{F}$ over $\underline{C}$ .

Definition and proposition 2.11. A cogroupoid $(R_0,R_1,\ldots)$ in $\hat{\underline{C}}_\wedge$ is called smooth if any one of the following equivalent conditions holds:

(i) $R_1$ is a formal power-series ring over $R_0$ via $s_1$ (and $s_2$ )

(ii) $\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}$ is smooth over $\underline{C}_\wedge$ .

A minimally smooth cogroupoid or a normalized smooth cogroupoid in $\hat{\underline{C}}_\wedge$ is a smooth cogroupoid $(R_0,R_1,\ldots)$ in $\hat{\underline{C}}_\wedge$ such that $\underline{h}_{(R_0,R_1,\ldots)}(k[\epsilon])$ is totally disconnected i.e., $s_1(k[\epsilon]) = s_2(k[\epsilon])$ .

Proof: $\epsilon : \underline{h}_{R_0} \to \underline{h}_{(R_0,R_1,\ldots)}$ is smooth over $\underline{C}_\wedge \Leftrightarrow$ given any surjective map $(S',f') \xrightarrow{(h,u)} (S, f)$ in $\underline{h}_{(R_0,R_1,\ldots)}$ , there exists a lifting $f''$ of $f$ to $S'$ together with an isomorphism $(S',f'') \xrightarrow{(1,u')} (S',f') \Leftrightarrow$ given $u \in h_{R_1}(S)$ and a lifting $f'$ or $s_1(u)$ to $S'$ , there exists a lifting $u'$ of $u$ to $S'$ such that $s_1(u') = f'$ , where $S' \to S$ is an arbitrary surjective map in $\underline{C}_\wedge \Leftrightarrow s_1 : \underline{h}_{R_1} \to \underline{h}_{R_0}$ is smooth over $\underline{C}_\wedge$ .

Proposition 2.12. Let $(R_0,R_1,\ldots)$ be a cogroupoid in $\hat{\underline{C}}_\wedge$ .

(a) If $(R_0,R_1,\ldots)$ is smooth, then $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous.

(b) If $s_1(k[\epsilon]) = s_2(k[\epsilon])$ , then the following statements are equivalent

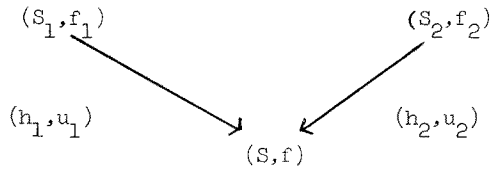
(i) $(R_0,R_1,\ldots)$ is smooth

(ii) $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous

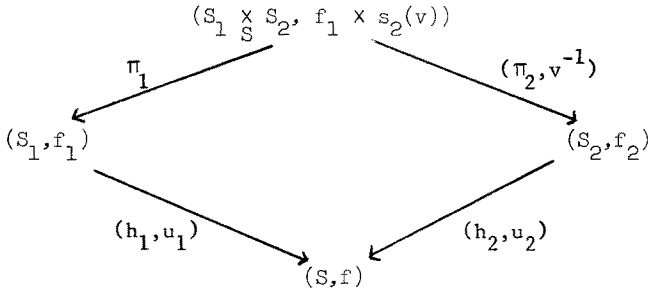
(iii) $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous

(c) If $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous, then it is equivalent to $\underline{h}_{(S_0,S_1,\ldots)}$ for some minimally smooth cogroupoid $S_*$ .

Proof: (a) Consider a diagram




in $\underline{h}_{(R_0,R_1,\ldots)}$ where $h_2 : S_2 \longrightarrow S$ is surjective in $\underline{C}_\Lambda$ . Since $s_1 : \underline{h}_{R_1} \longrightarrow \underline{h}_{R_0}$ is smooth by hypothesis, $u_1^{-1} u_2$ can be lifted to some $v \in h_{R_1}(S_2)$ such that $s_1(v) = f_2$ . Since $h_2 \cdot s_2(v) = h_1 \cdot f_1$ , $f_1 \times_S s_2(v) \in h_{R_0}(S_1 \times_S S_2)$ is defined and we obtain the commutative diagram




We claim that the above diagram defines a fibre-product $(S_1,f_1) \times_{(S,f)} (S_2,f_2)$ in $\underline{h}_{(R_0,R_1,\ldots)}$ . Indeed, for any object (T, g) in $\underline{h}_{(R_0,R_1,\ldots)}$ we have

$$\mathrm{Hom}((T,g),(S_1 \times_S S_2, f_1 \times s_2(v))) = \{(h_1' \times h_2', w_1 \times w_2) \in h_T(S_1 \times_S S_2) \times h_{R_1}(S_1 \times_S S_2) \mid s_1(w_1 \times w_2) = (h_1' \times h_2')\, g\ ,$$

$$s_2(w_1 \times w_2) = f_1 \times s_2(v)\} = \{(h_1' \times h_2', w_1, w_2) \in h_T(S_1 \times_S S_2) \times h_{R_1}(S_1) \times h_{R_1}(S_2) \mid h_1 w_1 = h_2 w_2,$$

$$s_1(w_1) = h_1'\, g,\ s_1(w_2) = h_2'\, g,\ s_2(w_1) = f_1,\ s_2(w_2) = s_2(v)\} =$$

$$\{(h_1' \times h_2', w_1, w_2) \in h_T(S_1 \times_S S_2) \times h_{R_1}(S_1) \times h_{R_1}(S_2) \mid u_1^{-1} u_2\, h_2(v^{-1} w_2) = h_1 w_1\ ,\ s_1(w_1) = h_1'\, g,$$

$s_2(w_1)=f_1,\ s_1(v^{-1}w_2)=h_2'g,\ s_2(v^{-1}w_2)=f_2\} \simeq \{(h_1'\times h_2', w_1, w_2') \in h_T(S_1 \amalg S_2) \times h_{R_1}(S_1) \times h_{R_1}(S_2) \mid$

$u_2 h_2 w_2' = u_1 h_1 w_1,\ s_1(w_1) = h_1'g,\ s_2(w_1) = f_1,\ s_1(w_2') = h_2'g,\ s_2(w_2') = f_2\} =$

$\mathrm{Hom}((T, g), (S_1, f_1)) \underset{\mathrm{Hom}((T,g), (S,f))}{\times} \mathrm{Hom}((T, g), (S_2, f_2))$ .

(b) It suffices to show (iii) $\Rightarrow$ (i) . Since $\underline{h}_{(R_0,R_1,\dots)}$ is semi-homogeneous and $\dim[\underline{h}_{(R_0,R_1,\dots)}(k[\varepsilon])] = \dim h_{R_0}(k[\varepsilon]) < \infty$ , $\underline{h}_{(R_0,R_1,\dots)}$ admits a minimally versal formal object. On the other hand, $(R_0, 1_{R_0})$ is a quasi-initial minimal object of $\widehat{\underline{h}_{(R_0,R_1,\dots)}}$ since $\underline{h}_{R_0}(k[\varepsilon]) \to [\underline{h}_{(R_0,R_1,\dots)}(k[\varepsilon])]$ is bijective, and therefore $(R_0, 1_{R_0})$ is a minimally versal formal object of $\underline{h}_{(R_0,R_1,\dots)}$ by 1.10(ii), which means that $\varepsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\dots)}$ is minimally smooth.

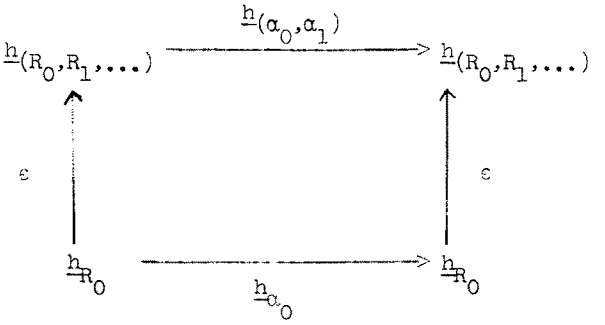
(c) Let $(S_0, \xi_0)$ be a minimally versal formal object for $\underline{h}_{(R_0,R_1,\dots)}$ (1.11). For any object $\xi : R_0 \to T$ in $\underline{h}_{(R_0, R_1,\dots)}(T)$ , the functor $\mathbf{Isom}(\xi, \xi)$ from $C_\wedge/T \otimes_\wedge T$ to (sets) which to any $R$ associates the sets of isomorphisms between the two images of $\xi$ on $R : R_0 \xrightarrow{\xi} T \longrightarrow T \otimes_\wedge T \longrightarrow R$ , is represented by $R_1 \hat{\otimes}_{R_0 \otimes_\wedge R_0} (T \otimes_\wedge T)$ . By a passage to limit, one get that $\mathbf{Isom}(\xi_0, \xi_0)$ is pro-represented by some $S_1$ , and $\underline{h}_{(R_0,R_1,\dots)}$ is equivalent to $\underline{h}_{(S_0,S_1,\dots)}$ . By (b), $S_*$ is minimally smooth.

Definition 2.13. A cofibered groupoid $A$ over $C_\wedge$ is called (minimally) pro-representable if it is $C_\wedge$-equivalent to $\underline{h}_{(R_0,R_1,\dots)}$ for some (normalized) cogroupoid $(R_0,R_1,\dots)$ in $\hat{C}_\wedge$ .

Proposition 2.14. If $(R_0, R_1,\dots)$ is a normalized cogroupoid in $\hat{C}_\wedge$ , then every homomorphism $(\alpha_0,\alpha_1) : (R_0,R_1,\dots) \to (R_0,R_1,\dots)$ which induces $C_\wedge$-equivalence $\underline{h}_{(\alpha_0,\alpha_1)} : \underline{h}_{(R_0,R_1,\dots)} \longrightarrow \underline{h}_{(R_0, R_1,\dots)}$ is an isomorphism. In

particular, for any cofibered groupoid $A$ over $C_\wedge$, a normalized prorepresentation, if it exists, is unique up to an isomorphism.

Proof: Consider the commutative diagram




where $\underline{h}_{(\alpha_0,\alpha_1)}$ is a $C_\wedge$-equivalence. Since $\underline{h}_{(R_0,R_1,\dots)}(k[\varepsilon])$ is totally disconnected, $[\underline{h}_{(\alpha_0,\alpha_1)}(k[\varepsilon])$ and $[\varepsilon(k[\varepsilon])]$ are both bijections, and hence so is $\underline{h}_{\alpha_0}(k[\varepsilon])$.

Consequently, $\alpha_0 : R_0 \to R_0$ is a surjective endomorphism and hence is an automorphism. It then follows that $\alpha_1$ is also an automorphism of $R_1$.

Lemma 2.15. Let $A$, $B$ be discrete groupoids over $C_\wedge$ and $\varphi : A \to B$ a minimally smooth functor. If $A$, $B$ are homogeneous, then $\varphi$ is a $C_\wedge$-equivalence.

Proof: We must show that $\varphi(S) : A(S) \longrightarrow B(S)$ is bijective for all $S \in ob(C_\wedge)$. We note that $\varphi(S)$ is surjective for all $S \in ob(C_\wedge)$ and bijective for $S = k[\varepsilon]$. We argue by induction on the length of $S$. Let $S' \xrightarrow{f} S$ be a small extension in $C_\wedge$. Since $A$, $B$ are homogeneous discrete groupoids, the canonical isomorphism $1 \times \tilde{f} : S' \underset{S}{\times} S' \longrightarrow S' \underset{k}{\times} k[\operatorname{Ker} f]$ induces the commutative diagram

$$\begin{array}{ccc} A(S') \underset{A(S)}{\times} A(S') & \xrightarrow{\sim} & A(S') \times A(k[\mathrm{Ker}\ f]) \\ \downarrow & & \downarrow \\ B(S') \underset{B(S)}{\times} B(S') & \xrightarrow{\sim} & B(S') \times B(k[\mathrm{Ker}\ f]) \end{array}$$

Therefore $A(S')$ , $B(S')$ are principal homogeneous sets over $A(S)$ , $B(S)$ with the structural group $A(k[\mathrm{Ker}\ f]) \xrightarrow{\sim} B(k[\mathrm{Ker}\ f])$ . Consequently if $\varphi(S)$ is bijective, then $\varphi(S')$ is injective and hence is bijective.

Corollary 2.16. Let $\underline{A}$ be a homogeneous groupoid over $\underline{C}_{\wedge}$ such that $\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_{\varepsilon})$ is of finite-dimensional. Then for any $\xi \in \mathrm{ob}(\hat{\underline{A}})$ , the functor $\mathrm{Isom}_{\xi} : \underline{C}_{\hat{R \otimes R}} \longrightarrow (\mathrm{Ens})$ is prorepresentable, where $R = P(\xi)$ .

Proof: Assume that $\mathrm{Isom}_{\xi}$ is homogeneous. We then have, by 1.11, a minimally smooth functor $\tilde{\varphi} : \underline{h}_{R_1} \to \underline{\mathrm{Isom}}_{\xi}$ , which is a $\underline{C}_{\hat{R \otimes R}}$-equivalence by 2.15 i.e., $\mathrm{Isom}_{\xi}$ is prorepresentable. Therefore it suffices to show that $\mathrm{Isom}_{\xi}$ is homogeneous. Consider a diagram

$$T_1 \longrightarrow T \longleftarrow T_2$$

in $\underline{C}_{\hat{R \otimes R}}$ where $T_2 \longrightarrow T$ is surjective. Since $\underline{A}$ is homogeneous, $\xi_i(T_1) \underset{\xi_i(T)}{\times} \xi_i(T_2)$ exists in $\underline{A}$ , and we have a canonical isomorphism $\xi_i(T_1 \underset{T}{\times} T_2) \simeq \xi_i(T_1) \underset{\xi_i(T)}{\times} \xi_i(T_2)$ for $i = 1,2$ (cf. 2.6(a)). It follows that $\mathrm{Isom}_{\xi}(T_1 \underset{T}{\times} T_2) \longrightarrow \mathrm{Isom}_{\xi}(T_1) \underset{\mathrm{Isom}_{\xi}(T)}{\times} \mathrm{Isom}_{\xi}(T_2)$ is an isomorphism.

Theorem 2.17. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_{\wedge}$ . Then it is prorepresentable by a (normalized) smooth cogroupoid if and only if

(1) $\underline{A}$ *is homogeneous*

(2) $[A(k[\varepsilon])]$ *and* $[\mathrm{Aut}(1_\varepsilon)]$ *are both finite-dimensional vector spaces over* $k$ .

*Proof*: Assume that $\underline{A}$ is prorepresentable by a smooth cogroupoid i.e., $\underline{A}$ is $\underline{C}_\wedge$-equivalent to $\underline{h} = \underline{h}_{(R_0,R_1,\ldots)}$ for some smooth cogroupoid $(R_0,R_1,\ldots)$ in $\hat{\underline{C}}_\wedge$ . Since $\underline{h}$ is homogeneous by 2.12(a), and

$$[\underline{h}(k[\varepsilon])] = \mathrm{Coker}\,(\mathrm{Hom}_{\wedge\text{-alg}}(R_1,k[\epsilon] \overset{s_1}{\underset{s_2}{\rightrightarrows}} \mathrm{Hom}_{\wedge\text{-alg}}(R_0,k[\varepsilon])) \text{ as well as}$$

$$\mathrm{Aut}_{\underline{h}}(1_\varepsilon) = \mathrm{Ker}(\mathrm{Hom}_{\wedge\text{-alg}}(R_1,k[\varepsilon]) \xrightarrow{s_1} \mathrm{Hom}_{\wedge\text{-alg}}(R_0,k[\varepsilon]))$$

$$= \mathrm{Ker}(\mathrm{Hom}_{\wedge\text{-alg}}(R_1,k[\varepsilon]) \xrightarrow{s_2} \mathrm{Hom}_{\wedge\text{-alg}}(R_0,k[\varepsilon])) \text{ are both finite-dimensional vector spaces over } k \text{ , we must have the same properties on } \underline{A} \text{ . Conversely assume (1) and (2) on } \underline{A} \text{ . We claim that } \underline{A} \text{ is prorepresentable by a normalized smooth cogroupoid. Since } \underline{A} \text{ is homogeneous and } [\underline{A}(k[\varepsilon])] \text{ is a finite-dimensional vector space over } k \text{ , } \underline{A} \text{ admits a minimally versal formal object } \xi \text{ . Set } P(\xi) = R_0 \text{ and consider the functor } \mathrm{Isom}_\xi : \underline{C}_{R_0 \hat{\otimes}_\wedge R_0} \to (\mathrm{Ens}) \text{ given by}$$

$$T \longmapsto \mathrm{Isom}_{\underline{A}(T)}(\xi_1(T),\xi_2(T))$$

where $\xi_i(T) = \xi(f_i)$ and $f_i : R_0 \to T$ denotes the composit maps

$$R_0 \overset{s_1}{\underset{s_2}{\rightrightarrows}} R_0 \hat{\otimes}_\wedge R_0 \longrightarrow T \ .$$

This functor is pro-representable by 2.16, i.e. there exists $R_1 \in \mathrm{ob}(\underline{\hat{C}}_{R_0 \hat{\otimes}_\wedge R_0})$ and an isomorphism $\xi_1(R_1) \xrightarrow{\varphi} \xi_2(R_1)$ such that $\tilde{\varphi} : h_{R_1} \to \mathrm{Isom}_\xi$ is an isomorphism over $\underline{C}_{R_0 \hat{\otimes}_\wedge R_0}$ . Then $(R_0,R_1,\ldots)$ defines a cogroupoid in $\hat{\underline{C}}_\wedge$ and the functor $(\xi,\varphi) : \underline{h}_{(R_0,R_1,\ldots)} \to \underline{A}$ is a $\underline{C}_\wedge$-equivalence. Since $\xi$ is a minimally versal formal object of $\underline{A}$ , the composit functor $\underline{h}_{(R_0,R_0,\ldots)} \overset{\varepsilon}{\to} \underline{h}_{(R_0,R_1,\ldots)} \xrightarrow{(\xi,\varphi)} \underline{A}$ is, by definition, minimally smooth and therefore so is $\varepsilon$ since $(\xi,\varphi)$ is a $\underline{C}_\wedge$-equivalence i.e., $(R_0,R_1,\ldots)$ is a normalized smooth cogroupoid in $\hat{\underline{C}}_\wedge$ .

Corollary 2.18. A functor $F : \underline{C}_\wedge \longrightarrow (Ens)$ is pro-representable if and only if it is homogeneous and $\dim F(k[\varepsilon]) < \infty$ .

Corollary 2.19. Every smooth cogroupoid in $\hat{\underline{C}}_\wedge$ can be normalized up to isomorphism.

Given a cogroupoid $(R_0,R_1,\dots)$ in $\hat{\underline{C}}_\wedge$ , we set $\bar{R}_1 = R_1/J$ where $J$ is the ideal in $R_1$ generated by $\{s_1(r) - s_2(r) \mid r \in R_0\}$ . Then $(R_0,\bar{R}_1,\dots)$ defines a formal group scheme over $R_0$ with respect to the induced composition rule. Now let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$ prorepresented by a normalized smooth cogroupoid $(R_0,R_1,\dots)$ via $(\xi,\varphi)$ . Let $\underline{C}_k \hookrightarrow \underline{C}_{R_0} \hookrightarrow \underline{C}_{R_0 \hat{\otimes}_\wedge R_0}$ be the embeddings corresponding to the surjective maps $R_0 \hat{\otimes}_\wedge R_0 \longrightarrow R_0 \longrightarrow k$ , and let $\mathrm{Aut}_\xi$ , $\mathrm{Aut}^\circ$ be the restriction functors of $\mathrm{Isom}_\xi$ on $\underline{C}_{R_0}$ , $\underline{C}_k$ respectively. In other words, $\mathrm{Aut}_\xi(S)$ = the automorphism group of $\xi(S)$ for every $S \in \mathrm{ob}(\underline{C}_{R_0})$ , and $\mathrm{Aut}^\circ(S)$ = the automorphism group of $1_S$ for every $S \in \mathrm{ob}(\underline{C}_k)$ , where $1_S$ = the direct image of $1 \in \underline{A}(k)$ under the map $k \to S$ . Since $R_1$ prorepresents the functor $\mathrm{Isom}_\xi$ on $\underline{C}_{R_0 \hat{\otimes}_\wedge R_0}$ , it follows that $\bar{R}_1$ prorepresents the functor $\mathrm{Aut}_\xi$ on $\underline{C}_{R_0}$ , and $k \hat{\otimes}_{R_0} \bar{R}_1$ prorepresents $\mathrm{Aut}^\circ$ on $\underline{C}_k$ . Though $\underline{A}$ is prorepresented by a smooth cogroupoid $(R_0,R_1,\dots)$ , neither $(R_0,\bar{R}_1,\dots)$ nor $(k,\ k \otimes_{R_0} \bar{R}_1,\dots)$ are in general smooth. Indeed we have

Corollary 2.20. Let $\underline{A}$ be a homogeneous cogroupoid over $\underline{C}_\wedge$ prorepresented by a normalized smooth cogroupoid $(R_0,R_1,\dots)$ via $(\xi,\varphi)$ .

(a) $(R_0,\bar{R}_1,\dots)$ and $(k,\ k \otimes_{R_0} \bar{R}_1,\dots)$ prorepresent the functors $\mathrm{Aut}_\xi$ and $\mathrm{Aut}^\circ$ respectively.

(b) The following statements are equivalent

(i) the formal group scheme $\underline{h}_{(R_0,\bar{R}_1,\dots)}$ is formally smooth over $R_0$ ;

(ii) for every surjective arrow $a' \to a$ in $\underline{A}$ ,

$$\mathrm{Aut}_{P(a')}(a') \longrightarrow \mathrm{Aut}_{P(a)}(a) \text{ is surjective,}$$

(iii) $R_0$ prorepresents the functor $[\underline{A}]$ over $\underline{C}_\wedge$ .

Proof: We have already noted (a). As for (b), (i) $\Leftrightarrow$ (ii) is clear whereas (ii) $\Leftrightarrow$ (iii) follows from 2.7 and 2.15, since $\tilde{\xi} : h_{R_0} \longrightarrow [\underline{A}]$ is minimally smooth.

Remark 2.21. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let $\underline{A}|\underline{C}_k$ be the pull-back of $p : \underline{A} \longrightarrow \underline{C}_\Lambda$ under $\underline{C}_k \hookrightarrow \underline{C}_\Lambda$. Then it is trivial to see that if $\underline{A}$ is prorepresented by $(R_0, R_1, \ldots)$ via $(\xi, \varphi)$, then $\underline{A}|\underline{C}_k$ is prorepresented by $(k \underset{\Lambda}{\otimes} R_0, k \underset{\Lambda}{\otimes} R_1, \ldots)$ via $(k \underset{\Lambda}{\otimes} \xi, k \underset{\Lambda}{\otimes} \varphi)$ where $k \underset{\Lambda}{\otimes} \xi$ = its direct image of $\xi$ under the map $R_0 \longrightarrow k \underset{\Lambda}{\otimes} R_0$, $k \underset{\Lambda}{\otimes} \varphi$ = the image of $\varphi$ under $\mathrm{Isom}_{\underline{A}(R_0)}(\xi(s_1), \xi(s_2)) \longrightarrow \mathrm{Isom}_{\underline{A}(k \underset{\Lambda}{\otimes} R_1)}(k \underset{\Lambda}{\otimes} \xi(s_1), k \underset{\Lambda}{\otimes} \xi(s_2))$. If $(R_0, R_1, \ldots)$ is normalized, then so is $(k \underset{\Lambda}{\otimes} R_0, k \underset{\Lambda}{\otimes} R_1, \ldots)$, and the relative dimension of $R_1$ over $R_0$ is equal to the relative dimension of $k \underset{\Lambda}{\otimes} R_1$ over $k \underset{\Lambda}{\otimes} R_0$ since $R_1$ is smooth over $R_0$.

Remark 2.22. Let $\underline{A}$ be prorepresented by smooth cogroupoid $(R_0, R_1, \ldots)$ and $(R_0', R_1', \ldots)$ where we assume that $(R_0, R_1, \ldots)$ is normalized. We then obtain a homomorphism $(i_0, i_1) : (R_0, R_1, \ldots) \longrightarrow (R_0', R_1', \ldots)$. Now $R_0'$ is a formal power-series ring over $R_0$ by 2.10(b), and the homomorphism $(i_0, i_1)$ induces an isomorphism $R_0' \underset{R_0}{\otimes} (R_0, R_1, \ldots) \xrightarrow{\sim} (R_0', R_1', \ldots)$.

## 3. The coherent sheaves $D_X^i$

One of the nice functors which is usually not pro-representable but nevertheless admits a formal versal object is the deformation functor. To this end we recall a few basic facts on commutative ring extensions and its obstruction theory. This section contains nothing new (possibly except in its systematic approach via Grothendieck cohomology on site). A purpose of this section is to make this expose as much self-contained. (Indeed we prove here more than we need later). The basic facts are stated in theorem 3.12. Throughout rings are meant to be commutative rings with unit.

Given a ring $R$ we denote by $\underline{C}_R$ the category of R-algebras. For each object $A$ in $\underline{C}_R$ we set

$\mathrm{Cov}(A)$ = the set of all surjective arrows $\varphi : A' \to A$ in $\underline{C}_R$ with target $A$.

It clearly defines a pretopology and in turn we obtain a site $\underline{C}_R$. For each object $A$ in $\underline{C}_R$, we shall denote by $\underline{C}_{A|R}$ the category of arrows in $\underline{C}_R$ with target $A$. We may and shall provide the category $\underline{C}_{A|R}$ with the topology induced from the forgetful functor $j_A : \underline{C}_{A|R} \to \underline{C}_R$. We note that the category $\underline{C}_{A|R}$ has a final object $(1_A : A \to A)$ and an initial object $(i_A : R \to A)$, and admits a fibre-product as well as coproduct. Henceforth we shall denote the object $(B \to A)$ in $\underline{C}_{A|R}$, by abuse of notation, simply by $B$ in case when there is no possible confusion. Given an abelian category $\underline{C}$, a $\underline{C}$-valued sheaf on $\underline{C}_{A|R}$ is, by definition, a functor $F : \underline{C}^{0}_{A|R} \to \underline{C}$ such that, for any surjective arrow $B' \to B$ in $\underline{C}_{A|R}$,

$$0 \to F(B) \to F(B') \rightrightarrows F(B' \underset{B}{\times} B')$$

is exact. The additive presheaves are those which preserve coproduct i.e., $F(B_1 \underset{R}{\otimes} B_2) = F(B_1) \oplus F(B_2)$. For example, an A-module $M$ defines the functor

$$\mathrm{Der}_R(-,M) : \underline{C}^{0}_{A|R} \to (A\text{-mod})$$

given by $B \to \mathrm{Der}_R(B,M)$ = R-linear derivations of $B$ with values in $M$ regarded as B-module via the structural map $B \to A$, and it is clearly an additive sheaf on $\underline{C}^{0}_{A|R}$, where (A-mod) denotes the category of A-modules. A presheaf $F$ on $\underline{C}_{A|R}$ with values in (A-mod) is called "finite type" if $F(B)$ is an A-module of finite type whenever $B$ is an R-algebra of finite type. For example, if $A$ is noetherian and $M$ is an A-module of finite type, then $\mathrm{Der}_R(-,M)$ is an additive sheaf of finite type on the site $\underline{C}_{A|R}$.

A few more words on notations and basic facts in homological algebra:

3.1(a) Throughout the rest of this section, we shall denote by $\underline{C}^{\sim}_{A|R}$ (or $\underline{C}^{\vee}_{A|R}$) the category of sheaves (or presheaves) with values in (A-mod). We have the (left exact) injection functor $i_{A|R} : \underline{C}^{\sim}_{A|R} \to \underline{C}^{\vee}_{A|R}$, and the associated-sheaf (exact)

functor $G_{A|R}: \underline{C}^{\vee}_{A|R} \to \underline{C}^{\sim}_{A|R}$ which is adjoint to the functor $i_{A|R}$. An object $B \in ob(\underline{C}_{A|R})$ induces a continuous functor $j : \underline{C}_{B|R} \to \underline{C}_{A|R}$ and in turn a commutative diagram

(*)
$$\begin{array}{ccc} \underline{C}^{\sim}_{A|R} & \xrightarrow{\ j^{\sim}\ } & \underline{C}^{\sim}_{B|R} \\ \downarrow{\scriptstyle i_{A|R}} & & \downarrow \\ \underline{C}^{\vee}_{A|R} & \xrightarrow[\ j^{\vee}\ ]{} & \underline{C}^{\vee}_{B|R} \end{array}$$

where the restriction functor $\tilde{j}$ (as well as $j^{\vee}$) is an exact functor, since the site $\underline{C}_{B|R}$ is nothing but the category of arrows in $\underline{C}_{A|R}$ having target $B$, with the induced topology. A ring homomorphism $S \to R$ induces a continuous functor $\beta : \underline{C}_{A|R} \to \underline{C}_{A|S}$, and in turn a commutative diagram

(**)
$$\begin{array}{ccc} \underline{C}^{\sim}_{A|S} & \xrightarrow{\ \tilde{\beta}\ } & \underline{C}^{\sim}_{A|R} \\ \downarrow{\scriptstyle i_{A|S}} & & \downarrow \\ \underline{C}^{\vee}_{A|S} & \xrightarrow[\ \beta^{\vee}\ ]{} & \underline{C}^{\vee}_{A|R} \end{array}$$

Note that $\tilde{\beta}$ is not necessarily an exact functor.

3.1(b) The right derived functor of the injection functor $i_{A|R}: \underline{C}^{\sim}_{A|R} \to \underline{C}^{\vee}_{A|R}$ will be denoted by $\underline{H}^{\circ}_{A|R}$. For each sheaf $F$ on $\underline{C}_{A|R}$, we set

$$H^{\circ}(A|R,F) = [H^{\circ}{}_{A|R}(F)](A)$$

For any $B' \to B$ in $Cov(B)$, let us define the functor $H^{\circ}((B' \to B),-) : \underline{C}^{\vee}_{A|R} \to (A\text{-mod})$ by $H^{\circ}((B' \to B),F) = \mathrm{Ker}(F(B') \rightrightarrows F(B' \times_B B'))$. The right derived functors of $H^{\circ}((B' \to B),-)$ will be denoted by $H^{\circ}((B' \to B),-)$. One knows that the $H^{\circ}((B' \to B),F)$ are the cohomology groups of the (Čech)

semi-simplicial module

$$F(B') \rightrightarrows F(B' \times_B B') \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; F(B' \times_B B' \times_B B') \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow\\ \rightarrow}\; \cdots$$

The right derived functor of $\overset{\vee 0}{\underline{H}}_{A|R} : \underline{C}_{A|R} \to \underline{C}_{A|R}$ will be denoted by $\overset{\vee}{\underline{H}}^{\circ}_{A|R}$, where

$$[\overset{\vee 0}{\underline{H}}_{A|R}(F)](B) = \varinjlim_{\{B' \to B\} \in \mathrm{Cov}(B)} H^0(\{B' \to B\}, F) \quad .$$

For each presheaf $F$ on $\underline{C}_{A|R}$, we set

$$\overset{\vee}{H}{}^{\cdot}(A|R,F) = [\overset{\vee}{H}{}^{\cdot}_{A|R}(F)](A) = \varinjlim_{\{A' \to A\} \in \mathrm{Cov}(A)} H^{\cdot}(\{A' \to A\}, F)$$

for any sheaf $F$ on $\underline{C}_{A|R}$, we have the usual spectral sequence

$$\overset{\vee}{H}{}^p(A|R, \underline{H}^q_{A|R}(F)) \Rightarrow H^{p+q}(A|R,F)$$

where we note that $\overset{\vee 0}{H}(A|R, \underline{H}^q(F)) = 0$ for all $q > 0$.

In particular

$$\begin{cases} H^1(A|R,F) = H^1(A|R,F) \\ 0 \to \overset{\vee 2}{H}(A|R,F) \to H^2(A|R,F) \to \overset{\vee 1}{H}(A|R, \underline{H}^1_{A\ R}(F)) \to \overset{\vee 3}{H}(A|R,F) \quad \text{is exact.} \end{cases}$$

3.1(c) If $B$ is an object in $\underline{C}_{A|R}$, the commutative diagram (*) in 2.1(a) together with the fact that $\tilde{j}$ as well as $j^{\vee}$ are exact functors yields the canonical isomorphism

$$H^{\cdot}(B|R, \tilde{j}F) = [\underline{H}^{\cdot}_{A|R}(F)](B)$$

On the other hand, a ring homomorphism $S \to R$ induces the commutative diagram (**) in 2.1(a). Now $\tilde{\beta}$ is not necessarily an exact functor. However, $\beta$ preserves a fibre-product, and thus we obtain a spectral sequence

$$\underline{H}^p_{A|R}(R^q\tilde{\beta}F) \Rightarrow \overset{\vee}{\beta}\underline{H}^{p+q}_{A|S}(F) \qquad \text{i.e.,} \quad H^p(A|R, R^q\tilde{\beta}F) \Rightarrow H^{p+q}(A|S,F)$$

Since we have the canonical isomorphism $\tilde{\beta} = G_{A|R} \circ \overset{\vee}{\beta} \circ i_{A|S}$ and $\beta^{\#} = G_{A|R} \circ \overset{\vee}{\beta}$ is an

exact functor, we have $R^{\cdot}\widetilde{\beta}F = \beta^{\#}\underline{H}^{\cdot}_{A|S}(F)$ . Thus the spectral sequence above can now be written as

$$H^p(A|R,\beta^{\#}\underline{H}^q_{A|S}(F)) \Rightarrow H^{p+q}(A|S,F) \quad .$$

Lemma 3.2. Given $B \in ob(C_{A|R})$ , choose a surjective arrow $P \to B$ in $C_{B|R}$ , where $P$ is a polynomial algebra over $R$ . Then for any $F \in ob(C^{\vee}_{A|R})$ , the natural map $H^{\cdot}(\{P \to B\},F) \to \underset{\vee}{H}^{\cdot}(F)\ (B)$ is an isomorphism.

Proof: It suffice to see that the natural map

$$H^0(\{P \to B\},F) \to \underset{\vee}{H}^0(F)\ (B) = \varinjlim_{\{B' \to B\} \in Cov(B)} H^0(\{B' \to B\},F)$$

is an isomorphism, which is clear since, for any $\{B' \to B\} \in Cov(B)$ , there exists an arrow $\{P \to B\} \to \{B' \to B\}$ .

Corollary 3.3.

(a) Assume that $A$ is noetherian. If $F \in ob(\widetilde{C_{A|R}})$ is of finite type, then $\underline{H}^n(F)$ is of finite type for all $n$ .

(b) For any polynomial algebra $P$ over $R$ , we have $H^n(P|R,F) = 0$ for all $n > 0$ and for all $F \in ob(\widetilde{C_{P|R}})$ .

(c) A diagram $0 \to F' \to F \to F'' \to 0$ in $\widetilde{C_{A|R}}$ is exact if and only if $0 \to F'(P) \to F(P) \to F''(P) \to 0$ is exact for every $P \in ob(C_{A|R})$ which is a polynomial algebra over $R$ .

Proof:

(a) In view of the spectral sequence

$$\underset{\vee}{H}^p(B|R,\underline{H}^q(F)) \Rightarrow H^{p+q}(B|R,F)$$

together with the fact that $\underset{\vee}{H}^0(\underline{H}^q(F)) = 0$ for all $q > 0$ , it suffices to show that $\underline{H}^n(G)$ is of finite type for all $n$ and for all presheaf $G$ of finite type. However $\underset{\vee}{H}^{\cdot}(B|R,G) = H^{\cdot}(\{P \to B\},G)$ , where $P$ is a polynomial algebra over $R$ and $\{P \to B\} \in Cov(B)$ , and furthermore $H^{\cdot}(\{P \to B\},G)$ is simply the

cohomology gotten from the complex

$$\cdots\cdots\cdots\cdots G(P \underset{B}{\times} P \underset{B}{\times} P) \overleftarrow{\leftarrow} G(P \underset{B}{\times} P) \overleftarrow{\leftarrow} G(P) \ .$$

If $B$ is an $R$-algebra of finite type, then we may take $P$ to be of finite type over $R$ so that $P \underset{B}{\times} \cdots \underset{B}{\times} P$ are all $R$-algebras of finite type. If $G$ is of finite type, then $G(P \underset{B}{\times} \cdots \underset{B}{\times} P)$ are all $A$-modules of finite type, so that $H^*(\{P \to B\},G)$ is an $A$-module of finite type since $A$ is noetherian.

(b) Let $P$ be a polynomial algebra over $R$. If $B \to P$ is any surjective $R$-algebra map, then it amits a section and hence $H^i(\{B \to P\},G) = 0$ for all $i > 0$, so that $\check{H}^i(P|R, G) = 0$ for all $i > 0$ and for all presheaves $G$ on $\underline{C}_{P|R}$. It follows that the spectral sequence $\check{H}^p(P|R, \underline{H}^q_{P|R}(F)) \Rightarrow H^{p+q}(P|R, F)$ degenerates and hence $\check{H}^0(P|R, \underline{H}^n(F)) \simeq H^n(P|R, F)$ for all $n$. However $\check{H}^0(P|R, \underline{H}^n(F)) = 0$ for all $n > 0$ whenever $F$ is a sheaf, and hence $H^n(P|R, F) = 0$ for all $n > 0$ and for all $F \in \mathrm{ob}(\underline{\tilde{C}}_{P|R})$.

(c) If $0 \to F' \to F \to F'' \to 0$ is an exact sequence in $\underline{\tilde{C}}_{A|R}$, then

$$0 \to F'(B) \to F(B) \to F''(B) \to H'(B|R, F') \to H'(B|R, F) \to \cdots$$

is exact. If $P \in \mathrm{ob}(\underline{C}_{A|R})$ is a polynomial algebra over $R$, then $H^1(P|R, F') = 0$ by 3.3(b) and hence

$$0 \to F' \to F \to F'' \to 0$$

is exact. As for the other direction, let $\underline{P}_{A|R}$ be the full subcategory of $\underline{C}_{A|R}$ in which

$$\mathrm{ob}(\underline{P}_{A|R}) = \{P \in \mathrm{ob}(\underline{C}_{A|R}) \mid P \text{ is a polynomial algebra over } R\} \ ,$$

provided with the induced topology, and let $p : \underline{P}_{A|R} \to \underline{C}_{A|R}$ be the inclusion functor. Since every object in $\underline{C}_{A|R}$ can be covered by an object in $\underline{P}_{A|R}$, it follows from a comparison lemma ( ) that $\tilde{p} : \underline{\tilde{C}}_{A|R} \to \underline{\tilde{P}}_{A|R}$ is an equivalence of categories. Consequently, if $0 \to F'(P) \to F(P) \to F''(P) \to 0$ is exact for every $P \in \mathrm{ob}(\underline{P}_{A|R})$, then $0 \to \tilde{p}F' \to \tilde{p}F \to \tilde{p}F'' \to 0$ is exact in $\underline{\tilde{P}}_{A|R}$ and hence $0 \to F' \to F \to F'' \to 0$ is exact in $\underline{\tilde{C}}_{A|R}$.

Remark 3.4. Every surjective arrow in $\underline{P}_{A|R}$ admits a section. It follows immediately that every presheaf on $\underline{P}_{A|R}$ is actually a sheaf i.e., $\underline{P}^{\sim}_{A|R} = \underline{P}_{A|R}$. Consequently we find that the restriction functor $\underline{C}^{\sim}_{A|R} \to \underline{P}^{\vee}_{A|R}$ is an equivalence of categories.

Lemma 3.5. Consider a diagram of R-algebras

$$\begin{array}{ccc} & & C \\ & & \downarrow \\ B' & \longrightarrow & B \\ & \text{surjective} & \end{array}$$

and let S be an R-algebra.

(i) If $\mathrm{Tor}_1^R(B,S) = 0$, then the canonical map

$$(B' \underset{B}{\times} C) \underset{R}{\otimes} S \to (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S)$$

is an isomorphism.

(ii) Assume that $B'$ is R-flat. If $\mathrm{Tor}_i^R(C,S) = 0$ for all $0 < i < n$ and $\mathrm{Tor}_i^R(B,S) = 0$ for all $0 < i < n$, then $\mathrm{Tor}_i^R(B' \underset{B}{\times} C, S) = 0$ for all $0 < i < n$.

Proof: Let $0 \to J \to B' \to B \to 0$ be exact.

(i) If $\mathrm{Tor}_i^R(B, S) = 0$, then

$$0 \to J \underset{R}{\otimes} S \to B' \underset{R}{\otimes} S \to B \underset{R}{\otimes} S \to 0$$

is exact so that

$$0 \to J \underset{R}{\otimes} S \to (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S) \to C \underset{R}{\otimes} S \to 0$$

is exact. We thus obtain an exact commutative diagram

$$\begin{array}{ccccccccc} & & J \underset{R}{\otimes} S & \longrightarrow & (B' \underset{B}{\times} C) \underset{R}{\otimes} S & \longrightarrow & C \underset{R}{\otimes} S & \longrightarrow & 0 \\ & & \| & & \downarrow & & \| & & \\ 0 & \longrightarrow & J \underset{R}{\otimes} S & \longrightarrow & (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S) & \longrightarrow & C \underset{R}{\otimes} S & \longrightarrow & 0 \end{array}$$

and therefore $(B' \underset{B}{\otimes} C) \underset{R}{\otimes} S \to (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S)$ is an isomorphism.

(ii) Since $B'$ is R-flat and $\mathrm{Tor}_i^R(B, S) = 0$ for all $0 < i \leq n$ by hypothesis, we have $\mathrm{Tor}_i^R(J, S) = 0$ for all $0 < i < n$. Then the exact sequence $0 \to J \to B' \underset{B}{\times} C \to C \to 0$ together with the hypothesis that $\mathrm{Tor}_i^R(C, S) = 0$ for all $0 < i < n$ entails that $\mathrm{Tor}_i^R(B' \underset{B}{\times} C, S) = 0$ for all $0 < i < n$.

Corollary 3.6.

(a) *Let* $A, S$ *be* R*-algebras. If* $\mathrm{Tor}_1^R(A, S) = 0$, *then the canonical map* $\check{H}^*(A \underset{R}{\otimes} S | S, G) \to \check{H}^*(A|R, SG)$ *is an isomorphism for any* $G \in \mathrm{ob}(\underset{A \underset{R}{\otimes} S | S}{\check{C}})$ *where the functor* $S : \underline{C}_{A|R} \to \underline{C}_{A \underset{R}{\otimes} S | S}$ *is given by* $X \longmapsto X \underset{R}{\otimes} S$. *In particular if* $S$ *is* R*-flat, then the canonical map* $\check{S}\check{H}^*(G) \to \check{H}(\check{S}G)$ *is an isomorphism for every* $G \in \mathrm{ob}(\underset{A \underset{R}{\otimes} S | S}{\check{C}})$.

(b) *Let* $P \to B$ *be a surjective* R*-algebra map, where* $P$ *is a polynomial algebra over* R. *If* $\mathrm{Tor}_i^R(B, S) = 0$ *for all* $0 < i \leq n$, *then* $\mathrm{Tor}_i^R(P \underset{B}{\times} P \underset{B}{\times} \dots \underset{B}{\times} P, S) = 0$ *for all* $0 < i < n$.

Proof:

(a) Given $B \in \mathrm{ob}(\underline{C}_{A|R})$, choose $\{P \to B\} \in \mathrm{Cov}(B)$ where $P$ is a polynomial algebra over R. If $\mathrm{Tor}_1^R(B, S) = 0$, the natural map $(\overbrace{P \underset{B}{\times} \dots \underset{B}{\times} P}^{n}) \underset{R}{\otimes} S \to (P \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} \dots \underset{(B \underset{R}{\otimes} S)}{\times} (P \underset{R}{\otimes} S)$ is an isomorphism for all $n$ by 3.5(i), and hence $\check{H}^*(B|R, \check{S}G) = H^*(\{P \to B\}, \check{S}G) = H^*(\{P \underset{R}{\otimes} S \to B \underset{R}{\otimes} S\}, G) = \check{H}^*(B \underset{R}{\otimes} S | S, G)$. If $S$ is R-flat, then $\mathrm{Tor}_1^R(X, S) = 0$ for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ and hence $\check{H}^*(X \underset{R}{\otimes} S | S, G) \to \check{H}^*(X|R, \check{S}G)$ is an isomorphism for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ i.e. $\check{S}\underline{\check{H}}^*(G) \to \underline{\check{H}}^*(\check{S}G)$ is an isomorphism.

(b) Set $P^{(m)} = \overbrace{P \underset{B}{\times} \dots \underset{B}{\times} P}^{m+1}$. If $m = 0$ there is nothing to prove since $P$ is R-flat. Assume that $\mathrm{Tor}_i^R(P^{(m-1)}, S) = 0$ for all $0 < i < n$. Since $P \to B$ is surjective and $P^{(m)} = P \underset{B}{\times} P^{(m-1)}$, it follows from 3.5(ii) that $\mathrm{Tor}_i^R(P^{(m)}, S) = 0$ for all $0 < i < n$.

Proposition 3.7.

(a) Let $A$, $S$ be $R$-algebras. If $\mathrm{Tor}_i^R(A, S) = 0$ for all $0 < i \leq n$, then for any sheaf $F$ on $\underline{C}_{A\underset{R}{\otimes}S|S}$, the canonical map $H^i(A\underset{R}{\otimes}S|S, F) \to H^i(A|R, \tilde{S}F)$ is an isomorphism for all $i \leq n$, where $S : \underline{C}_{A|R} \to \underline{C}_{A\underset{R}{\otimes}S|S}$ is given by $X \longmapsto X\underset{R}{\otimes}S$.

(b) Let $P$ be a polynomial algebra over $R$. Then for any additive sheaf $F$ on $\underline{C}_{P\underset{R}{\otimes}A|R}$, the canonical map $H^n(P\underset{R}{\otimes}A|R, F) \to H^n(A|R, F)$ is an isomorphism for all $n > 0$.

(c) Let $S \to R \to A$ be ring homomorphisms. Then for any additive sheaf $F$ on $\underline{C}_{A|S}$, we have an exact sequence

$$0 \to H^0(A|R, F_R) \to H^0(A|S, F) \to H^0(R|S, F) \to H^1(A|R, F_R) \to H^1(A|S, F) \to H^1(R|S, F) \to$$
$$\cdots\cdots \to H^n(A|R, F_R) \to H^n(A|S, F) \to H^n(R|S, F) \to H^{n+1}(A|R, F_R) \to \cdots.$$

where $F_R$ is a sheaf on $\underline{C}_{A|R}$ defined by

$$F_R(X) = \mathrm{Ker}(F(X) \to F(R))$$

for all $X \in \mathrm{ob}(\underline{C}_{A|R})$ via the forgetful functor $\underline{C}_{A|R} \to \underline{C}_{A|S}$.

Proof:

(a) Consider the continuous functor

$$S : \underline{C}_{A|R} \to \underline{C}_{A\underset{R}{\otimes}S|S}$$

where $S(X) = X\underset{R}{\otimes}S$. Since the functor $S$ sends a polynomial algebras over $R$ to polynomial algebras over $S$, $\tilde{S} : \underline{\tilde{C}}_{A\underset{R}{\otimes}S|S} \to \underline{\tilde{C}}_{A|R}$ is an exact functor by 3.3(c) and in turn we obtain the canonical map $\check{S}\underline{H}^{\cdot}_{A\underset{R}{\otimes}S|S}(F) \to \underline{H}^{\cdot}_{A|R}(\tilde{S}F)$, and in turn we obtain the commutative diagram of spectral sequences:

$$\begin{array}{ccc}
E_S^{p,q}(S(X)) = \check{H}^p(X\underset{R}{\otimes}S|S, \underline{H}^q_{A\underset{R}{\otimes}S|S}(F)) & \Longrightarrow & H^{p+q}(X\underset{R}{\otimes}S|S, F) \\
\downarrow & & \\
\check{H}^p(X|R, \check{S}\underline{H}^q_{A\underset{R}{\otimes}S|S}(F)) & & \downarrow \\
\downarrow & & \\
E_R^{p,q}(X) = \check{H}^p(X|R, \underline{H}^q_{A|R}(\tilde{S}F)) & \Longrightarrow & H^{p+q}(X|R, \tilde{S}F)
\end{array}$$

We note that $E_S^{p,q}(S(X)) \to \check{H}^p(X|R, \check{S}\underline{H}^q_{\underline{A}\otimes_R S|S}(F))$ is an isomorphism whenever $\mathrm{Tor}_1^R(X, S) = 0$ by 3.6(a). Assume now that $[\check{S}\underline{H}^q_{\underline{A}\otimes_R S|S}(F)](X) \to [\underline{H}^q_{\underline{A}|R}(\tilde{S}F)](X)$ is an isomorphism for all $q < n$ and for all $X \in \mathrm{ob}(\underline{C}_{\underline{A}|R})$ such that $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$. It follows immediately from 3.6(b) that $E_S^{p,q}(S(X)) \to E_R^{p,q}(X)$ is an isomorphism for all $q < n$ and for all $p$, provided $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$. Since $E_S^{0,n}(S(X)) \to E_R^{0,n}(X)$ is an isomorphism (since both sides are zero), we get that $H^n(X_{\otimes_R}S|S, F) \to H^n(X|R, \tilde{S}F)$ is an isomorphism whenever $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$ i.e., $[\check{S}\underline{H}^i_{\underline{A}\otimes_R S|S}(F)](X) \to [\underline{H}^i_{\underline{A}|R}(\tilde{S}F)](X)$ is an isomorphism for all $i \leq n$ whenever $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$.

(b) Let $P$ be a polynomial algebra over $R$ and let $f : \underline{C}_{\underline{A}|R} \to \underline{C}_{\underline{A}\otimes_R P|R}$ be the continuous functor given by $f(X) = X\otimes_R P$. Let $P' \to X$ be a surjective arrow in $\underline{C}_{\underline{A}|R}$ where $P'$ is a polynomial algebra over $R$. Then $P'\otimes_R P \to X\otimes_R P$ is a surjective arrow in $\underline{C}_{\underline{A}\otimes_R P|R}$, and $P'\otimes_R P$ is again a polynomial algebra over $R$. Since $P$ is flat over $R$, it follows from 3.5(i) that $\check{H}^*(X|R, \check{f}G) = H^*(\{P' \to X\}, \check{f}G) = H(\{P'\otimes_R P \to X\otimes_R P\}, G) = \check{H}^*(X\otimes_R P|R, G)$ for every $G \in \mathrm{ob}(\underline{C}^{\check{}}_{\underline{A}\otimes_R P|R})$. It follows that $\tilde{f} : \underline{C}^{\sim}_{\underline{A}\otimes_R P|R} \to \underline{C}_{\underline{A}|R}$ sends acyclic ones to acyclic ones. On the other hand, $f$ sends polynomial algebras over $R$ to polynomial algebras over $R$, and hence $\tilde{f} : \underline{C}^{\sim}_{\underline{A}\otimes_R P|R} \to \underline{C}^{\sim}_{\underline{A}|R}$ is an exact functor by 3.2(c). It follows that $H^*(A\otimes_R P|R, F) \to H^*(A|R, \tilde{f}F)$ is an isomorphism for every $F \in \mathrm{ob}(\underline{C}^{\sim}_{\underline{A}\otimes_R P|R})$. Now for each $X \in \mathrm{ob}(\underline{C}_{\underline{A}|R})$, we have canonical maps

$$(\tilde{f}F)(X) = F(X\otimes_R P) \to F(X)\oplus F(P) ,$$

and hence a canonical map $\tilde{f}F \to R\oplus F(P)$ where $F(P)$ denotes a constant sheaf (and $F$ on the right hand side denotes, strictly speaking, the restriction of $F$ on $\underline{C}_{\underline{A}|R}$). If $F$ is an additive sheaf, then $(\tilde{f}F)(X) = F(X\otimes_R P) \to F(X)\oplus F(P)$ is an isomorphism for every $X \in \mathrm{ob}(\underline{C}_{\underline{A}|R})$ and hence $\tilde{f}F \cong R\oplus F(P)$ is an isomorphism. Therefore

$$H^n(A|R, \tilde{f}F) \simeq H^n(A|R, F \otimes F(P)) = H^n(A|R, F)$$

for all $n > 0$ since $F(P)$, being a constant sheaf, is flask. It follows that $H^n(A \underset{R}{\otimes} P|R, F) \to H^n(A|R, F)$ is an isomorphism for all $n > 0$, provided $F$ is an additive sheaf on $\underline{C}_{A \underset{R}{\otimes} P|R}$.

(c) (communicated from Rinehart) Consider the forgetful functor $\beta : \underline{C}_{A|R} \to \underline{C}_{A|S}$. We have a spectral sequence

$$E^{p,q} = H^p(A|R, \beta^{\#}\underline{H}^q_{A|S}(F)) \Longrightarrow H^{p+q}(A|S, F)$$

where $\beta^{\#} = G_{A|R} \circ \beta^{\vee}$, $G_{A|R} : \underline{C}^{\vee}_{A|R} \to \underline{C}^{\sim}_{A|R}$ is "associated-sheaf" functor. Now let $F$ be an additive sheaf. We claim that $\beta^{\#}\underline{H}^q_{A|S}(F)$ is a constant sheaf for all $q > 0$. Indeed, if $P$ is a polynomial algebra over $R$, then $P = R \underset{S}{\otimes} P'$ for a polynomial algebra $P'$ over $S$ and hence

$$[\beta^{\#}\underline{H}^q_{A|S}(F)](P) = H^q(P|S, F) = H^q(R \underset{S}{\otimes} P'|S, F) \to H^q(R|S, F)$$

is an isomorphism for all $q > 0$ by 3.7(b), and surjective for $q = 0$. It follows from 3.3(c) that the canonical map $\beta^{\#}\underline{H}^q_{A|S}(F) \to H^q(R|S, F) =$ constant sheaf is an isomorphism for all $q > 0$, and is surjective for $q = 0$. It follows that $E^{p,q} = H^p(A|R, \beta^{\#}\underline{H}^q_{A|S}(F)) = 0$ for all $p, q > 0$, and hence the above spectral sequence yields the exact sequence

$$(*) \quad 0 \to E^{1,0} \to H^1 \to E^{0,1} \to E^{2,0} \to H^2 \to E^{0,2} \to E^{3,0} \to H^3 \to \cdots$$

where we note that $E^{0,q} = [\beta^{\#}\underline{H}^q_{A|S}(F)](A) \simeq H^q(R|S, F)$ for all $q > 0$ and $E^{p,0} = H^p(A|R, \tilde{\beta}F)$. On the other hand, the exact sequence $0 \to F_R \to \tilde{\beta}F \to F(R) \to 0$ gives us the exact sequence

$$(**) \quad 0 \to H^0(A|R, F_R) \to H^0(A|R, \tilde{\beta}F) \to H^0(A|R, F(R)) \to H^1(A|R, F_R) \to H^1(A|R, \tilde{\beta}F) \to 0$$

$$\qquad\qquad\qquad\qquad \| \qquad\qquad\qquad\qquad \|$$

$$\qquad\qquad\qquad H^0(A|S, F) \qquad\qquad H^0(R|S, F)$$

The exact sequences (*) and (**) combined gives the desired exact sequence.

Corollary 3.8.

(a) If A, B are R-algebras such that $\mathrm{Tor}_i^R(A, B) = 0$ for all $0 < i \leq n$, then for any additive sheaf $F \in \mathrm{ob}(\widetilde{C}_{A\underset{R}{\otimes}B|R})$, the canonical map

$$H^i(A\underset{R}{\otimes}B|R, F) \to H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism for all $i \leq n$.

(b) Let A be an R-algebra such that $\mathrm{Tor}_i^R(A, A) = 0$ for all $i > 0$. If $A\underset{R}{\otimes}A \to A$ is an isomorphism, then for any additive sheaf $F \in \mathrm{ob}(\widetilde{C}_{A|R})$, we have $H^i(A|R, F) = 0$ for all $i$.

Proof:

(a) Let $F \in \mathrm{ob}(\widetilde{C}_{A\underset{R}{\otimes}B|R})$ be additive. By 3.7(c), the ring homomorphisms $R \to A \to A\underset{R}{\otimes}B$ gives us the exact sequence

(*) $\quad \ldots \to H^i(A\underset{R}{\otimes}B\ A, F_A) \to H^i(A\underset{R}{\otimes}B\ R, F) \to H^i(A|R, F) \to \ldots.$

consider now the commutative diagram

$$\begin{array}{ccc} H^i(A\underset{R}{\otimes}B|A, F_A) & \longrightarrow & H^i(A\underset{R}{\otimes}B|R, F) \\ \downarrow & & \downarrow \\ H^i(B|R, \widetilde{A}F_A) & \xrightarrow{\simeq} & H^i(B|R, F) \end{array}$$

where the functor $A : \underline{C}_{B|R} \to \underline{C}_{A\underset{R}{\otimes}B|A}$ is given by $X \longmapsto A\underset{R}{\otimes}X$, and the map $H^i(B|R, \widetilde{A}F_A) \to H^i(B|R, F)$ is induced from the map $\widetilde{A}F_A \to F$, which is an isomorphism since F is additive. Note that the commutativity of the above diagram, because of its functorial nature, is reduced to the case $i = 0$.

If $\mathrm{Tor}_i^R(A, B) = 0$ for all $0 < i \leq n$, then $H^i(A\underset{R}{\otimes}B|A, F_A) \to H^i(B|R, \widetilde{A}F_A)$ is an isomorphism for all $i \leq n$ by 3.7(a), and hence the composite map

$$H^i(A\underset{R}{\otimes}B|A, F_A) \to H^i(A\underset{R}{\otimes}B|R, F) \to H^i(B|R, F)$$

and (because of the symmetry)

$$H^i(A\underset{R}{\otimes}B|B, F_B) \to H^i(A\underset{R}{\otimes}B|R, F) \to H^i(A|R, F)$$

are isomorphisms for all $i \leq n$. This implies that the exact sequence (*) breaks up to short split-exact sequences

$$0 \to H^i(A\underset{R}{\otimes}B|A, F_A) \to H^i(A\underset{R}{\otimes}B|R, F) \to H^i(A|R, F) \to 0$$

for all $i \leq n$, and that

$$H^i(A\underset{R}{\otimes}B|R, F) \to H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism for all $i \leq n$.

(b) Since $\mathrm{Tor}_i^R(A,A) = 0$ for all $i > 0$, it follows from 3.8(a) that

$$H^i(A\underset{R}{\otimes}A|R, F) \to H^i(A|R, F) \oplus H^i(A|R, F)$$

is an isomorphism for all $i$. On the other hand, $A\underset{R}{\otimes}A \to A$ is an isomorphism by hypothesis and hence $H^i(A|R, F) \to H^i(A\underset{R}{\otimes}A|R, F)$ is an isomorphism for all $i$. This means that the diagonal map $H^i(A|R, F) \to H^i(A|R, F) \oplus H^i(A|R, F)$ is an isomorphism for all $i$ i.e., $H^i(A|R, F) = 0$ for all $i$.

Corollary 3.9: Let $S$ be a multiplicative subset of an R-algebra $A$. If $F \in \mathrm{ob}\left(\underset{\sim}{C}_{S^{-1}A|R}\right)$ is additive such that $F_A$ is also additive, then the canonical map

$$H^i(S^{-1}A|R, F) \to H^i(A|R, F)$$

is an isomorphism for all $i$, where $F_A \in \mathrm{ob}\left(\underset{\sim}{C}_{S^{-1}A|A}\right)$ is defined by $F_A(X) = \mathrm{Ker}(F(X) \to F(A))$ for all $X \in \mathrm{ob}\left(\underset{\sim}{C}_{S^{-1}A|A}\right)$.

Proof: Since $F$ is additive, the ring homomorphisms $R \to A \to S^{-1}A$ gives us the exact sequence

$$\ldots. \to H^n(S^{-1}A|A, F_A) \to H^n(S^{-1}A|R, F) \to H^n(A|R, F) \to \ldots.$$

On the other hand, since $S^{-1}A\underset{A}{\otimes}S^{-1}A \to S^{-1}A$ is an isomorphism and $F_A$ is additive by hypothesis, it follows from 3.8(b) that $H^n(S^{-1}A|A, F_A) = 0$ for all $n$, and hence $H^n(S^{-1}A|R, F) \to H^n(A|R, F)$ is an isomorphism for all $n$.

Now let $A$ be an $R$-algebra. An $A$-module $M$ defines an additive sheaf $\mathrm{Der}_R(-, M) \in \mathrm{ob}(\underset{\sim}{C}_{A|R})$ by $X \to \mathrm{Der}_R(X, M)$. We now make the following definition:

Definition 3.10: Let $A$ be an $R$-algebra. For each $A$-module $M$ we set

$$D^i(A|R, M) = H^i(A|R, \mathrm{Der}_R(-, M)) \ .$$

We want to make an explicit computation of low-dimensional ones i.e., $D^i(A|R, M)$ for $i = 1, 2$ : Choose a surjective arrow $P \to A$ in $\underset{\sim}{C}_{A|R}$ where $P$ is a polynomial algebra over $R$, and let $0 \to I \to P \to A \to 0$ be exact. Then

Lemma 3.10:

$$D^1(A|R, M) = \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_P(I, M))$$
$$D^2(A|R, M) = H^1(A|R, \underset{\sim}{H}^1_{A|R}(\mathrm{Der}_R(-, M)))$$

Proof: Consider the simplicial algebra

$$(*) \qquad \cdots\cdots P_A^{(2)} \rightrightarrows P_A^{(1)} \rightrightarrows P_A^{(0)}$$

where $P_A^{(n)} = \overbrace{P \underset{A}{\times} \cdots \underset{A}{\times} P}^{n+1}$. For each $n$ we have the exact sequence

$$0 \to I^{(n)} \to P_A^{(n)} \to A \to 0$$

where $I^{(n)} = \overbrace{I \times I \times \cdots \times I}^{n+1}$, and it is easy to see that

(a):

$$0 \to \mathrm{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \to \mathrm{Der}_R(P_A^{(n)}, M) \xrightarrow{\mathrm{Der}(\Delta, M)} \mathrm{Der}_R(P, M) \to 0$$

is exact for all $n$, where $\bar{I} = I/I^2$ and $\Delta$ stands for the diagonal map. We also note the exact sequence

(b):

$$0 \to \mathrm{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \to \mathrm{Hom}_A(\bar{I}^{(n)}, M) \xrightarrow{\mathrm{Hom}(\Delta, M)} \mathrm{Hom}_A(I, M) \to 0$$

The exact sequences (a) and (b) yield the respective exact sequences of simplicial $A$-modules. Since $\{I \to 0\}$, $\{I \to I\}$ admits sections, we have

$$H^i(\{I \to 0\}, \mathrm{Hom}_A(-, M)) = H^i(\{I \to I\}, \mathrm{Hom}_A(-, M)) = 0$$

for all $i > 0$ , and it follows readily that

$$\begin{cases} 0 \to \mathrm{Der}_R(A, M) \to \mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(\bar{I}, M) \to H^1(\{P \to A\}, \mathrm{Der}_R(-, M)) \to 0 \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{is exact} \\ H^i(\{P \to A\}, \mathrm{Der}_R(-, M)) = 0 \quad \text{for all} \quad i > 1 \end{cases}$$

It follows from 3.1(b) that

$$\begin{aligned} D^1(A|R, M) &= \overset{\vee}{H}{}^1(A|R, \mathrm{Der}_R(-, M)) = H^1(\{P \to A\}, \mathrm{Der}_R(-, M)) \\ &= \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(\bar{I}, M)) \\ &= \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_P(I, M)) \ , \\ D^2(A|R, M) &= H^1(A|R, \underline{H}^1_{A|R}(\mathrm{Der}_R(-, M))) \ . \end{aligned}$$

Now choose a free P-module $F$ and a P-linear map $F \overset{\gamma}{\to} P$ such that $\gamma(F) = I$ . Associated to P-linear map $\gamma : F \to P$ , there is a Koszul complex, which is simply denoted, by abuse of notation, by $K_I$ .

Lemma 3.11: $D^2(A|R, M) \simeq \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H_1(K^I), M))$ .

Proof: We have

$$\begin{aligned} D^2(A|R, M) &= H^1(A|R, H^1_{A|R}(\mathrm{Der}_R(-, M))) \\ &= \mathrm{Ker}(H^1(P_A^{(1)}|R, M) \rightrightarrows H^1(P_A^{(2)}|R, M)) \end{aligned}$$

since $[\underline{H}^1_{A|R}(\mathrm{Der}_R(-, M))](P_A^0) = [\underline{H}^1_{A|R}(\mathrm{Der}_R(-, M))](P) = 0$ .

Let $0 \to K \to F \overset{\gamma}{\to} I \to 0$ be exact. If we set $P[F]$ to be the symmetric algebra over $P$ generated by $F$ , then $P[F]$ and $P[F] \otimes_P P[F]$ are polynomial algebras over $R$ and we obtain the surjective arrows

$$P[F] \to P_A^{(1)} = P \underset{A}{\times} P \qquad \text{given by} \qquad X \to (0, \gamma(X))$$

$$P[F] \otimes_P P[F] \to P_A^{(2)} = P \underset{A}{\times} P \underset{A}{\times} P \qquad X \otimes 1 \to (0, \gamma(X), 0), \ 1 \otimes X \to (0, 0, \gamma(X))$$

where $X \in F$ . If we set $J = \mathrm{Ker}(P[F] \to P_A^{(1)})$ and $J' = \mathrm{Ker}(P[F] \otimes_P P[F] \to P_A^{(2)})$ , then

$$J = (F) \cap (F_Y) = (K) + (F \cdot F_Y)$$

$$J' = (F \otimes 1 + 1 \otimes F) \cap (F_Y \otimes 1 + 1 \otimes F) \cap (F \otimes 1 + 1 \otimes F_Y)$$

$$= (K \otimes 1 + 1 \otimes K) + (F \cdot F_Y \otimes 1 + 1 \otimes F \cdot F_Y) + (F \otimes F)$$

where $F_Y$ = the ideal generated by $\{X - Y(X) \mid X \in F\}$ . Now $\pi_i : P_A^{(2)} \to P_A^{(1)}$ $i = 0, 1, 2$ can be lifted to $\hat{\pi}_i : P[F] \otimes_P P[F] \to P[F]$ where, for each $X \in F$ ,

$$\pi_0 = \begin{cases} X \otimes 1 \to Y(X) - X \\ 1 \otimes X \to X \end{cases} \qquad \pi_1 = \begin{cases} X \otimes 1 \to 0 \\ 1 \otimes X \to X \end{cases} \qquad \pi_2 = \begin{cases} X \otimes 1 \to X \\ 1 \otimes X \to 0 \end{cases} .$$

A trivial computation shows that $\pi_0 - \pi_1 + \pi_2$ sends $K \otimes 1 + 1 \otimes K + 1 \otimes F \cdot F_Y$ to zero, whereas it sends $(F \cdot F_Y \otimes 1 + F \otimes F)$ onto $(F \cdot F_Y)$ . Now for any $D \in \mathrm{Der}_R(P[F] \otimes_P P[F], M)$ , $D(F \cdot F_Y \otimes 1 + F \otimes F) = 0$ since $I \cdot M = 0$ , and consequently

$$D^2(A|R, M) = \mathrm{Ker}(D^1(P_A^{(1)}|R, M) \rightrightarrows D^1(P_A^{(2)}|R, M))$$

$$= \{[f] \in D^1(P_A^{(1)}|R, M) \mid f(F \cdot F_Y) = 0\}$$

$$= \mathrm{Coker}(\mathrm{Der}_R(P[F], M) \to \mathrm{Hom}_{P[F]}((F) \cap (F_Y)/(F \cdot F_Y), M))$$

$$= \mathrm{Coker}(\mathrm{Der}_P(P[F], M) \to \mathrm{Hom}_{P[F]}((F) \cap (F_Y)/(F) \cdot (F_Y), M)) .$$

Now

$$0 \to (F) \to P[F] \xrightarrow{P[0]} P \to 0$$

$$0 \to (F_Y) \to P[F] \xrightarrow{P[Y]} P \to 0 .$$

are exact and hence $(F) \cap (F_Y)/(F) \cdot (F_Y) = \mathrm{Tor}_1^{P[F]}(P, P_Y) = H_1(K_I)$ where $K_I$ is the Koszul complex associated to the P-linear map $F \xrightarrow{Y} I (\subset P)$ . We thus have

$$D^2(A|R, M) = \mathrm{Coker}(\mathrm{Hom}_P(F, M) \to \mathrm{Hom}_P(H_1(K_I), M))$$

$$= \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H^1(K_I), M)) .$$

We can now state all the basic facts needed for our purpose, summarizing the various results we have already established.

Theorem 3.12.

(1) If $0 \to M' \to M \to M'' \to 0$ is an exact sequence of A-modules, then

$$0 \to D^0(A|R, M') \to D^0(A|R, M) \to D^0(A|R, M'') \to D^1(A|R, M') \to$$
$$D^1(A|R, M) \to D^1(A|R, M'') \to \cdots\cdots \to D^n(A|R, M') \to$$
$$D^n(A|R, M) \to D^n(A|R, M'') \to \cdots$$

is exact.

(2) Let $S \to R \to A$ be ring homomorphisms. Then for any A-module $M$ we have the exact sequence

$$0 \to D^0(A\ R, M) \to D^0(A\ S, M) \to D^0(R\ S, M) \to D^1(A\ R, M) \to$$
$$D^1(A\ S, M) \to D^1(R\ S, M) \to \cdots\cdots \to D^n(A\ R, M) \to$$
$$D^n(A\ S, M) \to D^n(R\ S, M) \to \cdots.$$

(3) Let $A$, $S$ be R-algebras such that $\mathrm{Tor}_i^R(A, S) = 0$ for all $i > 0$. Then for any $A \underset{R}{\otimes} S$-module $M$, the canonical map $D^i(A \underset{R}{\otimes} S|S, M) \to D^i(A|R, M)$ is an isomorphism for all $i$. In particular, if $A$ is R-flat we have

$$D^i(A \underset{R}{\otimes} S|S, M) \cong D^i(A|R, M)$$

for all $i$ and for all R-algebras $S$.

(4) Let $S$ be a multiplicative subset of $A$. Then for any $S^{-1}A$-module $N$, the canonical map

$$D^i(S^{-1}A|R, N) \to D^i(A|R, N)$$

is an isomorphism for all $i$. Consequently, for any A-module $M$ we have the canonical isomorphism

$$D^i(S^{-1}A|R, S^{-1}M) \cong S^{-1}D^i(A|R, M)$$

for all $i$.

(5) If $A$, $B$ are R-algebras such that $\mathrm{Tor}_i^R(A, B) = 0$ for all $0 < i \leq n$, then for any $A \underset{R}{\otimes} B$-module $M$, the canonical map

$$D^i(A \underset{R}{\otimes} B|R, M) \to D^i(A|R, M) \oplus D^i(B|R, M)$$

is an isomorphism for all $i \leq n$. In particular, if $\mathrm{Tor}_i^R(A, A) = 0$ for all

$i > 0$ and

$$\Delta : \mathrm{Spec}(A) \to \mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(A)$$

is an open immersion (for instance A/R etale), then $D^i(A|R, M) = 0$ for all $i > 0$ and for all A-modules M .

(6) Let A be an R-algebra of essentially finite type over R , where R is noetherian, and let E be a flat A-module. Then we have a canonical isomorphism

$$E \otimes_A D^i(A|R, M) \xrightarrow{\sim} D^i(A|R, E \otimes_A M)$$

for all $i \geq 0$ and for all A-modules M .

(7) If R is noetherian and A is an R-algebra of essentially finite type, then $D^i(A|R, M)$ is A-module of finite type for all i , provided M is an A-module of finite type.

(8) Let P be a polynomial algebra over R and let $0 \to I \to P \to A \to 0$ be exact. Then $D^0(A|R, M) = \mathrm{Der}_R(A, M)$ ,

$$D^1(A|R, M) = \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)$$
$$= \text{the set of isomorphism classes of}$$
$$\text{R-algebra extensions of A by M},$$
$$D^2(A|R, M) = \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H_1(K_I), M)) .$$

Proof:

(1) If $0 \to M' \to M \to M'' \to 0$ is an exact sequence of A-modules, then for any polynomial algebra P over R , $0 \to \mathrm{Der}_R(P, M') \to \mathrm{Der}_R(P, M) \to \mathrm{Der}_R(P, M'') \to 0$ is exact and hence $0 \to \mathrm{Der}_R(-, M') \to \mathrm{Der}_R(-, M) \to \mathrm{Der}_R(-, M'') \to 0$ is an exact sequence of sheaves by 3.3(c).

(2) $\mathrm{Der}_S(-, M)$ is an additive sheaf on $\underline{C}_{A|S}$ , and $\mathrm{Der}_R(B, M) = \mathrm{Ker}(\mathrm{Der}_S(B, M) \to \mathrm{Der}_S(R, M))$ for every $B \in \mathrm{ob}(\underline{C}_{A|R})$ . Therefore our assertion follows from 3.7(c).

(3) Let $S : \underline{C}_{A|R} \to \underline{C}_{A \otimes_R S|S}$ be the functor given by $X \longmapsto X \otimes_R S$ . Then $\tilde{S}\mathrm{Der}_S(-, M) = \mathrm{Der}_R(-, M)$ and hence our assertion follows from 3.7(a).

(4) Follows from 3.9. Indeed, $F = \mathrm{Der}_R(-, N)$ is an additive sheaf on $\underline{C}_{S^{-1}A|R}$, and $F_A \in \mathrm{ob}\left(\underline{C}^{\sim}_{S^{-1}A|A}\right)$ is also additive since $F_A(X) = \mathrm{Ker}(F(X) \to F(A)) = \mathrm{Ker}(\mathrm{Der}_R(X, N) \to \mathrm{Der}_R(A, N)) = \mathrm{Der}_A(X, N)$ i.e., $F_A = \mathrm{Der}_A(-, N)$ .

(5) The first assertion follows from 3.8(a). Now assume that $\mathrm{Tor}_i^R(A, A) = 0$ for all $i > 0$ and that

$$\Delta : \mathrm{Spec}(A) \to \mathrm{Spec}(A) \underset{\mathrm{Spec}(R)}{\times} \mathrm{Spec}(A)$$

is an open immersion i.e., $(A \otimes_R A)_{\varphi^{-1}(\underline{m})} \to A_{\underline{m}}$ is an isomorphism for all maximal ideals $\underline{m}$ of $A$, where $\varphi : A \otimes_R A \to A$ is the multiplication map. Then the first assertion combined with 3.12(4) entails that the diagonal map

$$\Delta_{\underline{m}} : D^i(A|R, M)_{\underline{m}} \to D^i(A|R, M)_{\underline{m}} \oplus D^i(A|R, M)_{\underline{m}}$$

is an isomorphism for every maximal ideal $\underline{m}$ of $A$ and hence

$$\Delta : D^i(A|R, M) \to D^i(A|R, M) \oplus D^i(A|R, M)$$

is an isomorphism, which entails that $D^i(A|R, M) = 0$ for all $i$ .

(6) In view of 3.12(4), we may assume that $A$ is an R-algebra of finite type. Now consider the map

$$E \otimes_A \mathrm{Der}_R(-, M) \to \mathrm{Der}_R(-, E \otimes_A M)$$

in $\underline{C}^{\sim}_{A|R}$ . Since $E$ is A-flat, we have an isomorphism

$$E \otimes_A \mathrm{Der}_R(B, M) \xrightarrow{\sim} \mathrm{Der}_R(B, E \otimes_A M)$$

whenever $B \in \mathrm{ob}(\underline{C}_{A|R})$ is of finite type over $R$ . It follows from 3.3(b) that

$$D^i(A|R, E \otimes_A \mathrm{Der}_R(-, M)) \to D^i(A|R, E \otimes_A M)$$

is an isomorphism. On the other hand, the functor $E \otimes_A -$ : (A-mod) $\to$(A-mod) is exact, and therefore we have

$$E \otimes_A D^i(A|R, M) = D^i(A|R, E \otimes_A \mathrm{Der}_R(-, M)) \quad ,$$

and hence our assertion.

(7) follows from 3.3(a).

(8) In view of 3.10 and 3.11, it suffices to show that $\mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) = \mathrm{Exalcomm}(A|R, M)$, where $\mathrm{Exalcomm}(A|R, M)$ denotes the set of isomorphism classes of R-algebra extensions of $A$ by $M$. Let $0 \to M \to A' \to A \to 0$ be an R-algebra extension of $A$ by $M$. Since $P$ is a polynomial algebra over $R$, we obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0 \end{array}$$

where $I \to M$ is uniquely determined by the extensions $A'$ up to $\mathrm{Der}_R(P, M)$. We thus obtain a map

$$\Phi : \mathrm{Exalcomm}(A|R, M) \to \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) .$$

Conversely a P-linear map $I \to M$ yields an extension by push-out

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0 \end{array}$$

Two P-linear maps $I \rightrightarrows M$ which differ by $\mathrm{Der}_R(P, M)$ is trivially seen to yield isomorphic extensions and therefore we obtain a map

$$\Psi : \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) \to \mathrm{Exalcomm}(A|R, M) .$$

It is straightforward to verify that $\Phi$, $\Psi$ are inverse process to each other.

Corollary 3.13. (a) R-algebra $A$ is formally smooth over $R$ if and only if $D^1(A|R, M) = 0$ for every A-module $M$.

(b) Let $R$ be noetherian and $A$ an R-algebra of finite type. Then $A$ is a relative complete-intersection over $R$ if and only if $D^2(A|R, M) = 0$ for every A-module $M$.

(c) Let $R$ be noetherian and $A$ an R-algebra of finite type. Then for any multiplicative subset $S$ in $A$, we have a canonical isomorphism $D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M)$ for all $i \geq 0$.

(d) Let R be noetherian and A a local R-algebra of essentially finite type. We denote by $\hat{A}$ the m-adic completion of A where m = the maximal ideal of A . Then for any A-module E of finite type we have a canonical isomorphism

$$D^1(\hat{A}|R, \hat{A} \underset{A}{\otimes} E) \xrightarrow{\sim} \hat{A} \underset{A}{\otimes} D^1(A|R, E) .$$

Proof: (a) follows immediately from $D^1(A|R, M) = \mathrm{Exalcomm}(A|R, M)$ .

(b) Let P be a polynomial algebra over R in a finite number of variables and let $0 \to I \to P \to A \to 0$ be exact. Localizing if necessary, we may assume that I is generated by a P-regular sequence which would imply that $H_1(K_I) = 0$ and therefore $D^2(A|R, M) = \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H_1(K_I), M)) = 0$ . Conversely assume that $D^2(A|R, M) = 0$ for every A-module M .

Let $0 \to I \to P \to A \to 0$ be exact where P is a polynomial algebra over R in a finite number of variables. We have

$$0 = D^2(A|R, M) = \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H_1(K_I), M))$$

i.e., $H^1(K_I, M) \to \mathrm{Hom}_A(H_1(K_I), M)$ is surjective for all A-module M . Choose an exact sequence $0 \to K \to F \to I \to 0$ where F is a free P-module of finite type. We note that $H_1(K_I) = K/K_0$ where $K_0 = \mathrm{Im}(\overset{2}{\wedge} F \to \overset{1}{\wedge} F)$ . The fact that $D^2(A|R, H_1(K_I)) = 0$ i.e.,

$$H^1(K_I, H_1(K_I)) \to \mathrm{Hom}_A(H_1(K_I), H_1(K_I)) \text{ is surjective}$$

entails that $0 \to K/K_0 \to F/IF \to I/I^2 \to 0$ is exact and splits as A-modules and in particular $I/I^2$ is projective as A-module. Therefore, localizing A if necessary, we may assume that $I/I^2$ is A-free, and that $F/IF \to I/I^2$ is an isomorphism. Then the fact that $0 \to K/K_0 \to F/IF \to I/I^2 \to 0$ is exact and splits as A-modules entails that $H_1(K_I) = K/K_0 = 0$ . This means that I is generated by a P-regular sequence i.e., A is a relative complete intersection over R .

(c) follows immediately from 3.12(4) and (6).

(d) Firstly it follows from 3.12(6) that

$$D^i(A\ R, \hat{A} \underset{A}{\otimes} E) \xrightarrow{\sim} \hat{A} \underset{A}{\otimes} D^i(A|R, E)$$

is an isomorphism for all i , and therefore it suffices to show that

$D^1(\hat{A}|R, \hat{A}\otimes_A E) \to D^1(A|R, \hat{A}\otimes_A E)$ is an isomorphism. Our proof is based on the description of $D^1$ in terms of algebra extensions. Let $0 \to \hat{A}\otimes_A E \to A' \xrightarrow{\varphi} A \to 0$ be an R-algebra extension. The fact that $\hat{A}\otimes_A E$ is an $\hat{A}$-module of finite type entails readily that $A'$ is a local R-algebra, complete with respect to $\underline{m}'$-adic topology where $\underline{m}'$ = the maximal ideal of $A'$. Therefore if there is an R-algebra map $\psi : A \to A'$ such that $\varphi \circ \psi = \mathrm{id}_A$, then $\psi$ extends uniquely to $\hat{\psi} : \hat{A} \to A'$ such that $\varphi \circ \hat{\psi} = \mathrm{id}_{\hat{A}}$. This proves that

$$D^1(\hat{A}|R, \hat{A}\otimes_A E) \to D^1(A|R, \hat{A}\otimes_A E)$$

is injective. Now let $0 \to \hat{A}\otimes_A E \to B \xrightarrow{\varphi} A \to 0$ be an R-algebra extension. Firstly we claim that the topology on $\hat{E} = \hat{A}\otimes_A E$ induced from $\underline{m}_B$-adic topology on $B$ coincides with the topology on $\hat{E}$ as $\hat{A}$-module. Indeed, since $A$ is essentially of finite type over $R$, one can find a subalgebra $B_0$ of $B$ such that $B_0$ is a local R-algebra of essentially finite type over $R$ and that $\varphi(B_0) = A$. Set $E_0 = E \cap B_0$ and denote by $\underline{m}_B$, $\underline{m}_0$, $\underline{m}$ the maximal ideals of $B$, $B_0$, $A$ respectively. Then $\underline{m}_B = \underline{m}_0 + \hat{E}$ and hence $\underline{m}_B^{n+1} = \underline{m}_0^{n+1} + \underline{m}^n\hat{E}$ for all $n$. Since $B_0$ is noetherian, there exists an integer $r > 0$ by Artin-Rees Theorem such that

$$\hat{E} \cap \underline{m}_B^{n+1} = \underline{m}^n\hat{E} + \hat{E} \cap \underline{m}_0^{n+1} = \underline{m}^n\hat{E} + E_0 \cap \underline{m}_0^{n+1}$$
$$= \underline{m}^n\hat{E} + \underline{m}^{n+1-r}(E_0 \cap \underline{m}_0^r) \subset \underline{m}^{n+1-r}\hat{E}$$

for all $n \geq r$, which proves our assertion about the two topologies on $E$. In particular $\hat{E} = \hat{A}\otimes_A E$ is complete with respect to the topology induced from $\underline{m}_B$-adic topology on $B$, and hence we obtain the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \hat{E} & \longrightarrow & B & \longrightarrow & A & \longrightarrow & 0 \\ & & \| & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \hat{E} & \longrightarrow & \hat{B} & \longrightarrow & \hat{A} & \longrightarrow & 0 \end{array}$$

where $\hat{B}$ stands for $\underline{m}_B$-adic completion of $B$. This proves that $D^1$ $D^1(\hat{A}|R, \hat{E}) \to D^1(A|R, \hat{E})$ is surjective and hence we are done.

Corollary 3.14. Let $R$ be noetherian and $A$ an $R$-algebra of finite type, smooth everywhere except at one closed point $x \in \mathrm{Spec}(A)$. We then have isomorphisms

$$D^1(A|R, A) \simeq D^1(A_x|R, A_x) \simeq D^1(\hat{A}_x|R, \hat{A}_x) .$$

Proof: Since $A$ is smooth over $R$ except at $x \in \mathrm{Spec}(A)$, $D^1(A|R, A)$ is an $A$-module of finite type with its support on the set $\{x\}$, and therefore we have

$$D^1(A|R, A) = D^1(A|R, A)_x = \widehat{D^1(A|R, A)_x} .$$

Therefore our assertion follows from 3.13.

Remark 3.15. Let $R$ be topological ring and $A$ a topological $R$-algebra. We then define, for any $A$-module $M$,

$$D^i_{\mathrm{top}}(A|R, E) = \varinjlim_{\alpha} D^i(A_\alpha|R_\alpha, A_\alpha \otimes_A E)$$

where $A_\alpha = A/J_\alpha$, $R_\alpha = R/I_\alpha$, and $J_\alpha$, $I_\alpha$ are both open in $A$, $R$ respectively. Then $A$ is formally smooth topological $R$-algebra means that $D^1_{\mathrm{top}}(A|R, E) = 0$ for every $A$-module $E$.

Now let $X$ be a scheme over a ring $R$ and $\underline{E}$ a quasi-coherent $X$-module. We then define the sheaves $\underline{D}^i_{X|R}(\underline{E}) = \underline{D}^i(X|R, \underline{E})$ on $X$ via presheaves on $X$ given by

$$[\underline{D}^i_{X|R}(\underline{E})](U) = D^i(\Gamma(U, \underline{O}_X)|R, \Gamma(U, \underline{E}))$$

for every affine open set $U \subset X$. If $R$ is noetherian and $X$ is of finite type over $R$, then the property 3.13(a) shows that $\underline{D}^i_{X|R}(\underline{E})$ can actually be constructed canonically, as a quasi-coherent $X$-module.

In case when there is no possible confusion we shall abreviate by setting $\underline{D}^i_{X|R} = \underline{D}^i_{X|R}(\underline{O}_X)$. All the statements of 3.13 can now be translated in terms of these quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$, which we leave to the readers. We list below a few properties which is most relevant to our purpose.

Corollary 3.15. Let $R$ be noetherian and $X$ an $R$-scheme of finite type. Then

(a) For any coherent X-module $\underline{E}$, $\underline{D}^i_{X|R}(\underline{E})$ is again a coherent X-module for all $i$, and furthermore $\mathrm{Supp}\, \underline{D}^i_{X|R}(\underline{E}) \subset X^{\mathrm{Sing}}$ for all $i > 0$ where

$X^{\mathrm{Sing}} = \{x \in X |\ X$ is non-smooth over $R$ at the point $x \in X\}$.

(b) If $X$ is flat over $R$, then for any $R$-algebra $S$, we have the canonical isomorphism

$$\underline{D}^i_{X\otimes_R S|S}(\underline{E}) \tilde{\to} \underline{D}^i_{X|R}(\underline{E})$$

for all $i$.

(c) If $X$ is a relative complete intersection over $R$, then for every surjective ring homomorphism $S \to R$ we have the exact sequence

$$0 \to \underline{D}^1_{X|R}(\underline{E}) \to \underline{D}^1_{X|S}(\underline{E}) \to \underline{\mathrm{Hom}}_{\underline{O}_X}(I/I^2 \otimes_{\underline{O}_X}, \underline{E}) \to 0$$

where $I = \mathrm{Ker}(S \to R)$. If, furthermore, $R$ is local with the residue field $k$ and $X$ is $R$-flat, then we have the exact sequence

$$0 \to \underline{D}^1_{X_0|k}(\underline{O}_{X_0}) \to \underline{D}^1_{X|S}(\underline{O}_{X_0}) \to D^1(R|S, k) \otimes_k \underline{O}_{X_0} \to 0$$

where $X_0 = X \otimes_R k$.

Proof: (a) Follows from 3.12(6) and (8) whereas (b) follows from 3.12(3).

(c) Since $X$ is a relative complete-intersection over $R$, we have $\underline{D}^i_{X|R}(\underline{E}) = 0$ for all $i \geq 2$ by 3.12(10) and hence

$$0 \to \underline{D}^0_{X|R}(\underline{E}) \to \underline{D}^0_{X|S}(\underline{E}) \to \underline{D}^0_{R|S}(\underline{E}) \to \underline{D}^1_{X|R}(\underline{E}) \to \underline{D}^1_{X|S}(\underline{E}) \to \underline{D}^1_{R|S}(\underline{E}) \to 0$$

is exact by 3.12(2), where $\underline{D}^i_{R|S}(\underline{E})$ are quasi-coherent sheaves of $X$ defined by

$$[\underline{D}^i_{R|S}(\underline{E})](U) = D^i(R|S, \underline{E}(U))$$

for every affine open subset $U \subset X$. In particular, $\underline{D}^0_{R|S}(\underline{E}) = 0$ since $S \to R$ is surjective, and

$$\underline{D}^1_{R|S}(\underline{E}) = \underline{\mathrm{Hom}}_{\underline{O}_X}(I/I^2 \otimes_{\underline{O}_X}, \underline{E})$$

by 3.12(7), and hence the first exact sequence. If $X$ is R-flat, we have

$\underline{D}^i_{X_0|k}(\underline{E}) \tilde{\to} \underline{D}^i_{X|R}(\underline{E})$ for all $i$ and for all quasi-coherent $X_0$-module $\underline{E}$ and in particular

$$\underline{D}^1_{X_0|k}(\underline{O}_{X_0}) \tilde{\to} \underline{D}^1_{X|R}(\underline{O}_{X_0})$$

from which the second exact sequence follows.

Remark 3.16: The quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$ we have defined should appear as $E_2^{0,i}$ of a spectral sequence relating the local ones and the global ones. L. Illusie has obtained such a theory (Lecture Notes in Mathematics 239). His method is via "homotopical algebra" generalizing the methods of M. Andre and D. Quillen rather than "cohomological algebra" . It seems that the approach we have taken here can be globalized via Grothendieck cohomology on the category of schemes with an appropriate topology generalizing the one on the affine category.

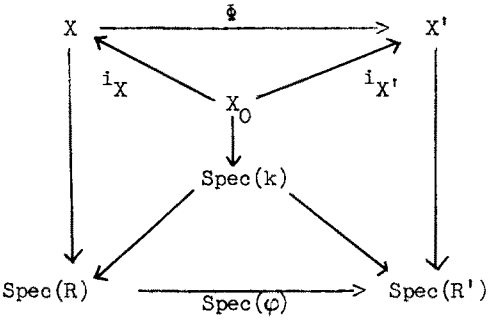
## 4. Formal moduli of deformations

One amongst the nice cofibered categories in which the isomorphism classes of objects are usually not prorepresentable, but nevertheless admit a formal versal object is the cofibered category of deformations. We now come back to the situations and notations of paragraph 1. Thus, throughout the section, $\Lambda$ is a fixed complete local noetherian ring with the residue field $k$ , and $\underline{C}_\Lambda$ is the category of artinian local $\Lambda$-algebras with the same residue field $k$ .

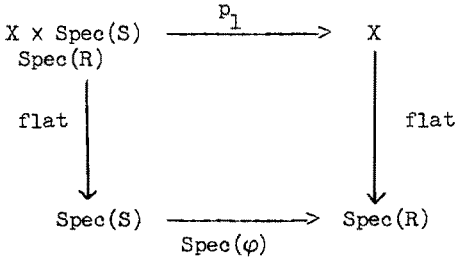
Let $X_0$ be a fixed scheme over the field $k$ . By an (infinitesimal) deformation of $X_0$ , we mean a pair $(R, X)$ where $R \in \mathrm{ob}(\underline{C}_\Lambda)$ and $X$ is a flat R-scheme together with a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that the induced morphism $X_0 \to X \times_{Spec(R)} Spec(k)$ is an isomorphism. We note that, since $R$ is an artinian local ring, $i_X : X_0 \to X$ is necessarily a closed immersion which is a homeomorphism on underlying topological spaces. If $(R, X)$, $(R', X')$ are deformations of $X_0$ to $R$, $R'$ respectively, we define an arrow $(R', X') \to (R, X)$ to consist of a map $R' \xrightarrow{\varphi} R$ in $\underline{C}_\Lambda$ and a morphism $\Phi : X \to X'$ of schemes with a commutative diagram




We thus obtain the category $\underline{M}_{X_0}$ of (infinitesimal) deformations of $X_0$ in which the objects and the arrows are defined as above. The category $\underline{M}_{X_0}$ is provided with a functor $p : \underline{M}_{X_0} \to \underline{C}_\Lambda$ defined by $(R, X) \to R$. Now let $(R, X) \in ob(\underline{M}_{X_0})$ and a map $R \xrightarrow{\varphi} S$ in $\underline{C}_\Lambda$ be given. We then obtain a commutative diagram




and $\varphi$-arrow $(\varphi, p_1) : (R, X) \to (S, X \times_{Spec(R)} Spec(S))$ is obviously cocartesian in $\underline{M}_{X_0}$. It follows that the category $\underline{M}_{X_0}$ is a cofibered category over $\underline{C}_\Lambda$ via $P : \underline{M}_{X_0} \to \underline{C}_\Lambda$. We observe the following facts:

4.1(a): Let $(1_R, \Phi) : (R, X_1) \to (R, X_2)$ be an arrow in $\underline{M}_{X_0}$ i.e., $\Phi : X_2 \to X_1$ is a morphism over $R$ such that $X_2 \otimes_R k \to X_1 \otimes_R k$ is an isomorphism. Since $R$ is artinian and $X_i$ is R-flat, it follows immediately that $\Phi$ is an isomorphism so that $(1_R, \Phi) : (R, X_1) \to (R, X_2)$ is an isomorphism in $\underline{M}_{X_0}$. Consequently, $\underline{M}_{X_0}$ is a cofibered category in groupoids over $\underline{C}_\Lambda$.

4.1(b): Suppose that $X_0$ is affine, say $X_0 = \mathrm{Spec}(A_0)$. Then for any object $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$, $X$ is also affine. Consequently, for any $R \in \mathrm{ob}(\underline{C}_\Lambda)$, the category $\underline{M}_{X_0}(R)$ is canonically equivalent, via the sections, to the category in which an object is a flat algebra $A$ over $R$ together with an R-algebra map $j_A : A \to A_0$ inducing an isomorphism $A \otimes_R k \xrightarrow{\sim} A_0$, and a map $\Phi : A_1 \to A_2$ is an R-algebra map with a commutative diagram

$$\begin{array}{ccc} A_1 & \xrightarrow{\Phi} & A_2 \\ {\scriptstyle j_{A_1}}\downarrow & & \downarrow{\scriptstyle j_{A_2}} \\ A_0 & = & A_0 \end{array}$$

4.1(c): Consider the cofibered category $\hat{\underline{M}}_{X_0}$ in groupoids over $\hat{\underline{C}}_\Lambda$. By definition, the category $\hat{\underline{M}}_{X_0}(R)$ for any $R \in \mathrm{ob}(\hat{\underline{C}}_\Lambda)$ is canonically equivalent to the category of formal schemes $\mathcal{X}$, adic over the formal spectrum $\mathrm{Spf}(R)$, such that $\mathcal{X}_n = \mathcal{X} \otimes_R R/\underline{m}^{n+1}$ is flat over $R/\underline{m}^{n+1}$ for every $n \geq 0$ (or what amounts to the same, if $X_0$ is noetherian, such that $\mathcal{X}$ is flat over $\mathrm{Spf}(R)$). Consequently, if $X_0$ is noetherian, $\hat{\underline{M}}_{X_0}$ is canonically $\hat{\underline{C}}_\Lambda$-equivalent to the category of deformations of $X_0$ over the objects in $\hat{\underline{C}}_\Lambda$, in the sense of formal schemes.

4.1(d): Let $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$ i.e., we have a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that $X_0 \to X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$ is an isomorphism. Now let $U_0 \subset X_0$ be a fixed open subscheme of $X_0$. Since $i_X : X_0 \to X$ is a closed immersion which is a homeomorphism on the underlying topological spaces, we obtain a commutative diagram

$$\begin{array}{ccc} U_0 & \xrightarrow{i_X|U_0} & X|U_0 \\ \downarrow & & \downarrow \text{flat} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that $U_0 \to (X|U_0) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k) = X \otimes_R k|U_0$ is an isomorphism, so that $(R, X|U_0)$ is a deformation of $U_0$ to $R$. Furthermore, for any arrow $(\varphi, \Phi) : (R, X) \to (R', X')$ in $\underline{M}_{X_0}$, $(\varphi, \Phi|U_0) : (R, X|U_0) \to (R', X'|U_0)$ is an arrow in $\underline{M}_{U_0}$. Consequently we have the restriction functor $\underline{M}_{X_0} \to \underline{M}_{U_0}$ over $\underline{C}_\Lambda$. One may also consider a localization as well as a formalization at a point. Namely let $x_0$ be a fixed point in $X_0$. If $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$, then $\mathrm{Spec}(\underline{O}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\underline{O}_{X_0,x_0})$ to $R$ and $\mathrm{Spec}(\hat{\underline{O}}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_0})$ over $R$, where $\hat{}$ stands for the completion with respect to the linear topology given by the powers of the maximal ideal. We thus obtain functors $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0,x_0})} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_0})}$.

Firstly we establish that $\underline{M}_{X_0}$ is a homogeneous cofibered category over $\underline{C}_\Lambda$.

Lemma 4.2. *Consider a diagram*

$$\begin{array}{ccc} & & R' \\ & & \downarrow \\ R & \xrightarrow[\text{surjective}]{} & \bar{R} \end{array}$$

*in* $\underline{C}_\Lambda$, *and let* $E$, $E'$ *be flat modules over* $R$, $R'$ *respectively. Then for any* $R'$*-homomorphism* $E' \to \bar{E} = E \otimes_R \bar{R}$, *the fibre-product* $E \times_{\bar{E}} E'$ *is flat as* $R \times_{\bar{R}} R'$*-module*

**Proof:** Let $0 \to I \to R \to \bar{R} \to 0$ be exact so that $0 \to I \to R \times_{\bar{R}} R' \to R' \to 0$ is exact. Since $E$ is R-flat, $0 \to I \otimes_R E \to E \to \bar{E} \to 0$ is exact and hence we obtain the exact sequence

$$(*) \qquad 0 \to I \otimes_R E \to E \times_{\bar{E}} E' \to E' \to 0 \ .$$

Set $S = R \times_{\bar{R}} R$ . Since $I$ is nilpotent, $E \times_{\bar{E}} E'$ is flat as an S-module if and only if $(E \times_{\bar{E}} E') \otimes_S R'$ is flat as an R'-module and $\mathrm{Tor}_1^S (E \times_{\bar{E}} E', R') = 0$ . However, we note that $I \otimes_R E = I \otimes_S (E \times_{\bar{E}} E')$ and $E' = R' \otimes_S (E \times_{\bar{E}} E')$ and therefore the exact sequence (*) can be rewritten as

$$0 \to I \otimes_S (E \times_{\bar{E}} E') \to E \times_{\bar{E}} E' \to R' \otimes_S (E \times_{\bar{E}} E') \to 0 \ .$$

This means that $\mathrm{Tor}_1^S (R', E \times_{\bar{E}} E') = 0$ and $R' \otimes_S (E \times_{\bar{E}} E') = E'$ is flat as R'-module, and therefore $E \times_{\bar{E}} E'$ is S-flat.

Corollary 4.3. *Given a diagram*

$$\begin{array}{ccc} (R, X) & & (R', X') \\ & {\scriptstyle (\varphi, \Phi)} \searrow \quad \swarrow {\scriptstyle (\varphi', \Phi')} & \\ & (\bar{R}, \bar{X}) & \end{array}$$

*in* $\underline{M}_{X_0}$ , *the amalgamated sum* $X \coprod_{\bar{X}} X'$ *(in the sense of schemes) is a flat scheme over* $R \times_{\bar{R}} R'$ , *provided* $\varphi : R \to \bar{R}$ *is surjective. Consequently,* $\underline{M}_{X_0}$ *is a homogeneous cofibered category in groupoids over* $\underline{C}_\Lambda$ .

**Proof:** For each affine open $U_0 \subset X_0$ , we have by definition that $X \coprod_{\bar{X}} X' |_{U_0} \simeq \mathrm{Spec}(\Gamma(U_0, X) \times_{\Gamma(U_0, \bar{X})} \Gamma(U_0, X'))$ . Consequently, if $\varphi : R \to \bar{R}$ is surjective, $X \coprod_{\bar{X}} X'$ is a flat scheme over $R \times_{\bar{R}} R'$ by 4.2, and the morphism $i_X \coprod_{i_{\bar{X}}} i_{X'} : X_0 \to X \coprod_{\bar{X}} X'$ induces an isomorphism $X_0 \to (X \coprod_{\bar{X}} X') \otimes_S k$ , where $S = R \times_{\bar{R}} R'$ . Therefore $(R \times_{\bar{R}} R', X \coprod_{\bar{X}} X') \in \mathrm{ob}(\underline{M}_{X_0})$ and we obviously have that $(R \times_{\bar{R}} R', X \coprod_{\bar{X}} X') = (R, X) \times_{(\bar{R}, \bar{X})} (R', X')$ in the category $\underline{M}_{X_0}$ , which

means that $\underline{M}_{X_0}$ is homogeneous over $\underline{C}_\Lambda$ .

Lemma 4.4. For any scheme $X_0$ over k , we have a canonical exact sequence

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to H^0(X_0, \underline{D}^1_{X_0})$$

where $\underline{D}^i_{X_0}$ are as defined in paragraph 3.

Proof: (Special case) Assume that $X_0$ is affine and let $\Gamma(X_0, \underline{O}_{X_0}) = A_0$ . Then as it was noted in 4.1(b), the category $\underline{M}_{X_0}(k[\epsilon])$ is canonically equivalent to the category $\underline{E}$ , where

ob($\underline{E}$) : object in $\underline{E}$ is a flat algebra A over $k[\epsilon]$

together with a $k[\epsilon]$ -algebra map $j_A : A \to A_0$ inducing the isomorphism $k \otimes_{k[\epsilon]} j_A : k \otimes_{k[\epsilon]} A \xrightarrow{\sim} A_0$ . Alternatively, an object in $\underline{E}$ is a k-algebra A together with an exact sequence $0 \to A_0 \xrightarrow{\epsilon} A \xrightarrow{j_A} A_0 \to 0$ in which $\epsilon(A_0)^2 = 0$ as an ideal in A .

Fl($\underline{E}$) : An arrow $A_1 \to A_2$ in $\underline{E}$ with respect to first description of the object in $\underline{E}$ is a $k[\epsilon]$-algebra map $\Phi : A_1 \to A_2$ such that $j_{A_1} = j_{A_2} \circ \Phi$ . Therefore, an arrow $A_1 \to A_2$ in $\underline{E}$ with respect to the alternative description of objects in $\underline{E}$ is a k-algebra map $\Phi : A_1 \to A_2$ yielding a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_0 & \xrightarrow{\epsilon} & A_1 & \xrightarrow{j_{A_1}} & A_0 & \longrightarrow & 0 \\ & & \| & & \downarrow \Phi & & \| & & \\ 0 & \longrightarrow & A_0 & \longrightarrow & A_2 & \xrightarrow{j_{A_2}} & A_0 & \longrightarrow & 0 \end{array} .$$

It follows that $[\underline{M}_{X_0}(k[\epsilon])]$ is canonically isomorphic, via $\Gamma(X_0, -)$ , to the set of isomorphism classes of k-algebra extensions of $A_0$ by $A_0$ . In other words, we have a canonical bijection $[\underline{M}_{X_0}(k[\epsilon])] \xrightarrow{\sim} H^0(X_0, \underline{D}^1_{X_0})$ via $\Gamma(X_0, -)$, which is trivially seen to respect the vector space structures on both sides.

(General case) For each affine open $U_0 \subset X_0$ , we get a canonical map $[\underline{M}_{X_0}(k[\epsilon])] \to H^0(U_0, \underline{D}^1_{X_0})$ defined as the composite map $[\underline{M}_{X_0}(k[\epsilon])] \to [\underline{M}_{U_0}(k[\epsilon])] \xrightarrow{\sim} H^0(U_0, \underline{D}^1_{X_0})$ and hence we obtain a canonical map $[\underline{M}_{X_0}(k[\epsilon])] \to H^0(X_0, \underline{D}^1_{X_0})$ . Its kernel is the set of isomorphism classes of locally-trivial deformations of $X_0$ to $k[\epsilon]$ , i.e., the set of isomorphism classes of deformations of $X_0$ to $k[\epsilon]$ , locally (on $X$ ) isomorphic to $X_0 \otimes_k k[\epsilon]$ . Since $\underline{D}^0_{X_0} = \underline{Der}_k(\underline{O}_{X_0}, \underline{O}_{X_0}) = \underline{Aut}(X_0 \otimes_k k[\epsilon] | X_0)$ = the sheaf of germs of automorphisms of $X_0 \otimes_k k[\epsilon]$ inducing the identity on the subscheme $X_0$ , it is canonically isomorphic to $H^1(X_0, \underline{D}^0_{X_0})$ , so that we obtain the desired exact sequence.

Theorem 4.5. Let $X_0$ be a scheme of finite type over $k$ for which either (i) $X_0$ is proper over $k$ or (ii) $X_0$ is affine but $X_0^{sing} = \{x \in X_0 | X_0$ is non-smooth over $k$ at the point $x\}$ is a finite set. Then

(a) $\underline{M}_{X_0}$ admits a formal versal object (called a formal versal deformation of $X_0$ ) over $\underline{C}_\Lambda$ . If $(R, \mathfrak{X})$ is a formal versal deformation of $X_0$ , then for any $(S, \mathfrak{Y}) \in ob(\hat{\underline{M}}_{X_0})$ , there is a map $(\varphi, \Phi) : (R, \mathfrak{X}) \to (S, \mathfrak{Y})$ in $\hat{\underline{M}}_{X_0}$ such that the diagram below is cartesian

$$\begin{array}{ccc} \mathfrak{Y} & \xrightarrow{\Phi} & \mathfrak{X} \\ \downarrow & & \downarrow \\ \mathrm{Spf}(S) & \xrightarrow[\mathrm{Spf}(\varphi)]{} & \mathrm{Spf}(R) \end{array} .$$

Furthermore a formal versal deformation $(R, \mathfrak{X})$ of $X_0$ prorepresents the functor $[\underline{M}_{X_0}]$ if and only if, for every surjective map $R' \to R$ in $\underline{C}_\Lambda$ and a deformation $(R', X')$ of $X_0$ to $R'$ , the homomorphism

$$\mathrm{Aut}_{R'}(X' | X_0) \to \mathrm{Aut}_R(X' \otimes_{R'} R | X_0)$$

is surjective.

(b) If $X_0$ is proper over $k$ , then $\underline{M}_{X_0}$ is smoothly pseudo-representable

over $\underline{C}_\Lambda$ .

Proof: (a) Since $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$ by 4.3, it suffices to see that $[\underline{M}(k[\epsilon])]$ is a finite-dimensional vector space over $k$ by 1.11 and 2.9(b).

Now consider the exact sequence

$$(*) \qquad 0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to H^0(X_0, \underline{D}^1_{X_0}) \ .$$

If $X_0$ is proper over $k$ , then $H^1(X_0, \underline{D}^0_{X_0})$ and $H^0(X_0, \underline{D}^1_{X_0})$ are both finite dimensional since $\underline{D}^i_{X_0}$ are coherent for all $i \geq 0$ by 2.13(a). Now assume that $X_0$ is affine such that $X_0^{sing}$ is a finite set. Since $\underline{D}^1_{X_0}$ is a coherent sheaf on $X_0$ such that $\operatorname{Supp} \underline{D}^1_{X_0} \subset X_0^{sing}$ by 2.13(a), we find that $H^0(X_0, \underline{D}^1_{X_0})$ is a finite-dimensional vector space over $k$ , whereas $H^1(X_0, \underline{D}^0_{X_0}) = 0$ since $X_0$ is affine. Therefore, in either case, it follows from the exact sequence (*) that $[\underline{M}_{X_0}(k[\epsilon])]$ is a finite dimensional vector space over $k$ . The second statement of (a) follows from 2.7.

(b) If $X_0$ is proper, $\operatorname{Aut}(X_0 \otimes_k k[\epsilon] | X_0) = H^0(X_0, \underline{D}^0_{X_0})$ is a finite dimensional vector space and hence our statement follows from 2.13.

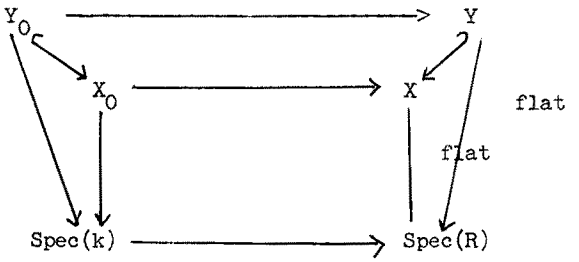
Remark 4.6(a): Let $X_0$ be affine. Then a formal versal deformation of $X_0$ exists if and only if $\underline{D}^1_{X_0}$ has its support on a finite number of points. However this does not imply that non-smooth points of $X_0$ are all isolated. There are affine schemes $X_0$ over $k$ having a non-isolated non-smooth point but $\operatorname{supp} \underline{D}^1_{X_0}$ is a finite set.

Remark 4.6(b): Henceforth a deformation of $X_0$ is meant by an object in $\underline{M}_{X_0}$ . In case when there is a possible confusion, an object in $\underline{M}_{X_0}$ will be referred as an infinitesimal deformation of $X_0$ .

Remark 4.6(c): Let $X_0$ be a scheme over $k$ . We have fixed a complete local noetherian ring $\Lambda$ with the residue field $k$ , and considered the deformations

of $X_0$ over the category $\underline{C}_\Lambda$ . (In practice one usually takes $\Lambda = k$ or $W(k)$ if $k$ is a perfect field of positive characteristic). Now $\underline{C}_k \subset \underline{C}_\Lambda$ is a fully faithful imbedding, and that $\underline{M}_{X_0} | \underline{C}_k$ is simply the category of deformations of $X_0$ over $S$ where $S \in ob(\underline{C}_k)$ . Therefore if $(R, X)$ is a formal versal deformation of $X_0$ over $\underline{C}_\Lambda$ , then $(k \otimes_\Lambda R, k \otimes_\Lambda X)$ is a formal versal deformation of $X_0$ over $\underline{C}_k$ . It follows, for example, that $\dim_{\underline{C}_\Lambda} \underline{M}_{X_0} = \dim k \otimes_\Lambda R$ does not depend on the choice of $\Lambda$ but depends only on $X_0$ .

One can generalize the above theorem in many ways. Let us for instance consider a pair $Y_0 \to X_0$ where $X_0$ is a scheme of finite type over $k$ and $Y_0$ a fixed closed subscheme of $X_0$ . For each $R \in ob(\underline{C}_\Lambda)$ , a deformation of $(X_0, Y_0)$ to $R$ is meant to be a commutative diagram



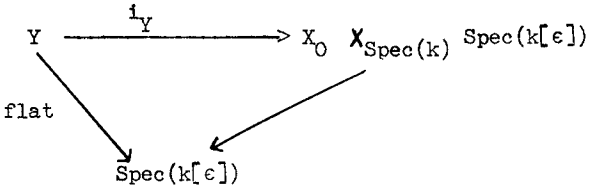

such that $(X_0, Y_0) \to (X \times_{Spec(R)} Spec(k), Y \times_{Spec(R)} Spec(k))$ is an isomorphism. Defining a morphism $(R'; X', Y') \to (R; X, Y)$ of deformations of $(X_0, Y_0)$ to be the obvious ones, we obtain the cofibered category $\underline{M}_{X_0, Y_0}$ in groupoids over $\underline{C}_\Lambda$ . (This is a process often used in rigidification of automorphisms in moduli problems of deformations.) One sees immediately, by the same argument as in 4.3 based on 4.2, that $\underline{M}_{X_0, Y_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$ .

Lemma 4.7. $0 \to H^0(Y_0, \underline{Hom}_{\underline{O}_{Y_0}}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0})) \to [\underline{M}_{X_0, Y_0}(k[\epsilon])] \to [\underline{M}_{X_0}(k[\epsilon])]$

is exact, where $\underline{I}$ is the ideal sheaf of $\underline{O}_{X_0}$ defining $Y_0$ i.e.

$0 \to \underline{I} \to \underline{O}_{X_0} \to \underline{O}_{Y_0} \to 0$ *is exact.*

*Proof:* Let $\underline{B}$ be the full subcategory of $\underline{M}_{X_0,Y_0}(k[\varepsilon])$ whose objects are of the form $(k[\varepsilon];\ X \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]),\ Y)$ . We then have $[B] = \mathrm{Ker}\,([\underline{M}_{X_0,Y_0}(k[\varepsilon])] \to [\underline{M}_{X_0}(k[\varepsilon])])$ . The category $\underline{B}$ is the groupoid of deformations of $Y_0$ to $k[\varepsilon]$ "inside $X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon])$ ", i.e., an object in $\underline{B}$ is a commutative diagram




such that $\mathrm{Spec}(k) \times_{\mathrm{Spec}(k[\varepsilon])} i_Y = i_{Y_0}$ , and an arrow $(Y_1, i_{Y_1}) \to (Y_2, i_{Y_2})$ is a $k[\varepsilon]$-morphism $\Phi : Y_1 \to Y_2$ (necessarily an isomorphism) such that $\mathrm{Spec}(k) \times_{\mathrm{Spec}(k[\varepsilon])} \Phi = \mathrm{id}_{Y_0}$ and $i_{Y_2} \circ \Phi = i_{Y_1}$ . Alternatively, an object in $\underline{B}$ is a pair $(Y, j_Y)$ , where $Y$ is a deformation of $Y_0$ to $k[\varepsilon]$ and $j_Y : Y \to X_0$ is a morphism over $k$ such that the composite morphism $Y_0 \to Y \xrightarrow{j_Y} X_0$ is $i_{Y_0}$ , and an arrow $(Y_1, j_{Y_1}) \to (Y_2, j_{Y_2})$ is a morphism $\Phi : Y_1 \to Y_2$ such that $k \otimes_{k[\varepsilon]} \Phi = \mathrm{id}_{Y_0}$ and $j_{Y_2} \circ \Phi = j_{Y_1}$ . Therefore the association

$$(Y, j_Y) \longmapsto \begin{array}{ccccccccc} 0 & \longrightarrow & \underline{I} & \longrightarrow & \underline{O}_{X_0} & \longrightarrow & \underline{O}_{Y_0} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow \jmath_Y & & \| & & \\ 0 & \longrightarrow & \underline{O}_{Y_0} & \xrightarrow{\varepsilon} & \underline{O}_Y & \longrightarrow & \underline{O}_{Y_0} & \longrightarrow & 0 \end{array}$$

gives rise to an isomorphism

$$[B] \xrightarrow{J|\underline{I}} \mathrm{Hom}_{\underline{O}_{X_0}}(\underline{I}, \underline{O}_{Y_0}) = \mathrm{Hom}_{\underline{O}_{Y_0}}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0}) = H^0(Y_0, \underline{\mathrm{Hom}}_{\underline{O}_{Y_0}}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0})) \ .$$

Remark 4.8. For a more general formula implying the above lemma, see SGA3 III 4.4. One may also note the following fact. One can define for any S-scheme $X$ and a quasi-coherent X-module $\underline{E}$ a quasi-coherent X-module $\underline{D}^i_{X|S}(\underline{E})$ for all $i \geq 0$, generalizing the case when $S$ is affine. As usual we simply write $\underline{D}^i_{X|S}(\underline{O}_X) = \underline{D}^i_{X|S}$. We then have

$$H^0(Y_0, \underline{\mathrm{Hom}}_{Y_0}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0})) = H^0(Y_0, \underline{D}^1_{Y_0|X})$$

where $0 \to \underline{I} \to \underline{O}_{X_0} \to \underline{O}_{Y_0} \to 0$ is exact.

Theorem 4.9. *Let $X_0$ be k-scheme of finite type and $Y_0$ a closed subscheme of $X_0$. Assume that either* (i) *$X_0$ is proper over k or* (ii) *$X_0$ is affine and smooth over k outside a finite number of points and $Y_0$ is proper over k. We then have*

(a) *$\underline{M}_{X_0, Y_0}$ admits a formal versal deformation over $C_\wedge$, and it pro-represents the functor $[\underline{M}_{X_0, Y_0}]$ if and only if, for every surjective map $R' \to R$ in $C_\wedge$ and a deformation $(R'; X', Y')$,*

$$\mathrm{Aut}_{R'}((X', Y')|X_0) \to \mathrm{Aut}_R(X' \otimes_{R'} R, Y' \otimes_{R'} R)|X_0)$$

*is surjective.*

(b) *If furthermore $H^0(X_0, \underline{D}^0_{Y_0|k} \underset{\underline{D}^0_{X_0|k}(\underline{O}_{Y_0})}{\times} \underline{D}^0_{X_0|k})$ is finite dimensional, then $\underline{M}_{X_0, Y_0}$ is smoothly pseudo-representable.*

Proof: It is obvious from 4.4 and 4.6 together with 1.11, 2.7, and 2.13.

We now consider the passage to localization and completion in the deformation theory of schemes. Firstly on a notation: Let $X_0$ be a k-scheme of finite type. Given a fixed point $x_0 \in X_0$, we have natural functors

$$\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0}, x_0)} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0}, x_0)}$$

and in turn we obtain the functors

$$\underline{\hat{M}}_{X_0} \to \underline{\hat{M}}_{\mathrm{Spec}(\underline{O}_{X_0},\, x_0)} \to \underline{\hat{M}}_{\mathrm{Spec}(\underline{\hat{O}}_{X_0},\, x_0)} \ .$$

As it was remarked in 4.1(c), $\underline{M}_{X_0}$ is canonically equivalent to the category in which an object is a pair $(R, \mathcal{X})$ where $R \in \mathrm{ob}(\underline{C}_\Lambda)$ and $\mathcal{X}$ is a flat R-adic formal scheme together with a fixed isomorphism $\mathcal{X}_0 \leftrightarrow X_0$ . Henceforth we shall write, when there is no possible confusion, the images of $(R, \mathcal{X})$ under

$\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0},\, x_0)}$ and $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0},\, x_0)}$ by $(R, \mathcal{X}_{(x_0)})$ and $(R, \mathcal{X}_{(x_0)})$

respectively. Therefore if $\mathcal{X} = \varinjlim_v \mathcal{X}_v$ where $\mathcal{X}_v = \mathcal{X} \times_{\mathrm{Spf}(R)} \mathrm{Spec}(R/\underline{m}^{v+1})$ then

$$\mathcal{X}_{(x_0)} = \varinjlim_v \mathrm{Spec}(\underline{O}_{\mathcal{X}_v},\, x_0)$$

$$\mathcal{X}_{(\hat{x}_0)} = \varinjlim_v \mathrm{Spec}(\hat{\underline{O}}_{\mathcal{X}_v},\, x_0) \ .$$

Theorem 4.10. Let $X_0$ be an affine scheme of finite type over $k$ .

(a) If $X_0$ is locally a complete-intersection, then $\underline{M}_{X_0}$ is smooth over $\underline{C}_\Lambda$ .

(b) Assume that, for a fixed closed point $x \in X_0$ , $X_0 - \{x\}$ is smooth over $k$ . Then

$$\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0},\, x)} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0},\, x)}$$

are both minimally smooth functors. In particular, if $(R, \mathcal{X})$ is a formal versal deformation of $X_0$ , then $(R, \mathcal{X}_{(x)})$ and $(R, \mathcal{X}_{(\hat{x})})$ are formal versal deformations of $\mathrm{Spec}(\underline{O}_{X_0},\, x)$ and $\mathrm{Spec}(\hat{\underline{O}}_{X_0},\, x)$ respectively.

Proof: In this proof we go back to the notations of the cohomology of commutative algebras defined in the paragraph 3. We set $X_0 = \mathrm{Spec}(A_0)$

(a) It suffices to show that $P : [\underline{M}_{X_0}] \to \underline{C}_\Lambda$ is smooth.

Let $(R, \mathrm{Spec}(A))$ be a deformation of $\mathrm{Spec}(A_0)$ to $R$, where $R \in \mathrm{ob}(\underline{C}_\Lambda)$. Given a small extension $R' \to R$ in $\underline{C}_\Lambda$ (i.e., $0 \to k \to R' \to R \to 0$ is exact), we must show that $(R, \mathrm{Spec}(A))$ can be lifted to $R'$. The exact sequence of $D^i$ with coefficient $A_0$ applied to the ring homomorphisms $R' \to R \to A$ gives us the exact sequence

$$\ldots \to D^1(A|R, A_0) \to D^1(A|R', A_0) \to D^1(R|R', k) \underset{k}{\otimes} A_0 \to D^2(A|R, A_0) \to \ldots$$

Since $R' \to R$ is a small extension, $D^1(R'|R, k)$ is 1-dimensional vector space over $k$ generated by the element (denoted by $[R']$) represented by the extension $0 \to k \to R' \to R \to 0$. On the other hand, $A$ is R-flat and hence $D^i(A|R, A_0) = D^i(A_0|k, A_0)$ for all $i > 0$ and in particular $D^2(A|R, A_0) = 0$ by 3.12(11) since $A_0$ is locally a complete-intersection. Consequently we get the exact sequence

$$\to D^1(A_0|k, A_0) \to D^1(A|R', A_0) \to D^1(R|R', k) \underset{k}{\otimes} A_0 \to 0$$

This means that there exists an R'-algebra extension of $A$ by $A_0$, denoted by $A'$, such that $[A'] \to [R'] \otimes 1$ under the map $D^1(A|R', A_0) \to D^1(R|R', k) \underset{k}{\otimes} A_0$, which means that the deformation $(R, \mathrm{Spec}(A))$ can be lifted to a deformation $(R', \mathrm{Spec}(A'))$.

(b) By 3.14 we have the isomorphisms

$$D^1(A_0|k, A_0) \simeq D^1((A_0)_x|k, (A_0)_x) \simeq D^1(\widehat{(A_0)}_x|k, \widehat{(A_0)}_x)$$

i.e., $[\underline{M}_{X_0}(k[\epsilon])] \simeq [\underline{M}_{\mathrm{Spec}(\underline{O}_{X_0}, x)}(k[\epsilon])] \simeq [\underline{M}_{\mathrm{Spec}(\widehat{\underline{O}_{X_0}}, x)}(k[\epsilon])]$.

It suffices to show, by 2.7(a), that

$$[\underline{M}_{X_0}] \to [\underline{M}_{\mathrm{Spec}(\underline{O}_{X_0}, x)}] \to [\underline{M}_{\mathrm{Spec}(\widehat{\underline{O}_{X_0}}, x)}]$$

are both smooth functors. Let $(R, \mathrm{Spec}(A))$ be a deformation of $X_0 = \mathrm{Spec}(A_0)$, and let $0 \to k \to R' \to R \to 0$ be a small extension in $\underline{C}_\Lambda$. As above, the ring homomorphisms $R' \to R \to A$ gives us the exact commutative diagram

$$\begin{array}{ccccccccc}
\to D^1(A_0|k,A_0) & \to & D^1(A|R',A_0) & \xrightarrow{\beta} & D^1(R|R',k)\otimes_k A_0 & \to & D^2(A_0|k,A_0) & \to \cdots \\
\downarrow\approx & & \downarrow & & \downarrow & & \downarrow\approx & \\
\to D^1(A_0|k,(A_0)_x) & \to & D^1(A|R',(A_0)_x) & \to & D^1(R|R',k)\otimes_k (A_0)_x & \to & D^2(A_0|k,(A_0)_x) & \to \cdots \\
\uparrow & & \uparrow & & \uparrow & & \uparrow & \\
\to D^1((A_0)_x|k,(A_0)_x) & \to & D^1(A_x|R',(A_0)_x) & \xrightarrow{p_x} & D^1(R|R',k)\otimes_k (A_0)_x & \to & D^2((A_0)_x|k,(A_0)_x) & \to \cdots
\end{array}$$

where the indicated isomorphism comes from the fact that $A_0$ is smooth over $k$ except at one closed point $x$ so that

$$D^i(A_0|k,\ (A_0)_x) = D^i(A_0|k,\ A_0)_x = D^i(A_0|k,\ A_0)$$

for all $i > 0$ . Now suppose that we are given an arrow $(R',\ \mathrm{Spec}(B)) \to (R,\ \mathrm{Spec}(A_x))$ in $[\underline{M}_{\mathrm{Spec}((A_0)_x)}]$ . It means that we are given a class $[B] \in D^1(A_x|R',\ (A_0)_x)$ such that $p_x([B]) = [R']$ . Then a trivial diagram chasing shows that there exists $[A'] \in D^1(A|R',\ A_0)$ such that $p([A']) = [R']$ and $A'_x = B$ i.e., $(R',\ \mathrm{Spec}(A'))$ is a deformation of $\mathrm{Spec}(A_0)$ to $R'$ such that $\mathrm{Spec}(A')_x = \mathrm{Spec}(B)$ . This proves that $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0},\ x)}$ is smooth. The smoothness of the functor $\underline{M}_{\mathrm{Spec}((A_0)_x)} \to \underline{M}_{\mathrm{Spec}(\widehat{(A_0)_x})}$ comes from the similar diagram together with the fact that

$$D^i((A_0)_x|k,\ \widehat{(A_0)_x}) = \widehat{D^i((A_0)_x|k,\ (A_0)_x)} = D^i((A_0)_x|k,\ (A_0)_x)$$

for all $i > 0$ since $x$ is an only possible non-smooth point.

Theorem 4.11. Let $X_0$ be a (separated) k-scheme of finite type, smooth outside a finite closed set $X_0^{\mathrm{Sing}} = \{x_1,\ldots,x_r\}$ of $X_0$ . Let $U_0^1,\ldots,U_0^r$ be affine open neighbourhoods of $x_1,\ldots,x_r$ such that $U_0^i \cap X_0^{\mathrm{Sing}} = \{x_i\}$ . If $H^2(X_0,\ \underline{D}^0_{X_0}) = 0$ , then the natural functor $\underline{M}_{X_0} \to \underline{M}_{U_0^1} \times \ldots \times \underline{M}_{U_0^r}$ is smooth.

Proof: We can complete $U_0^1,\ \ldots\ ,U_0^r$ into an affine open covering of $X_0$ by choosing $U_0^{r+1},\ldots,U_0^N$ such that $U_0^j \cap X_0^{\mathrm{Sing}} = \emptyset$ for all $j > r$ . Now let $R' \to R$ be a small extension in $\underline{C}_\wedge$ , and let $(R',\ U'^1),\ldots,(R',\ U'^r)$ be deformations of $U_0^1,\ldots,U_0^r$ to $R'$ such that $U'^i \otimes_{R'} R = X|U_0^i$ , $1 \le i \le r$ , for

some deformation $(R, X)$ of $X_0$ to $R$. We must show that there exists a deformation $(R', X')$ of $X_0$ to $R'$ such that $X' \otimes_{R'} R = X$ and $X'|U_0^i = U'^i$ for $1 \leq i \leq r$. Since $U_0^j$ for $j > r$ is smooth over $k$, $X|U_0^j$ can be lifted, by 4.9(a), to a deformation $(R', U'^j)$ i.e., there exist deformations $(R', U'^j)$ of $U_0^j$ to $R'$ such that $U'^j \otimes_{R'} R = X|U_0^j$ for all $j > r$. Since, for any pair $i \neq j$, $U_0^i \cap U_0^j$ is affine and smooth over $k$, we obtain an isomorphism

$$\tau_{ij} : U'^i|U_0^i \cap U_0^j \xrightarrow{\sim} U'^j|U_0^i \cap U_0^j$$

such that $\tau_{ij} \otimes_{R'} R = id_{X|U_0^i \cap U_0^j}$, and in turn defines a cohomology class in $H^2(X_0, \underline{D}^0_{X_0})$ given by $U_0^i \cap U_0^j \cap U_0^k \to \tau_{ij}\tau_{jk}\tau_{ik}^{-1}$. Since $H^2(X_0, \underline{D}^0_{X_0}) = 0$ by hypothesis, the obstruction for glueing these to obtain a global deformation vanishes, and hence we get a deformation $(R', X')$ of $X_0$ to $R'$ such that $X' \otimes_{R'} R = X$ and $X'|U_0^j \simeq U'^j$. This completes our proof.

Remark 4.12. The previous proof assumes $X_0$ to be separated, which is unnecessary. In fact, for a fixed deformation $(R, X)$ of $X_0$ and deformations $(R', U'^j)$ of $U_0^i$, the deformations $(R', V')$ of open set $V$ of $X$ to $R'$ compatible with the deformation $(R', U'^j)$ already given, form a "gerbe" (in Giraud's terminology) whose structure sheaf is $\underline{D}^0_{X_0}$. Hence the obstruction to the existence of a section of this "gerbe" (i.e., a global deformation $(R', X')$ of $X$ compatible with the given datum $(R', U'^j)$) is in $H^2(X_0, \underline{D}^0_{X_0})$ which is zero by assumption.

Corollary 4.13. Let $X_0$ be a k-scheme of finite type, smooth outside a finite subset $X_0^{Sing} = \{x_1, \ldots, x_r\}$ of $X_0$ such that $H^2(X_0, \underline{D}^0_{X_0}) = 0$. Then

(a) $\underline{M}_{X_0} \to \underline{M}_{Spec(\widehat{\underline{O}}_{X_0, x_1})} \times \ldots \times \underline{M}_{Spec(\widehat{\underline{O}}_{X_0, x_r})}$ is smooth.

In particular, $\dim \underline{M}_{X_0} = \dim H^1(X_0, \underline{D}^0_{X_0}) + \sum_{c=1}^{r} \dim \underline{M}_{Spec(\widehat{\underline{O}}_{X_0, x_0})}$.

(b) If furthermore $X_0$ is locally a complete-intersection, then $\underline{M}_{X_0}$ is smooth.

Proof:

(a) The first statement follows from 4.10 and 4.9(b). Now let $(R, \mathfrak{X})$ and $(R_1, \mathfrak{X}_1),\ldots,(R_r, \mathfrak{X}_r)$ be formal versal deformations of $X_0$ and $\mathrm{Spec}(\underline{O}_{X_0,x_1}),\ldots,\mathrm{Spec}(\underline{O}_{X_0,x_r})$ respectively. We then obtain a commutative diagram

$$\begin{array}{ccc} \underline{h}_R & \longrightarrow & \underline{h}_{R_1} \hat{\otimes}_{\Lambda} \ldots \hat{\otimes}_{\Lambda} R_r \\ \downarrow & & \downarrow \\ \underline{M}_{X_0} & \longrightarrow & \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_1})} \times \ldots \times \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_r})} \end{array}$$

Since the composite functor

$$\underline{h}_R \to \underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_1})} \times \ldots \times \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_r})}$$

is smooth whereas

$$\underline{h}_{R_1} \hat{\otimes}_{\Lambda} \ldots \hat{\otimes}_{\Lambda} R_r \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_1})} \times \ldots \times \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_r})}$$

is minimally smooth, it follows from 2.7(a) that

$\underline{h}_R \to \underline{h}_{R_1} \hat{\otimes}_{\Lambda} \ldots \hat{\otimes}_{\Lambda} R_r$ is smooth i.e., $R$ is a formal power-series ring over $R_1 \hat{\otimes}_{\Lambda} \ldots \hat{\otimes}_{\Lambda} R_r$. Since

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to [\underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_1})}(k[\epsilon])] \ldots [\underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_r})}(k[\epsilon])]$$

is exact by 4.4 i.e.,

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\epsilon]) \to \mathrm{Hom}_{\Lambda\text{-alg}}(R_1 \hat{\otimes}_{\Lambda} \ldots \hat{\otimes}_{\Lambda} R_r, k[\epsilon])$$

is exact, it follows that $R$ is a formal power-series ring over $R_1 \hat{\otimes}_{\Lambda} \ldots \hat{\otimes}_{\Lambda} R_r$ in d-variables where $d = \dim_k H^1(X_0, \underline{D}^0_{X_0})$.

Therefore

$$\begin{aligned} \dim \underline{M}_{X_0} &= \dim k \otimes_{\Lambda} R = d + \dim k \otimes_{\Lambda} (R_1 \hat{\otimes}_{\Lambda} \ldots \hat{\otimes}_{\Lambda} R_r) \\ &= d + \dim(k \otimes_{\Lambda} R_1) \hat{\otimes}_k \ldots \hat{\otimes}_k (k \otimes_{\Lambda} R_r) \\ &= d + \sum_{i=1}^{r} \dim k \otimes_{\Lambda} R_i = d + \sum_{i=1}^{r} \dim \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_i})} \end{aligned}$$

(b) follows from 4.10 and 4.9(a).

4.14. We close this paragraph with an explicit construction of a formal versal deformation in a simple situation. In fact, if $X_0$ is affine and is a complete-intersection, it is trivial to write down explicitely. Thus let

$$A_0 = k[x_1,\ldots,x_n]/(f_1,\ldots,f_r) \quad S/J$$

where $S = k[x_1,\ldots,x_n]$ , $J = (f_1,\ldots,f_r)$ , and $f_1,\ldots,f_r$ is an S-regular sequence. We then obtain the exact sequence

$$\ldots \to \mathrm{Der}_k(S, A_0) \to \mathrm{Hom}_{A_0}(J/J^2, A_0) \to D^1(A_0|k, A_0) \to 0$$

We note that $J/J^2$ is $A_0$-free module of rank $r$ generated by $f_1 (\mathrm{mod}\ J^2),\ldots,f_r (\mathrm{mod}\ J^2)$ , since $f_1,\ldots,f_r$ is an S-regular sequence. We now assume that $D^1(A_0|k, A_0)$ is a finite-dimensional vector space over $k$ , say $A$ has isolated non-smooth points only. Let $d = \dim_k D^1(A_0|k, A_0)$ and choose $p_j : J/J^2 \to A_0$ $(j = 1, 2,\ldots,d)$ which form a k-basis of $D^1(A_0|k, A_0)$ . Now choose $P_{jk} \in \Lambda[x_1,\ldots,x_n]$ such that $p_j(f_i) = P_{ji} \ (\mathrm{mod}\ \underline{m}_\Lambda + J)$ where $\underline{m}_\Lambda$ is the maximal ideal of $\Lambda$ . We then consider

$$R = \Lambda[[t_1,\ldots,t_d]]$$
$$B = R[x_1,\ldots,x_n]/(G_1,\ldots,G_r)$$

where $G_j = F_j - \sum_{i=1}^{d} t_i P_{ji}$ and $F_j$ is a polynomial with coefficients in $\Lambda$ gotten from $f_j$ by lifting the coefficients from $k$ to $\Lambda$ . Since $R[x_1,\ldots,x_n]$ is R-flat and $G_1,\ldots,G_r$ modulo $\underline{m}_R$ (which becomes $f_1,\ldots,f_r$) is a regular sequence in $k \otimes_R R[x_1,\ldots,x_n]$ , it follows that $B$ is R-flat, and $k \otimes_R B = A_0$ . Furthermore,

$$\mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \to D^1(A_0|k, A_0) = [\underline{M}_{\mathrm{Spec}(A_0)}(k[\varepsilon])]$$

is bijective by our construction of $B$ . Therefore, if we set

$$\mathrm{Spf}(B) = \varinjlim_{\nu} \mathrm{Spec}(R/\underline{m}^{\nu+1} \underset{R}{\otimes} B)$$

then $(R, \mathrm{Spf}(B))$ is an object in $\widehat{\underline{M}}_{\mathrm{Spec}(A_0)}$ and the functor $\underline{h}_R \to \underline{M}_{\mathrm{Spec}(A_0)}$ induced by $(R, \mathrm{Spf}(B))$ is smooth by 1.6, since $R$ is a formal power-series ring over $\Lambda$. In other words, $(R, \mathrm{Spf}(B))$ is a formal versal object of $\underline{M}_{\mathrm{Spec}(A_0)}$.

We note, by 4.10(b), that if we set

$$B = R[[X_1,\dots,X_n]]/(G_1,\dots,G_r) \ ,$$

then $(R, \mathrm{Spf}(B))$ is a formal versal deformation of $\mathrm{Spec}(A_0)$ where $A_0 = k[[X_1,\dots,X_n]]/(f_1,\dots,f_r)$.

Remark 4.15. Let $X_0$ be a k-scheme of finite type with isolated non-smooth points only, and let $(R, \mathcal{X})$ be a formal versal deformation of $X_0$. The above construction shows that if $X_0$ is affine and locally a complete-intersection, then $\mathcal{X}$ is a formalization of an R-scheme of finite type. This fact can easily be generalized to the case when $X_0$ is not necessarily affine but $H^2(X_0, \underline{D}^0_{X_0}) = 0$.

Remark 4.16. The global theory of the relative cotangent complex (the latter theory being reduced to the truncated complex of length 1, which is sufficient however for many purposes of deformation theory) allows to generalize and simplify certain results, such as 4.10. Namely let $S$ be a scheme, $S_0 \subset S' \subset S$ two subschemes of $S$ defined by ideals $J \supset I$ such that $JI = 0$, $X'$ a scheme flat over $S'$, $L_\cdot = L^{X_0/S_0}$ the relative cotangent complex of $X_0$ over $S_0$, which is an element of the derived category $D(X_0, \underline{O}_X)$. We consider the category $\underline{F}$ of all flat schemes $X$ over $S$ which extend $X'/S'$, i.e. together with an $S'$-isomorphism $X \underset{S}{\times} S' \simeq X'$. Then

(a) For an object $X'$ of $\underline{F}$, the group of automorphisms is canonically isomorphic to $\mathrm{Ext}^0(X_0; L_\cdot, I_{X_0})$ (where $I_{X_0}$ is the inverse image of $I$, viewed as a module on $S_0$, on $X_0$).

(b) The set of isomorphism classes of objects of $\underline{F}$ is empty or a torsor (= a principal homogeneous space) under the group $\mathrm{Ext}^1(X_0; L.,I_{X_0})$ .

(c) One can define a canonical element $c \in \mathrm{Ext}^2(X_0; L.,I_{X_0})$ whose vanishing is necessary and sufficient for $\underline{F}$ to be non empty.

In order to study the $\mathrm{Ext}^i(X_0; L.,I_{X_0})$ , it is often convenient to write

$$\mathrm{Ext}^i(X_0;L.,I_{X_0}) = H^i(X_0, \underline{\mathrm{RHom}}(L.,I_{X_0})) \quad .$$

and to notice that the complex $\underline{\mathrm{RHom}}(L.,I_{X_0})$ has as cohomology sheaves the sheaves $\underline{D}^i_{X_0}(I_{X_0})$ introduced in §3. For $i = 0, 1, 2,$ these sheaves may be viewed respectively as the sheaf of automorphisms of any object $X$ of $\underline{F}$ (understood: automorphisms inducing the identity on $X'$) , the sheaf of indeterminacy of isomorphism classes of local solutions to the flat deformation problem, and the sheaf of obstructions to the existence of local solutions. On the other hand, the standart spectral sequence gives us

$$\mathrm{Ext}^*(X_0 \quad L.,I_{X_0}) \leftarrow E_2^{p,q} = H^p(X_0, \underline{D}^q_{X_0}(I_{X_0})) \quad ,$$

which yields the well-known isomorphism

$$\mathrm{Aut}(X/X') \simeq H^0(X_0,\underline{D}^0_{X_0}(I_{X_0})) \simeq H^0(X_0,\underline{\mathrm{Hom}}(\underline{\Omega}^1_{X_0/S_0},I_{X_0})) = \mathrm{Hom}(\underline{\Omega}^1_{X_0/S_0},I_{X_0}) \quad ,$$

and the exact sequence in low degrees

$$0 - H^1(X_0,\underline{D}^0_{X_0}) - \mathrm{Ext}^1 - H^0(X_0,\underline{D}^1_{X_0}) - H^2(X_0,\underline{D}^0_{X_0}) - \mathrm{Ext}^2$$

This shows in particular that if $H^2(X_0,\underline{D}^0_{X_0}) = 0$ , and if there exists any local flat deformation $X$ of $X'/S'$ , then there exists a flat global deformation which corresponds to given local deformations. More precisely and generally, for a given local deformations (i.e. a section of the "sheaf of isomorphism classes of local flat deformations" ), the argument 4.10 shows that the obstruction to finding a global flat deformation compatible with these is in $H^2(X_0, \underline{D}^0_{X_0})$ , hence such a global flat deformation exists if $H^2(X_0, \underline{D}^0_{X_0}) = 0$ : this is essentially 4.10.

On the other hand, the spectral sequence shows that the $\mathrm{Ext}^2$ is zero whenever we have the relations

(*) $H^0(X_0, \underline{D}^2_{X_0}) = 0$ , $H^1(X_0, \underline{D}^1_{X_0}) = 0$ , $H^2(X_0, \underline{D}^0_{X_0}) = 0$ .

The first relation is always satisfied if $X_0$ is a relative complete intersection over $S_0$ (as then $\underline{D}^2_{X_0} = 0$ for any module $M$ on $X_0$ ), the second is satisfied provided the support of $\underline{D}^1_{X_0}$ is discrete, as is the case if $X_0$ is smooth over $S_0$ except on a discrete subset of $X_0$ over $S_0$ . When the previous three relations (*) are all satisfied, then by (c) above, $\underline{F}$ is non empty, which generalizes 4.12(b).

More geometrically, and without using the global theory of the relative cotangentcomplex (but only the globalizing of the functors $\underline{D}^i_{X_0}$ ), one can get a simple geometric interpretation of the three successive obstructions to deforming flatly $X'$ to $X$ , lying in the three left hand terms of (*). Namely by the local theory, the first obstruction $c_1$ in $H^0(X_0, \underline{D}^2_{X_0})$ can be defined as the obstruction to finding local solutions, when this obstruction vanishes, the sheaf $\underline{F}$ of isomorphism classes of local solutions of the problem is (by the local theory) in a canonical way a torsor under $\underline{D}^1_{X_0}$ , and the second obstruction $c_2$ , lying in $H^1(X_0, \underline{D}^1_{X_0})$ , is the obstruction to finding a section of this torsor, i.e. of finding a "coherent system of isomorphism classes of local solutions" ; if this second obstruction also vanishes, then, choosing arbitrarily a section of $\underline{F}$ , one gets, as noted above, an obstruction $c_3$ in $H^2(X_0, \underline{D}^0_{X_0})$ to finding a global solution corresponding to the given system of (isomorphism classes of) local solutions. Using the fact that the section of $F$ is determined up to adding a section of $\underline{D}^1_{X_0}$ , we get a map $d : H^0(X_0, \underline{D}^1_{X_0}) \to H^2(X_0, \underline{D}^0_{X_0})$ , and a solution to the deformation problem exists if and only if $c_1 = 0$ , $c_2 = 0$ , and $c_3$ is zero as an element of Coker $d$ (each $c_i$ being defined when the previous ones are zero). Of course, the groups $H^0(X_0, \underline{D}^2_{X_0})$ , $H^1(X_0, \underline{D}^1_{X_0})$ and

$H^2(X_0, \underline{D}^0_{X_0})/\mathrm{Im}\, d$ are just the initial terms $E_2^{0,2}$, $E_2^{1,1}$, and $E_3^{2,0}$ of the spectral sequence above, $d$ being the differential (up to sign ?) $d_2 : E_2^{0,1} \to E_2^{2,0}$.

## 5. Formal Jacobian subschemes

Let $X_0$ be a scheme of finite type over a field $k$, and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0$. Then $\mathfrak{X}$ is not an ordinary scheme over $R$ but an $R$-adic formal scheme. The purpose of this section is to clarify "the non-smooth locus of $\mathfrak{X}$ over $R$".

We fistly recall the definition of Fitting ideals associated to a module of finite type: Let $A$ be a commutative ring and $E$ an $A$-module of finite type. Choose a presentation

$$K \to F \to E \to 0$$

where $F$ is locally-free of constant rank $n$. We define

$$I_A^p(E) = \mathrm{Im}({}^n\bar{\wedge}^p K \otimes {}^n\bar{\wedge}^p F^* \to A) \qquad \text{for } p = 0, 1, 2, \ldots$$

The ideals $I_A^p(E)$ thus defined does not depend on the choice of a presentation and therefore are invariants of $A$-module $E$, called the Fitting ideals of $A$-module $E$. The following properties readily follow from the definition.

5.1. (a) For any ring homomorphism $A \to B$, the canonical map

$$B \otimes_A A/I_A^p(E) \xrightarrow{\sim} B/I_B^p(B \otimes_A E)$$

is an isomorphism for all $P$. In particular we have

$$I^p_{S^{-1}A}(S^{-1}E) = S^{-1}I_A^p(E)$$

for any multiplicative subset $S$ in $A$.

(b) We have $I_A^0(E) \subset I_A^1(E) \subset I_A^2(E) \subset \ldots$ , and, for each point $x \in \mathrm{Spec}(A)$ , $I_A^p(E)_x = A_x$ for all $p \geq \dim_{\kappa(x)} E(x)$ , where $E(x) = \kappa(x) \otimes_A E$ .

(c) $x \in \mathrm{Spec}(A) - \mathrm{Supp}\ A/I_A^p(E)$ if and only if, for any (or for some) presentation

$$K \to F \to E \to 0$$

where $F \approx A^n$ , $K_x$ contains $A_x^{n-p}$ which is isomorphic to a direct summand of $F_x$ . In particular, we have that

$$\mathrm{Supp}\ A/I_A^0(E) = \mathrm{Supp}\ E \ .$$

(d)

$$I_A^r(E_1 \oplus E_2) = \sum_{p+q=r} I_A^p(E_1) \cdot I_A^q(E_2)$$

(e) If $A$ is a noetherian adic ring, then

$$A/I_A^p(E) = \varprojlim_v A_v/I_{A_v}^p(E_v)$$

where $A = \varprojlim_v A_v$ and $E_v = E \otimes_A A_v$ .

In view of the property 5.1(a), our definition of Fitting ideals is readily globalized. Thus let $X$ be any scheme, and $\underline{E}$ a quasi-coherent X-module of finite presentation. We then obtain a sequence of quasi-coherent ideal sheaves $\underline{I}_X^p(\underline{E})$ for $p \geq 0$ defined by

$$\Gamma(U, \underline{I}_X^p(\underline{E})) \supset I^p_{\Gamma(U, \underline{O}_X)}(\Gamma(U, \underline{E}))$$

for every affine open subset $U \subset X$ .

Now let $X \to Y$ be a morphism of schemes. The relative Kähler-differentials $\underline{\Omega}_{X|Y} (= \underline{\Omega}^1_{X|Y})$ is a quasi-coherent X-module. If the morphism $X \to Y$ is of essentially finite presentation, then $\underline{\Omega}_{X|Y}$ is a quasi-coherent X-module of finite presentation, and in turn we may consider the ideal sheaves $\underline{I}_X^p(\underline{\Omega}_{X|Y})$ ..

Definition 5.2. Let $X \to Y$ be a morphism of essentially finite presentation. For each integer $p \geq 0$ , we define $J_X^p(X|Y)$ to be the closed subscheme of $X$

defined by the ideal sheaf $\underline{I}^p_X(X|Y) = \underline{I}^p_X(\underline{\Omega}_{X|Y})$ , called the Jacobian subschemes of $X$ over $Y$ . For each point $x \in X$ , we set $X_x = \mathrm{Spec}(\underline{O}_{X,x})$ .

Theorem 5.3. Let $X \to Y$ be a morphism of essentially finite presentation. Then

(1) For any base change $Y \leftarrow Y'$ , the canonical map

$J^p_{X'}(X'|Y') \simeq J^p_X(X|Y) \underset{Y}{\times} Y'$ is an isomorphism for all $p$ where $X' = X \underset{Y}{\times} Y'$ .

(2) For each point $x \in X$ , we have the canonical isomorphism

$J^p_{X_x}(X_x|Y) \simeq J^p_X(X|Y)_x$ .

(3) $J^0_X(X|Y) \supset J^1_X(X|Y) \supset J^2_X(X|Y) \supset \ldots$, and $x \notin J^p_X(X|Y)$ for all $p \geq \dim_{\kappa(x)} \Omega_{X|Y}(x)$ .

(4) $x \notin J^p_X(X|Y)$ if and only if there exists an open neighbourhood $U$ of $x$ which admits a closed immersion into a $Y$-scheme $U'$ such that $U'$ is smooth over $Y$ at $x'$ and $\dim_{x'} U'_{f(x)} = p$ where $x'$ is the image of $x$ under $U \to U'$ .

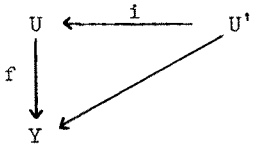
(5) $\underline{I}^r_{X_1 \underset{Y}{\times} X_2}(X_1 \underset{Y}{\times} X_2|Y) = \sum_{p+q=r} \pi_1^* \underline{I}^p_{X_1}(X_1|Y) \cdot \pi_2^* \underline{I}^q_{X_2}(X_2|Y)$

where $\pi_i : X_1 \underset{Y}{\times} X_2 \to X_i$ is the projection.

Proof: (3) follows from 5.1(b) whereas (1) follows from 5.1(a) together with the fact that $P^* \underline{\Omega}_{X|Y} \simeq \underline{\Omega}_{X'|Y'}$ is an isomorphism where $P : X \underset{Y}{\times} Y' \to X$ is the projection. (2) also follows from 5.1(a) since $\underline{O}_{X,x} \underset{\underline{O}_X}{\otimes} \underline{\Omega}_{X|Y} \simeq \underline{\Omega}_{X_x|Y}$ is an isomorphism (5) follows from 5.1(d) and the isomorphism

$$\pi_1^* \underline{\Omega}_{X_1|Y} \oplus \pi_2^* \underline{\Omega}_{X_2|Y} \simeq \underline{\Omega}_{X_1 \underset{Y}{\times} X_2|Y}$$

(4) Assume the existence of a diagram



where $U'$ is smooth over $Y$ at $x' = i(x)$ and $\dim_{x'}U'_{f(x)} = p$ . Then $\dim_{\kappa(x')}\underline{\Omega}_{U'|Y}(x') = p$ and $i^*\underline{\Omega}_{U'|Y} \to \underline{\Omega}_{U|Y} \to 0$ is exact, so that $\dim_{\kappa(x)}\underline{\Omega}_{U|Y}(x) \leq p$ which implies that $x \notin J^p_U(U|Y)$ by 5.3(3). Since $J^p_U(U|Y) = J^p_X(X|Y)|U$ , this means that $x \notin J^p_X(X|Y)$ . Now assume conversely that $x \notin J^p_X(X|Y)$ .

The question being local, we may assume that $X$, $Y$ are both affine, say $X = \mathrm{Spec}(B)$ , $Y = \mathrm{Spec}(A)$ . Choose a presentation over $A$

$$0 \to J \to P \to B \to 0$$

where $P = A[x_1,\ldots,x_n]$ . Then $J/J^2 \xrightarrow{d} B \otimes_P \Omega_{P|A} \to \Omega_{B|A} \to 0$ is exact and $B \otimes_P \Omega_{P|A}$ is $B$-free of rank $n$ , and therefore $I^p_B(\Omega_{B|A}) = \mathrm{Im}({}^n\wedge^p J/J^2 \otimes_B (B \otimes_P {}^n\wedge^p \Omega_{P|A})^* \to B)$ . Therefore, $x \notin J^p_X(X|Y) \Leftrightarrow$ there exists $f_1,\ldots,f_{n-p}$ in $J$ such that $[\det\left(\frac{f_i}{x_j}\right)](x) \neq 0$ for some choice of $n-p$ variables among $x_1, x_2,\ldots,x_n$ there exists $f_1, f_2,\ldots,f_{n-p}$ in $J$ such that $\mathrm{Spec}(P/(f_1,\ldots,f_{n-p})) \to \mathrm{Spec}(A)$ is smooth at $x'$ where $x'$ is the image of $x$ under $\mathrm{Spec}(B) \to \mathrm{Spec}(P/(f_1,\ldots,f_{n-p}))$ . Thus it suffices to take $U' = \mathrm{Spec}(P/(f_1,\ldots,f_{n-p}))$ .

The following is a generalization of the usual Jacobian criteria for relative situation.

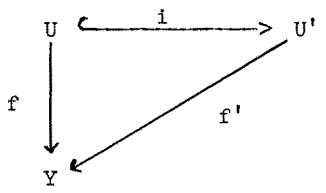
Theorem 5.4. Let $X \xrightarrow{f} Y$ be a morphism of essentially finite presentation. Assume that there exists a fixed integer $n \geq 0$ with either conditions:

(1) $f$ is flat and $\dim_x X_{f(x)} = n$ for every point $x \in X$ .

(2) $f$ is a relative complete-intersection with virtual dimension $n$ (SGA6 VIII 1.9).

Then $x \notin J^n_X(X|Y)$ if and only if $f$ is smooth at $x$ .

Proof: We have, by 5.3(4) that $x \notin J^n_X(X|Y)$ if and only if there exists a diagram




such that $f'$ is smooth at $x'$ and $\dim_{x'}U'_{f(x)} = n$, where $i$ is a closed immersion and $x' = i(x)$. Thus it suffices to show that either conditions (1) or (2) entails the isomorphism of $i$ at the point $x$. Set $A = \underline{O}_{Y,f(x)}$, $B' = O_{U',x'}$, $B = O_{U,x}$ and let

$$0 \to I \to B' \to B \to 0$$

be exact. We must show that $I = 0$. Assume (1). Since $B$ is A-flat,

$$0 \to I/\underline{m}I \to B'/\underline{m}B' \to B/\underline{m}B \to 0$$

is exact where $\underline{m}$ is the maximal ideal of $A$. By hypothesis, $\dim_x U_{f(x)} = \dim_{x'}U'_{f(x)}$ so that $\dim B'/\underline{m}B' = \dim B/\underline{m}B$. However $B'/\underline{m}B'$ is a regular local ring and therefore it entails that $I/\underline{m}I = 0$ so that $I = 0$. Now assume (2), and consider the exact sequence

$$I/I^2 \xrightarrow{d} B \otimes_{B'} \Omega_{B'|A} \to \Omega_{B|A} \to 0$$

Since $X$ is a relative complete-intersection over $Y$ and $U' \to Y$ is smooth at $x$, it follows that $d : I/I^2 \to B \otimes_{B'} \Omega_{B'|A}$ is injective at $x$ and $I/I^2$ and $B \otimes_{B'} \Omega_{B'|A}$ are both free at $x$. Therefore by the definition of virtual dimension of relative complete-intersection we must have that

$$n = \dim_{\kappa(x)}\Omega_{B'|A}(x') - \dim_{\kappa(x)}I/I^2(x) = n - \dim_{\kappa(x)}I/I^2(x)$$

i.e., $I/I^2 = 0$ i.e., $I = 0$.

Definition 5.5. Let $X \xrightarrow{f} Y$ be a morphism of essentially finite presentation. In case when there exists a fixed integer $n$ with the property (1) or (2) in 5.4,

we set $\underline{I}_X(X|Y) = \underline{I}_X^n(X|Y)$ and $J_X(X|Y) = J_X^n(X|Y)$ .

We can extend this notion to formal schemes in an obvious manner.

Thus let $\mathcal{X}$ be a noetherian formal scheme and $\underline{E}$ a coherent $\mathcal{X}$-module. Then we can define the ideal sheaf $\underline{I}_{\mathcal{X}}^P(\underline{E})$ by setting

$$\underline{I}_{\mathcal{X}}^P(\underline{E})(U) = I_{\mathcal{X}(U)}^P(\underline{E}(U))$$

for every affine open $U \subset \mathcal{X}$ . If $\mathcal{X} = \varinjlim_v X_v$ , then $\underline{E} = \varinjlim_v \underline{E}_v$ where $\underline{E}_v = \underline{E} \otimes_{\underline{O}_{\mathcal{X}}} \underline{O}_{X_v}$ and it follows from the fact that $\mathcal{X}$ is noetherian that

$$\underline{I}_{\mathcal{X}}^P(\underline{E}) = \varprojlim_v \underline{I}_{X_v}^P(\underline{E}_v)$$

Now let $S$ be a noetherian formal scheme and $\mathcal{X}$ an S-adic formal scheme of essentially finite type. Let $\underline{I}_{\mathcal{X}}$ be the ideal sheaf defining the closed immersion $\Delta : \mathcal{X} \to \mathcal{X} \underset{S}{\times} \mathcal{X}$ (= the fibre-product in the category of formal schemes) i.e.,

$$0 \to \underline{I}_{\mathcal{X}} \to \underline{O}_{\mathcal{X} \underset{S}{\times} \mathcal{X}} \to \underline{O}_{\mathcal{X}} \to 0$$

is exact. We define $\underline{\Omega}_{\mathcal{X}|S} = \underline{I}_{\mathcal{X}}/\underline{I}_{\mathcal{X}}^2$ . Since $\mathcal{X}$ is an S-adic formal scheme of essentially finite type, it follows that $\mathcal{X} = \varinjlim_v X_v$ where $X_v = \mathcal{X} \underset{S}{\times} S_v$ , and hence $\underline{\Omega}_{\mathcal{X}|S} \simeq \varprojlim_v \underline{\Omega}_{X_v|S_v}$ is a coherent $\mathcal{X}$-module. We note that for any formal S-scheme $S'$ of finite type, we have a canonical isomorphism

$$\underline{\Omega}_{\mathcal{X} \underset{S}{\times} S'|S'} \simeq \underline{O}_{S'} \otimes_{\underline{O}_S} \underline{\Omega}_{\mathcal{X}|S} \quad .$$

Definition 5.6. Let $S$ be a noetherian formal scheme and $\mathcal{X}$ an S-adic formal scheme of essentially finite type. We define the ideal sheaf $\underline{I}_{\mathcal{X}}^P(\mathcal{X}|S)$ by

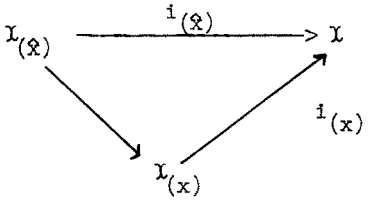
$$\underline{I}_{\mathcal{X}}^P(\mathcal{X}|S) = \underline{I}_{\mathcal{X}}^P(\underline{\Omega}_{\mathcal{X}|S})$$

and we set $J_{\mathcal{X}}^P(\mathcal{X}|S)$ the formal closed subscheme of $\mathcal{X}$ defined by $\underline{I}_{\mathcal{X}}^P(\mathcal{X}|S)$ . Since $\Omega_{\mathcal{X}|S} = \varprojlim_n \Omega_{X_n|S_n}$ we have

$$I^p_{\mathcal{X}}(\mathcal{X}|S) = \varprojlim_n I^p_{X_n}(X_n|S_n)$$

$$J^p_{\mathcal{X}}(\mathcal{X}|S) = \varinjlim_n J^p_{X_n}(X_n|S_n) \quad .$$

$\mathcal{X}_{(x)}$ is an S-adic formal scheme of essentially finite type and we have a commutative diagram




We note that if $x$ is an isolated point of $\mathcal{X}$, then $i_{(x)}$, $i_{(\hat{x})}$ are both isomorphisms to the connected component of $\mathcal{X}$ containing the point $x$.

In particular, if $|\mathcal{X}| = \{x_1,\ldots,x_m\}$ is a finite discrete space, then we have a canonical identification $\mathcal{X} = \coprod_{1\le i\le m} \mathcal{X}_{(x_i)}$ (disjoint union).

Corollary 5.7. Let $S$ be a noetherian formal scheme, and $\mathcal{X}$ an $S$-adic formal scheme of essentially finite type. We then have

(1) $J^p_{\mathcal{X}}(\mathcal{X}|S)$ commutes with base change i.e., $J^p_{\mathcal{X}'}(\mathcal{X}'|S') = J^p_{\mathcal{X}}(\mathcal{X}|S) \hat{\otimes}_S S'$ where $\mathcal{X}' = \mathcal{X} \hat{\otimes}_S S'$.

(2) The connected components of $J^p_{\mathcal{X}}(\mathcal{X}|S)$ is in bijective correspondence with that of $J^p_{X_0}(X_0|S_0)$. For each point $x \in |J^p_{X_0}(X_0|S_0)|$ we have a canonical isomorphism $J^p_{\mathcal{X}}(\mathcal{X}|S)_{(x)} \simeq J^p_{\mathcal{X}_{(x)}}(\mathcal{X}_{(x)}|S)$. In particular if

$$|J^p_{X_0}(X_0|S_0)| = \{x_1,\ldots,x_m\}$$

is discrete, then

$$J^p_{\mathcal{X}}(\mathcal{X}|S) = \coprod_{1\le i\le m} J^p_{\mathcal{X}_{(x_i)}}(\mathcal{X}_{(x_i)}|S)$$

(disjoint union).

(3) Let $S$ be an ordinary noetherian scheme and $X$ an $S$-scheme of essentially finite type. Given a closed subscheme $S_0 \subset S$, let $\hat{S}$ be the formal completion of $S$ along $S_0$.

If we set

$\hat{X} = X \underset{S}{\times} \hat{S}$ = the formal completion of $X$ along $X_0 = X \underset{S}{\times} S_0$, then

$J^p_{\hat{X}}(\hat{X}|\hat{S}) = J^p_X(X|S) \underset{S}{\times} \hat{S}$ = the formal completion of $J^p_X(X|S)$ along $J^p_{X_0}(X_0|S_0)$.

(4) If there exists a fixed integer $n \geq 0$ with the property 5.4(1) or (2) for $X_\nu \to S_\nu$ for all $\nu$, then $x \notin J^n_{\mathcal{X}}(\mathcal{X}|S)$ if and only if $\mathcal{X}$ is formally smooth over $S$ at $x$.

Proof: (1) follows from $\underline{\Omega}_{\mathcal{X}'|S'} \simeq \underline{O}_{S'} \underset{\underline{O}_S}{\hat{\otimes}} \underline{\Omega}_{\mathcal{X}|S}$ whereas (2) follows from $\underline{O}_{\mathcal{X}_{(x)}} \underset{\underline{O}_{\mathcal{X}}}{\hat{\otimes}} \underline{\Omega}_{\mathcal{X}|S} \simeq \underline{\Omega}_{\mathcal{X}_{(x)}|S}$.

(3) $J^p_{\hat{X}}(\hat{X}|\hat{S}) = \varinjlim_\nu J^p_{X \underset{S}{\times} S_\nu}(X \underset{S}{\times} S_\nu | S_\nu) = \varinjlim_\nu J^p_X(X|S) \underset{S}{\times} S_\nu = J^p_X(X|S) \underset{S}{\times} \hat{S}$.

(4) Since $\mathcal{X}$ is an $S$-adic formal scheme, $\mathcal{X} \to S$ is formally smooth at $x \Leftrightarrow X_\nu \to S_\nu$ is formally smooth at $x$ for all $\nu \Leftrightarrow x \notin J^n_{X_\nu}(X_\nu|S_\nu)$ for all $\nu$ $\Leftrightarrow x \notin \varinjlim_\nu J^n_{X_\nu}(X_\nu|S_\nu) = J^n_{\mathcal{X}}(\mathcal{X}|S)$.

Remark 5.8. Let $\mathcal{X}$ be an $S$-adic formal scheme of essentially finite type. Given a point $x \in \mathcal{X}$ we also consider the $S$-adic formal scheme $\mathcal{X}_{(\hat{x})} = \varinjlim_\nu \operatorname{Spec}(\widehat{\underline{O}_{X_\nu,x}})$. We thus have the natural morphisms $\mathcal{X}_{(\hat{x})} \to \mathcal{X}_{(x)} \to \mathcal{X}$. Now $\widehat{\underline{O}_{X_0,x}}$ is in general not essentially finite type over $S_{(y)}$ where $y$ = the image of $x$ under $\mathcal{X} \to S$. For simplicity, let us set $A_\nu = \underline{O}_{X_\nu,x}$, $R_\nu = \underline{O}_{S_\nu,y}$. Now the natural map $\Omega_{A_\nu|R_\nu} \to \widehat{\Omega}_{A_\nu|R_\nu}$ induces an isomorphism $\widehat{\Omega}_{A_\nu|R_\nu} \to \widehat{\Omega}_{A_\nu|R_\nu}$, and therefore $\varprojlim_\nu \widehat{\Omega}_{A_\nu|R_\nu} \simeq \varprojlim_\nu \widehat{\Omega}_{A_\nu|R_\nu}$ where $\widehat{\ }$ stands for the separable completion of a module with respect to the topology induced from the ring. Therefore if we define $\widehat{\Omega}_{\mathcal{X}_{(\hat{x})}|S} = \varprojlim_\nu \widehat{\Omega}_{\underline{O}_{X_\nu,x}|\underline{O}_{S_\nu}}$, then $\widehat{\Omega}_{\mathcal{X}_{(\hat{x})}|S}$ is a coherent $\mathcal{X}_{(\hat{x})}$-module. Thus if

we set $J^p_{\mathcal{X}_{(\mathfrak{X})}}(\mathcal{X}_{(\mathfrak{X})}|S)$ to be the formal closed subscheme of $\mathcal{X}_{(\mathfrak{X})}$ defined by the ideal sheaf $I^p_{\mathcal{X}_{(\mathfrak{X})}}(\mathcal{X}_{(\mathfrak{X})}|S) = I^p_{\mathcal{X}_{(\mathfrak{X})}}(\Omega_{\mathcal{X}_{(\mathfrak{X})}|S})$ , then we have

$$J^p_{\mathcal{X}_{(x)}}(\mathcal{X}_{(x)}|S)_{(\mathfrak{X})} \simeq J^p_{\mathcal{X}_{(\mathfrak{X})}}(\mathcal{X}_{(\mathfrak{X})}|S) \ .$$

We note that if $x$ is an isolated point of $J^p_{\mathcal{X}}(\mathcal{X}|S)$ , then $J^p_{\mathcal{X}_{(\mathfrak{X})}}(\mathcal{X}_{(\mathfrak{X})}|S) \to J^p_{\mathcal{X}_{(x)}}(\mathcal{X}_{(x)}|S)$ is an isomorphism.

To consider the "image" of $J^p_{\mathcal{X}}(\mathcal{X}|S)$ under the morphism $\mathcal{X} \overset{f}{\to} S$ , one may simply take the formal subscheme of $S$ defined by the ideal sheaves $\mathrm{Ker}(\underline{O}_S \to \underline{O}_{\mathcal{X}}/I^p_{\mathcal{X}}(\mathcal{X}|S))$ . However, it is more convenient for our purpose to use the Fitting ideal again.

Definition 5.9. Let $S$ be a noetherian formal scheme and $f : \mathcal{X} \to S$ an $S$-adic formal scheme of essentially finite type. Assume that $J^p_{\mathcal{X}}(\mathcal{X}|S)$ is proper over $S$ , so that $f_* \underline{O}_{J^p_{\mathcal{X}}(\mathcal{X}|S)}$ is a coherent $\underline{O}_S$-module. We then define the ideal sheaf $\underline{I}^p_S(\mathcal{X}|S)$ of $\underline{O}_S$ by

$$\underline{I}^p_S(\mathcal{X}|S) = \underline{I}^0_S(f_* \underline{O}_{J^p_{\mathcal{X}}(\mathcal{X}|S)})$$

and set $J^p_S(\mathcal{X}|S)$ to be the closed formal subscheme of $S$ defined by the ideal sheaf $\underline{I}^p_S(\mathcal{X}|S)$ . In case when there exists a fixed integer $n$ having the property (1) or (2) in 5.4 for $X_v \to S_v$ for all $v$ , we set

$$\underline{I}_{\mathcal{X}}(\mathcal{X}|S) = \underline{I}^n_{\mathcal{X}}(\mathcal{X}|S) \ , \qquad J_{\mathcal{X}}(\mathcal{X}|S) = J^n_{\mathcal{X}}(\mathcal{X}|S)$$

$$\underline{I}_S(\mathcal{X}|S) = \underline{I}^n_S(\mathcal{X}|S) \ , \qquad J_S(\mathcal{X}|S) = J^n_S(\mathcal{X}|S)$$

Corollary 5.10. Let $S$ be a noetherian formal scheme and $\mathcal{X} \overset{f}{\to} S$ an $S$-adic formal scheme, essentially of finite type such that $J^p_{\mathcal{X}}(\mathcal{X}|S)$ is proper over $S$ . Then

(1) $|J^p_S(\mathcal{X}|S)|$ = the image of $|J^p_{\mathcal{X}}(\mathcal{X}|S)|$ under $f : \mathcal{X} \to S$ .

(2) $J^p_S(\mathcal{X}|S)$ commute with base change i.e., $J^p_{S'}(\mathcal{X}'|S') = J^p_S(\mathcal{X}|S) \times_S S'$ .

(3) If $J^p_{\mathcal{X}}(\mathcal{X}|S) = \coprod_{1\le i\le m} \mathcal{Y}^{(i)}$ (disjoint union), then

$\underline{I}^p_S(\mathcal{X}|S) = \prod_{1\le i\le m} \underline{I}^0_S(f_*\mathcal{O}_{\mathcal{Y}(i)})$ . In particular if

$$|J^p_{X_0}(X_0|S_0)| = \{x_1,\dots,x_m\}$$

is discrete, then

$$\underline{I}^p_S(\mathcal{X}|S) = \prod_{1\le i\le m} \underline{I}^p_S(\mathcal{X}_{(x_i)}|S) = \prod_{1\le i\le m} \underline{I}^p_S(\mathcal{X}_{(\mathcal{X}_i)}|S)$$

(4) Let $S$ be an ordinary noetherian scheme and $X$ an $S$-scheme of essentially finite type. Given a closed subscheme $S_0 \to S$ , let $\hat{S}$ be the formal completion of $S$ along $S_0$ . If we set $\hat{X} = X \times_S \hat{S}$ = the formal completion of $X$ along $X_0 = X \times_S S_0$ , then $J^p_{\hat{S}}(\hat{X}|\hat{S}) = J^p_S(X|S) \times_S \hat{S}$ = the formal completion of $J^p_S(X|S)$ along $J^p_{S_0}(X_0|S_0)$ .

Proof: Supp $f_*\mathcal{O}_{J^p_{\mathcal{X}}(\mathcal{X}|S)} = |J^p_S(\mathcal{X}|S)|$ by 5.1(c), and hence (1) follows. (2) and (4) are clear from 5.1(a), and (3) follows from 5.1(d), and the isomorphism $J^p_{\mathcal{X}_{(\mathcal{X})}}(\mathcal{X}_{(\mathcal{X})}|S) \simeq J^p_{\mathcal{X}_{(x)}}(\mathcal{X}_{(x)}|S)$ for an isolated point $x$ in $J^p_{\mathcal{X}}(\mathcal{X}|S)$ .

Remark 5.11. Our definition of Jacobian ideal, adopted from the classical one over base field, commutes with base change and gives a smoothness criteria under a mild condition of 5.4. However one can define, probably in many different ways, a canonical subscheme which gives the non-smooth locus in a general setting. We shall outline one here: Given an A-algebra $B$ , define $I_{B|A} = \bigcap_E \mathrm{Ann}\, D^1(B|A,E)$ where $E$ runs through all $B$-modules of finite type. This ideal $I_{B|A}$ commutes with localizations by 3.13(c), and therefore, given a morphism $X \xrightarrow{f} Y$ , we obtain the ideal sheaf $\underline{I}_{X|Y}$ . If $f$ is a morphism of essentially finite type, then $f$ is smooth at $x$ if and only if $x \notin \mathrm{Supp}\, \underline{\Omega}^1_X/\underline{\Omega}^1_{X|Y}$ by 3.13(a). This notion is in an obvious way carried over to adic formal schemes of essentially

finite type and thus it is equally suitable for deformation theory.

## 6. Non-degenerate quadratic singularities

The objective in this section is to investigate the deformations of a k-scheme having non-degenerate quadratic singularities only. In view of 4.13, it becomes purely a local problem in an appropriate situation.

We go back to the notations of §4. Thus $\Lambda$ is a fixed complete noetherian local ring with the residue field $k$, and $\underline{C}_\Lambda$ stands for the category of local artinian $\Lambda$-algebras with the same residue field $k$.

Let $X$ be a k-scheme of finite type, and let $x$ be a closed point of $X$. We recall that the point $x$ is said to be a non-degenerate quadratic singularity rational over $k$ if $\hat{O}_{X,x} = k[[x_1,\ldots,x_n]]/(f)$ with $(\partial f/\partial x_1,\ldots,\partial f/\partial x_n) = (x_1,\ldots,x_n)$. One can always assume $f$ to be a quadratic form. Indeed we have

Lemma 6.1. *Let* $f \in k[[x_1,\ldots,x_n]]$. *If* $(\partial f/\partial x_1,\ldots,\partial f/\partial x_n) = M$ $(= (x_1,\ldots,x_n))$, *then one can find* $x_1',\ldots,x_n'$ *with* $x_i = x_i' \pmod{M^2}$ *such that* $f = q(x_1',\ldots,x_n')$ *where* $q$ *is a quadratic form.*

Proof: We do this by a successive approximation. Write $f = f_2 + h_3$ where $f_2$ is a quadratic form and $h_3 \in M^3$. Since $(\partial f/\partial x_1,\ldots,\partial f/\partial x_n) = M$, there exists $\epsilon_1^{(1)},\ldots,\epsilon_n^{(1)}$ in $M^2$ such that $\Sigma\, \epsilon_i^{(1)}\, \partial f/\partial x_i \equiv h_3 \pmod{M^4}$. If we set $X_i^{(1)} = X_i - \epsilon_i^{(1)}$, then $f(x_1^{(1)},\ldots,x_n^{(1)}) =$

$f(x_1,\ldots,x_n) - \Sigma\, \epsilon_i^{(1)}\, \partial f/\partial x_i + \ldots \equiv f_2(x_1,\ldots,x_n) \pmod{M^4}$ i.e.,

$f(x_1^{(1)},\ldots,x_n^{(1)}) \equiv f_2(x_1,\ldots,x_n) + h_4$ with $h_4 \in M^4$. Now

$(\partial f/\partial x_1^{(1)},\ldots,\partial f/\partial x_n^{(1)}) = (\partial f/\partial x_1,\ldots,\partial f/\partial x_n) = M$ and therefore we can choose

$\epsilon_1^{(2)},\ldots,\epsilon_n^{(2)}$ in $M^3$ such that $\Sigma\ \epsilon_i^{(2)}\ \partial f/\partial x_u^{(1)} \equiv h_4 \pmod{M^5}$ . Then $f(x_1^{(1)} - \epsilon_1^{(2)},\ldots,x_n^{(1)} - \epsilon_n^{(2)}) = f(x_1^{(1)},\ldots,x_n^{(1)}) - \Sigma\ \epsilon_i^{(2)}\partial f/\partial x_i^{(1)} + \ldots$ $\equiv f_2(x_1,\ldots,x_n) \pmod{M^5}$ . Continuing this process we can find $\epsilon_i^{(1)}$, $\epsilon_i^{(2)}$, $\epsilon_i^{(3)},\ldots$ $(i = 1,2,\ldots,n)$ with $\epsilon_i^{(v)} \in M^{v+1}$ such that if we set $x_i^{(v)} = x_i - (\epsilon_i^{(1)} + \ldots + \epsilon_i^{(v)})$ then $f(x_1^{(v)},\ldots,x_n^{(v)}) \equiv f_2(x_1,\ldots,x_n) \pmod{M^{v+3}}$ . Thus if we set $\epsilon_i = \Sigma_v\ \epsilon_i^{(v)}$ then $f(x_1-\epsilon_1,\ldots,x_n-\epsilon_n) = f_2(x_1,\ldots,x_n)$ i.e., $f(x_1,\ldots,x_n) = f_2(x_1+\epsilon_1,\ldots,x_n+\epsilon_n)$ .

Lemma 6.2. Let $A_0$ be a local k-algebra, essentially of finite type such that

$$\hat{A}_0 \simeq k[[x_1,\ldots,x_n]]/(f)$$

where $f$ is a non-degenerate quadratic form, and let $(R, \mathcal{X})$ be a formal versal deformation of $X_0 = \mathrm{Spec}(A_0)$ . Then

(1) $J_{\mathcal{X}}(\mathcal{X}|R) \to J_R(\mathcal{X}|R) \to \mathrm{Spec}(\Lambda)$ are both isomorphisms.

(2) $J_R(\mathcal{X}|R)$ is a principal ideal in $R$ generated by a free formal generator of $R$ over $\Lambda$ i.e. $I_R(\mathcal{X}|R) = (t)$ and $R = \Lambda[[t]]$ .

(3) $\mathcal{X} \to \mathrm{Spec}(\Lambda)$ is formally smooth.

Proof: We set $\mathcal{X} = \mathrm{Spf}(A)$ . By 4.10(b), $(R, \mathrm{Spf}(\hat{A}))$ is a formal versal deformation of $\hat{X}_0 = \mathrm{Spec}(\hat{A}_0)$ where $\hat{A}_0 = \varprojlim_v A \otimes_R R_v$ .

Now $\hat{A}_0 \simeq k[[x_1,\ldots,x_n]]/(f)$ and $(\partial f/\partial x_1,\ldots,\partial f/\partial x_n) = (x_1,\ldots,x_n)$ in $k[[x_1,\ldots,x_n]]$ . It follows readily that $[\underset{\hat{X}_0}{M}(k[\epsilon])] \simeq k$ and, by 4.14, we get

$$R \simeq \Lambda[[t]] \text{ and } \hat{A} \simeq R[[x_1,\ldots,x_n]]/(F - t)$$

where $F$ is a quadratic polynomial in $\Lambda[x_1,\ldots,x_n]$ $(\subset R[[x_1,\ldots,x_n]])$ obtained from $f$ by lifting the coefficients to $\Lambda$ . Now

$$0 \to \hat{A} \xrightarrow{df} \hat{A}dx_1 \oplus \ldots \oplus \hat{A}dx_n \longrightarrow \hat{\Omega}_{\hat{A}|R} \to 0$$

is exact where $df : \hat{A} \to \hat{A}dx_1 \oplus \ldots \oplus \hat{A}dx_n$ is given by $1 \to (\partial f/\partial x_1)dx_1 + \ldots + (\partial f/\partial x_n)dx_n$, and therefore $I_{\hat{A}}(\hat{A}|R) = (\partial F/\partial x_1, \ldots, \partial F/\partial x_n) = (x_1, \ldots, x_n)$, and hence $\hat{A}/I_{\hat{A}}(\hat{A}|R) = \hat{A}/(x_1, \ldots, x_n) = R[[x_1, \ldots, x_n]]/(x_1, \ldots, x_n, F - t) \simeq \Lambda$ i.e., the canonical map $\Lambda \to \hat{A}/I_{\hat{A}}(\hat{A}|R)$ is an isomorphism. It follows that $\hat{A} \to A/I_A(A|R)$ is an isomorphism, and consequently $\Lambda \to R/I_R(A|R)$ is also an isomorphism. This proves (1) and (2). On the other hand, (3) follows from $\hat{A} \simeq R[[x_1, \ldots, x_n]]/(F - t) \simeq \Lambda[[t, x_1, \ldots, x_n]]/(F - t) \simeq \Lambda[[x_1, \ldots, x_n]]$ as $\Lambda$-algebras.

Corollary 6.3. Let $X_0$ be a k-scheme of finite type, and let $x$ in $X_0$ be a non-degenerate quadratic singularity rational over $k$.

(1) Let $(R, \mathcal{X})$, $(R, \mathcal{Y})$ be any two deformations of $X_0$ where $R \in ob(\hat{C}_\Lambda)$. Assume that $k = k^2$. Then $\mathcal{X}_{(\hat{x})} \simeq \mathcal{Y}_{(\hat{x})}$ as R-adic formal schemes if and only if $I_R(\mathcal{X}_{(x)}|R) = I_R(\mathcal{Y}_{(x)}|R)$.

(2) Let $R \in ob(\hat{C}_\Lambda)$ be a regular local ring, and let $(R, \mathcal{X})$ be a deformation of $X_0$. Then $\mathcal{O}_{\mathcal{X},x}$ is a regular local ring if and only if $I_R(\mathcal{X}_{(x)}|R) \notin \mathfrak{m}_R^2$ i.e., if and only if $I_R(\mathcal{X}_{(x)}|R)$ is a regular divisor in $Spf(R)$.

Proof: Let $(R_0, Z)$ be a formal versal deformation of $Spec(\hat{\mathcal{O}}_{X_0,x})$. Since $x$ in $X_0$ is a non-degenerate quadratic singularity rational over $k$, we may assume that $R_0 \simeq \Lambda[[t]]$ and $Z \simeq Spf(R_0[[x_1, \ldots, x_n]]/(F - t))$ as in 6.2.

(1) $(R, \mathcal{X}_{(\hat{x})})$, $(R, \mathcal{Y}_{(\hat{x})})$ are deformations of $Spec(\hat{\mathcal{O}}_{X_0,x})$, and therefore they are induced from $(R_0, Z)$ by continuous local $\Lambda$-algebra maps $\varphi, \psi : R_0 \to R$ respectively, i.e., if we set $a = \varphi(t)$, $b = \psi(t)$, then

$$\mathcal{X}_{(\hat{x})} \simeq Spf(R[[X_1, \ldots, X_n]]/(R - a)),$$

$$\mathcal{Y}_{(\hat{x})} \simeq Spf(R[[X_1, \ldots, X_n]]/(F - b))$$

as R-adic formal schemes. Assume that $I_R(\mathcal{X}_{(x)}|R) = I_R(\mathcal{Y}_{(x)}|R)$ i.e.,

(a) = (b) in $R$ . Then $a = ab$ for some unit $u$ in $R$ . Since $k = k^2$ and $R$ is complete, we can write $u = v^2$ for some unit $v$ in $R$ , i.e., $a = v^2 b$ . Then $(F - b) = (v^2 F - a)$ , so that $X_i \to vX_i$ establishes an R-algebra isomorphism $R[[X_1,\ldots,X_n]]/(F - a) \xrightarrow{\sim} R[[X_1,\ldots,X_n]]/(F - b)$ since $F$ is a quadratic form. The other direction is trivial.

(2) There exists a continuous local $\Lambda$-algebra map $\varphi : R_0 = \Lambda[[t]] \to R$ such that $\mathcal{X}_{(\xi)} \simeq \mathrm{Spf}(R[[X_1,\ldots,X_n]]/(F - \varphi(t)))$ . We note that $I_R(\mathcal{X}_{(x)}|R) = I_R(\mathcal{X}_{(\xi)}|R) = (\varphi(t))$ , and that $R[[X_1,\ldots,X_n]]/(F - \varphi(t)) = \hat{\mathcal{O}}_{\mathcal{X},x}$ is regular if and only if $\mathcal{O}_{\mathcal{X},x}$ is regular. Now $R$ is regular and hence $R[[X_1,\ldots,X_n]]$ is also a regular local ring. Thus if we set $M$ to be the maximal ideal of $R[[X_1,\ldots,X_n]]$ , then $R[[X_1,\ldots,X_n]]/(F - \varphi(t))$ is regular if and only if $F - \varphi(t)$ is not in $M^2$ , i.e. if and only if $\varphi(t) \notin \mathfrak{m}_R^2$ , i.e., if and only if $I_R(\mathcal{X}_{(x)}|R) \not\subset \mathfrak{m}_R^2$ .

Theorem 6.4. Let $X_0$ be a k-scheme of finite type with isolated non-smooth points only, and let $(R, \mathcal{X})$ be a formal versal deformation of $X_0$ . Denote by $\{z_1,\ldots,z_m\}$ the set of non-smooth points in $X_0$ . Assume that $H^2(X_0, D^0_{X_0}) = 0$ , and all $z_i$'s are non-degenerate quadratic singularity rational over $k$ . Then

(1) $R$ is formally smooth over $\Lambda$ i.e., $R$ is a formal power-series ring over $\Lambda$ ;

(2) $J_{\mathcal{X}}(\mathcal{X}|R) = \coprod_{1\le i\le m} J_{\mathcal{X}_{(z_i)}}(\mathcal{X}_{(z_i)}|R)$ (disjoint union), and

$J_{\mathcal{X}_{(z_i)}}(\mathcal{X}_{(z_i)}|R) \xrightarrow{\sim} J_R(\mathcal{X}_{(z_i)}|R)$ is an isomorphism;

(3) $I_R(\mathcal{X}_{(z_i)}|R)$ is a principal ideal and in turn so is $I_R(\mathcal{X}|R) = \prod_{1\le i\le m} I_R(\mathcal{X}_{(z_i)}|R)$ . Furthermore if we set $I_R(\mathcal{X}_{(z_i)}|R) = (t_i)$ , then $t_1,\ldots,t_m$ can be extended to a system of free formal generators of $R$ over $\Lambda$ (as formal power-series ring over $\Lambda$ ).

(4) $\mathcal{X} \to \mathrm{Spec}(\Lambda)$ is smooth, i.e. $J_{\mathcal{X}}(\mathcal{X}|\Lambda) = \varphi$ .

(5) If $X_0$ is a complete irreducible curve, then $\dim \underline{M}_{X_0} = 3g - 3 + a$ where $g = \dim_k H^1(X_0, \underline{O}_{X_0})$ and $a = \dim_k H^0(X_0, \underline{D}^0_{X_0})$ .

Proof: (1) follows from (2) and (3): The first statement of (2) follows from 5.7(2). Now let $(R_i, \mathcal{Y}_i)$ be a formal versal deformation of $\mathrm{Spec}(\hat{\underline{O}}_{X_0,z_i})$ . Since $(R, \mathcal{X}_{(z_i)})$ is a deformation of $\mathrm{Spec}(\hat{\underline{O}}_{X_0,z_i})$ , there exists a continuous local $\Lambda$-algebra maps $\varphi_i = R_i \to R$ with a cartesian diagram

$$\begin{array}{ccc} \mathcal{X}_{(z_i)} & \longrightarrow & \mathcal{Y}_i \\ \downarrow & & \downarrow \\ \mathrm{Spf}(R) & \xrightarrow[\mathrm{Spf}(\varphi_i)]{} & \mathrm{Spf}(R_i) \end{array}$$

Therefore we have

$$J_{\mathcal{X}_{(z_i)}}(\mathcal{X}_{(z_i)}|R) = J_{\mathcal{X}_{(z_i)}}(\mathcal{X}_{(z_i)}|R) \simeq J_{\mathcal{Y}_i}(\mathcal{Y}_i|R_i) \underset{\mathrm{Spf}(R_i)}{\times} \mathrm{Spf}(R)$$

$$J_R(\mathcal{X}_{(z_i)}|R) = J_R(\mathcal{X}_{(z_i)}|R) \simeq J_{R_i}(\mathcal{Y}_i|R_i) \underset{\mathrm{Spf}(R_i)}{\times} \mathrm{Spf}(R) .$$

Since $J_{\mathcal{Y}_i}(\mathcal{Y}_i|R_i) \to J_{R_i}(\mathcal{Y}_i|R_i)$ is an isomorphism by 6.2(1), we have that the canonical morphism $J_{\mathcal{X}_{(z_i)}}(\mathcal{X}_{(z_i)}|R) \to J_R(\mathcal{X}_{(z_i)}|R)$ is an isomorphism. Now $I_{R_i}(\mathcal{Y}_i|R_i) = (u_i) = (u_i)$ with $R_i = \Lambda[[u_i]]$ by 6.2(2), and therefore

$$I_R(\mathcal{X}_{(z_i)}|R) = I_R(\mathcal{X}_{(z_i)}|R) = (t_i)$$

where $t_i = \varphi_i(u_i)$ , and therefore $I_R(\mathcal{X}|R) = \prod_{1 \le i \le m} I_R(\mathcal{X}_{(z_i)}|R) = (\prod_{i=1}^{n} t_i)$ .

Now $\underline{M}_{X_0} \to \underline{M}_{X_0,(z_1)} \times \cdots \times \underline{M}_{X_0,(z_m)}$ is smooth by and therefore $R_1 \hat{\otimes}_\Lambda \cdots \hat{\otimes}_\Lambda R_m \to R$ is formally smooth by 4.13, which implies that $t_1, \ldots, t_m$ is extended to a system of free formal generators of $R$ over $\Lambda$ .

(4) The projection morphism $\mathcal{Y}_{(\mathfrak{L}_i)} \simeq \mathcal{U}_i \times_{\mathrm{Spf}(R_i)} \mathrm{Spf}(R) \to \mathcal{U}_i$ is formally smooth since $\mathrm{Spf}(R) \xrightarrow{\mathrm{Spf}(\varphi_i)} \mathrm{Spf}(R_i)$ is formally smooth. On the other hand, $\mathcal{U}_i \to \mathrm{Spec}(\Lambda)$ is formally smooth by 6.2(3) and therefore $\mathcal{Y}_{(\mathfrak{L}_i)} \to \mathrm{Spec}(\Lambda)$ is formally smooth. Consequently $\mathcal{Y} \to \mathrm{Spec}(\Lambda)$ is formally smooth since $\{z_1, \ldots, z_m\}$ is the set of non-smooth points of $X_0$ .

(5) Since $R$ is formally smooth over $\Lambda$ and

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to H^0(X_0, \underline{D}^1_{X_0}) \to 0$$

is exact by 4.4 and 4.11, we have

$$\begin{aligned} \dim \underline{M}_{X_0} &= \dim k \otimes R = \dim_k H^1(X_0, \underline{D}^0_{X_0}) + \dim H^0(X_0, \underline{D}^1_{X_0}) \\ &= \dim_k H^1(X_0, \underline{D}^0_{X_0}) + m \ . \end{aligned}$$

Then it suffices to show that

$$\dim_k H^1(X_0, \underline{D}^0_{X_0}) + m = 3g - 3 + \dim_k H^0(X_0, \underline{D}^0_{X_0})$$

i.e., $\chi(\underline{D}^0_{X_0}) = 3 - 3g + m$

where we set $\chi(\underline{E}) = \dim_k H^0(X_0, \underline{E}) - \dim_k H^1(X_0, \underline{E})$ for any coherent $X_0$-module $\underline{E}$ . Since $\lambda(\underline{D}^0_{X_0})$ does not change when one replaces the base field $k$ by its algebraic closure, we may assume that $k$ is algebraically closed. Set $A = \underline{O}_{X_0, z}$ , where $z$ is a non-smooth point of $X_0$ , and let $A'$ be its normalization. By hypothesis, we have $A \simeq k[[x,y]]/(f)$ where $f$ is a non-degenerate quadratic form. Since $k$ is algebraically closed, we may assume that $f = xy$ , i.e., $A \simeq k[[x,y]]/(xy)$ . From this we conclude immediately that $\ell(A'/A) = 1$ and $\mathrm{Der}_k(A, \underline{m}_A) = \mathrm{Der}_k(A, A)$ . Let $C$ be the conductor of $A$ in $A'$ . Since $A$ is a complete-intersection we must have that $\ell(A/C) = \ell(A'/A) = 1$ so that $C = \underline{m}_A$ . We thus obtain that $A' = \{a' \in K \mid a'\underline{m}_A \subset \underline{m}_A\}$ where $K$ is the fraction field of $A$ . We now claim that $\mathrm{Der}_k(A, A) \subset \mathrm{Der}_k(A', A')$ as A-modules contained in $\mathrm{Der}_k(K, K)$ . Indeed, let $X \in \mathrm{Der}_k(A, A)$ . Then for any $a' \in A'$ and

$x \in \underline{m}_A$ we have $xa' \in \underline{m}_A$ so that $X(xa') \in \underline{m}_A$ and $a'X(x) \in \underline{m}_A$ since $\mathrm{Der}_k(A, \underline{m}_A) = \mathrm{Der}_k(A, A)$ . Consequently $X(xa') = a'X(x) + xX(a')$ entails that $xX(a') \in \underline{m}_A$ for all $x \in \underline{m}_A$ , and therefore $X(a') \in A'$ . Furthermore, the exact sequence

$$0 \to \mathrm{Der}_k(A, A) \to \mathrm{Der}_k(A', A') \to A'/\underline{m}_A A' \to 0$$

entails the exact sequence

$$0 \to \mathrm{Der}_k(A, A) \to \mathrm{Der}_k(A', A') \to A'/C \to 0$$

Therefore if we set $X_0'$ to be the normalization of $X_0$ , then

$$0 \to \underline{D}^0_{X_0} \to \underline{D}^0_{X_0'} \to \underline{O}_{X_0'}/C \to 0$$

is exact. Since $X_0$ is locally a complete-intersection, we have $\ell(\underline{O}_{X_0'}/C) = 2\ell(\underline{O}_{X_0'}/\underline{O}_{X_0}) = 2m$ , whereas $\chi(\underline{D}^0_{X_0'}) = 3 - 3(g-m)$ by Riemann-Roch theorem. Consequently

$$\chi(\underline{D}^0_{X_0}) = \chi(\underline{D}^0_{X_0'}) - \ell(\underline{O}_{X_0'}/C) = 3 - 3(g-m) - 2m = 3 - 3g + m \ .$$

This completes our proof.

Remark 6.5. In the formula 6.4(5), the number $a = H^0(X_0, \underline{D}^0_{X_0})$ can be easily computed in terms of Riemann-Roch theorem. Thus

$$a = \begin{cases} 0 & \text{if } g \geq 2 \\ 1 & \text{if } g = 1 \\ 3 & \text{if } g = 0 \end{cases}$$

Finally we add a remark on characteristic 2. If the characteristic of the base field $k$ is $2$ , there is no non-degenerate quadratic form on $n$ variables if $n$ is odd. In other words there does not exist a non-degenerate quadratic singularity of even dimension if $\mathrm{char}\ k = 2$ . In this case one may consider a quadratic form of maximum non-degeneracy:

Definition 6.6. Let $X$ be a $k$-scheme of finite type. A closed point $z \in X$ is called an ordinary quadratic singularity rational over $k$ if the following holds:

(i) If either char $k \neq 2$ or char $k = 2$ and dim $\mathcal{O}_{X,z}$ is odd, then a is a non-degenerate quadratic singularity rational over $k$.

(ii) If char $k = 2$ and dim $\mathcal{O}_{X,z}$ is even, then $\bar{k} \underset{k}{\otimes} \hat{\mathcal{O}}_{X,z} \simeq \bar{k}[[X_0, X_1, \ldots, X_{2n}]]/(f)$ with

$$f = X_0^2 + X_1X_2 + X_3X_4 + \ldots + X_{2n-1}X_{2n}$$

where $\bar{k}$ is the algebraic closure of $k$.

Lemma 6.7. Let $k$ be a field of characteristic 2 and set $A_0 = k[[X_0, X_1, \ldots, X_{2n}]]/(f)$ where

$$f = X_0^2 + X_1X_2 + X_3X_4 + \ldots + X_{2n-1}X_{2n} ,$$

and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \mathrm{Spec}(A_0)$. Then $R = \Lambda[[t, t_1]]$, where $t, t_1$ are free formal generators of $R$ over $\Lambda$, such that $I_R(\mathfrak{X}|R) = (t^2)$, and $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ is formally smooth.

Proof: We set $\mathfrak{X} = \mathrm{Spf}(A)$. Since

$$(f, \frac{\partial f}{\partial X_0}, \ldots, \frac{\partial f}{\partial X_{2n}}) = (X_0^2, X_1, \ldots, X_{2n}) ,$$

it follows that $D^1(A_0|k, A_0) \simeq k[X_0]/(X_0^2)$ and therefore $[\underline{M}_{X_0}(k[\epsilon])]$ is 2-dimensional vector space over $k$. Furthermore, it follows from 4.14 that

$$R \simeq \Lambda[[t, t_1]] \text{ and } A \simeq R[[X_0, X_1, \ldots, X_{2n}]]/(F)$$

where $F = f + t_1 + tX_0$. Now the exact sequence

$$0 \to A \xrightarrow{dF} AdX_0 \oplus \ldots \oplus AdX_{2n} \to \hat{\Omega}_{A|R} \to 0$$

where

$$dF = 1 \to \sum_{0 \leq i \leq 2n} \frac{\partial f}{\partial X_i} dX_i + dX_0 ,$$

entails that

$$I_A(A|R) = (t, x_1, x_2, \ldots, x_{2n}) = (x_0^2 + t_1, t, x_1, \ldots, x_{2n})$$

so that

$$A/I_A(A|R) \simeq A/(x_0^2 + t_1, t, x_1, \ldots, x_{2n}) \simeq \Lambda[[t_1, X_0]]/(x_0^2 - t_1) \ .$$

Since $\Lambda[[t_1, X_0]]/(x_0^2 - t_1) \simeq R/(t) \oplus R/(t)$ as R-modules, it follows that $I_R(A|R) = (t^2)$ , which proves our first assertion. On the other hand,

$$A \simeq \Lambda[[t, t_1, x_o, x_1, \ldots, x_{2n}]]/(x_0^2 + X_1X_2 + \ldots + X_{2n-1}X_{2n} + tX_0 + t_1)$$

$$\simeq \Lambda[[t, X_0, X_1, \ldots, X_{2n}]]$$

and therefore A is formally smooth over $\Lambda$ .

The above statement 6.7 yields a similar result as 6.4, proof of which we leave to the readers.

Corollary 6.8. Let $X_0$ be a k-scheme of finite type with isolated non-smooth points only such that $H^2(X_0, \underline{D}^0_{X_0}) = 0$ , and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0$ . Denote by $\{z_1, \ldots, z_m\}$ the set of non-smooth points in $X_0$ , and assume that char k = 2 , dim $X_0$ is even, and all $z_i$'s are ordinary quadratic singularities. Then

(1) R is formally smooth over $\Lambda$

(2) $I_R(\mathfrak{X}_{(z_i)}|R)$ is a principal ideal and in turn so is $I_R(\mathfrak{X}|R) = \prod_{1 \le i \le m} I_R(\mathfrak{X}_{(z_i)}|R)$ . Furthermore, $I_R(\mathfrak{X}_{(z_i)}|R) = (t_i^2)$ and $t_1, \ldots, t_m$ can be extended to a system of free formal generators of R over $\Lambda$ .

(3) $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ is formally smooth

(4) If k is algebraically closed and $X_0$ is a complete irreducible curve, then $\dim \underline{M}_{X_0} = 3g - 3 + a$ where $g = \dim_k H^1(X_0, \underline{O}_{X_0})$ and $a = \dim_k H^0(X_0, \underline{D}^0_{X_0})$ .